

Environment and Ecology

Teaching tables, Confused Pairs, Common Traps, Current Affairs, and other high-yield blocks.

UPPCS + UKPCS Study Library · chapter order follows Subject Notes.

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Topic 1 — Environment Basics

Current Affairs

Year	Fact	Why asked
2023	Amrit Dharohar Scheme — wetlands conservation; launched on WED 2023	Wetlands + governance
2023	National Green Hydrogen Mission	SD / climate policy overlap
2019	NCAP (राष्ट्रीय स्वच्छ वायु कार्यक्रम) — 132 cities; 20–30% PM2.5/PM10 reduction target by 2024	Urban (नगरीय) environment governance

Confused Pairs

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
Environment	Ecology	Total biotic and abiotic surroundings influencing an organism vs the scientific <i>study</i> of interactions between organisms and environment	पर्यावरण / पारिस्थितिकी
Ecology	Ecosystem	Scientific discipline/field of study (Ernst Haeckel 1866) vs functional, structural ecological <i>unit</i> of nature (A.G. Tansley 1935)	पारिस्थितिकी / पारिस्थितिकी तंत्र
Habitat	Ecological Niche	Physical address / geographical place where an organism lives vs functional profession / ecological role of a species in an ecosystem	वास स्थान / पारिस्थितिक निकेत (Niche)
Biotic Components	Abiotic Components	Living organisms (producers, consumers, decomposers) vs non-living physical and chemical factors (light, temperature, soil, water)	जैविक घटक / अजैविक घटक
Lentic Ecosystem	Lotic Ecosystem	Standing or stagnant fresh water bodies (ponds, lakes, bogs, swamps) vs running or flowing fresh water bodies (rivers, streams, brooks)	स्थिर जल (लैटिक) / प्रवाही जल (लोटिक)
Liebig's Law of the Minimum	Shelford's Law of Tolerance	Population growth is controlled by the scarcest limiting resource vs species survival requires environmental factors to remain between minimum and maximum tolerance limits	लाइबिग का न्यूनतम नियम / शेल्फोर्ड का सहनशीलता नियम
Internal Environment	External Environment	Biochemical and physiological conditions within an organism (homeostasis) vs physical surroundings external to the organism	आंतरिक पर्यावरण / बाह्य पर्यावरण
Physical Environment	Biological Environment	Inanimate non-living components (lithosphere, hydrosphere, atmosphere) vs living communities of flora, fauna, and microbes	भौतिक पर्यावरण / जैविक पर्यावरण
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1.1 Environment – Meaning

Source	Definition
NCERT (Class 12)	Sum of all biotic + abiotic factors influencing living organisms ← most asked
UNEP	Physical, chemical, and biotic factors acting on organism/community, determining form and survival
EPA 1986 (पर्यावरण संरक्षण 1986) (Legal)	Water, air, land + inter-relationship among them and with humans, other living beings, plants, and property
General	External conditions — physical, chemical, biological, social, economic, cultural — affecting life

1.2 Environmental Components

Abiotic factor category	Includes	Hindi
Edaphic	Soil texture, minerals, pH, moisture	मृदा संबंधी
Climatic	Temperature, rainfall, wind, humidity	जलवायु संबंधी
Topographic	Altitude, slope, aspect	स्थलाकृति संबंधी
Biotic group	Also called	Function
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Producers	Autotrophs	Convert solar energy → chemical energy
Consumers	Heterotrophs	Feed on other organisms
Decomposers	Saprotrophs	Break down dead matter; recycle nutrients
Detritivores	—	Feed on detritus (earthworms, termites)

1.3 Classification of Environment

Basis	Categories
Standard triad	Physical / Biological / Cultural (Operational is not a category)
Origin	Natural vs Anthropogenic (human-modified — cities, dams, cropland)
Components	Abiotic vs Biotic
Domain	Terrestrial (~29% surface) vs Aquatic (~71%)
Organism level	Internal (inside body — physiology) vs External (outside — ecology default)
Scale	Micro / Meso / Macro
Human use	Rural / Urban / Industrial

Physical, biological, and cultural environment

Aquatic type	Description	Indian example
Marine	Oceans/seas — high salinity	Arabian Sea, Bay of Bengal
Lentic	Standing freshwater	Loktak Lake (लोकटक), Dal Lake
Lotic	Flowing freshwater	Ganga, Yamuna
Estuarine	River meets sea — brackish	Sundarbans (सुंदरबन) delta (डेल्टा)
Wetland	Transitional — neither fully land nor water	Chilika, Keoladeo (केवलादेव)
Marine zone	Location	Key fact
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Neritic	Shallow shelf (0–200 m)	Fisheries, coral, mangroves
Pelagic	Open water column	Plankton, tuna
Benthic	Sea floor	Crabs, decomposers
Littoral	Intertidal shore	Barnacles, molluscs

India's 10 biogeographic zones

#	Zone	feature
1	Trans-Himalaya	Cold desert — Ladakh (लद्दाख), high-altitude sparse vegetation
2	Himalaya (हिमालय)	Alpine to subtropical belts along the main Himalayan range

#	Zone	feature
3	Desert	Thar (थार) — Rajasthan and adjoining Gujarat
4	Semi-Arid	Dry scrub/savanna (सवाना) fringe of Gujarat–Maharashtra–central India
5	Western Ghats (पश्चिमी घाट)	Biodiversity hotspot (जैव विविधता हॉटस्पॉट); high endemism
6	Deccan (दक्कन) Peninsula	Plateau forests and dry deciduous tracts
7	Gangetic Plain	Fertile alluvial (जलोढ़) plains — covers most of UP
8	Coasts	Mangroves, estuaries, coral reefs
9	North-East India	Indo-Malayan biodiversity corridor
10	Islands	Andaman (अंडमान) & Nicobar (निकोबार) and Lakshadweep (लक्षद्वीप)

1.4 Human–Environment Interaction

Model	Proponent	Idea	Status
Environmental Determinism	Ratzel	Environment dictates culture/society	Largely rejected
Possibilism	Vidal de la Blache	Environment offers possibilities ; humans choose	Accepted
Neo-determinism	Griffith Taylor	Humans modify environment within limits	Balanced view
Concept	Definition / fact		
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Carrying capacity	Maximum load an environment can sustain indefinitely. Resource extraction must stay below regeneration. Waste must stay within absorption capacity		
IPAT	Impact = P opulation × A ffluence × T echnology		
Ecological footprint (पारिस्थितिक पदचिह्न)	Land/water area needed to support consumption — measured in global hectares (gha)		
Tragedy of the Commons	Shared resources overexploited when individual interest > collective good — Garrett Hardin, 1968		
Sustainable Development	“Meet present needs without compromising future generations’ ability to meet theirs” — Brundtland Report, 1987 (<i>Our Common Future</i>)		

Common Traps — Don’t Fall For These

Trap	Correct fact
Art 48 vs 48A	Environment = 48A (48 = agriculture/cows)
Humus = abiotic?	No — organic, biotic-origin
Mangrove = terrestrial?	No — coastal/transitional
MoEFCC est. 1972?	No — 1985 (1972 = Stockholm)
Lentic = flowing?	No — Lentic = standing
Wildlife Day = 22 March?	No — 3 March
Agenda 21 = 1995?	No — Rio 1992

Trap	Correct fact
Brundtland = <i>Limits to Growth</i> ?	No — Brundtland = <i>Our Common Future</i> (1987)

Topic 2 — Ecology & Ecosystem

Current Affairs (this topic)

- Ecosystem-services and productivity stems stay conceptual — refresh MEA / IPBES wording if a CA stem cites (साइट्स) them.
- Keep Tansley / Haeckel / Odum name traps stable; they are not CA-dependent.

Confused Pairs

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
Ecology (Haeckel 1866)	Ecosystem (Tansley 1935)	Scientific academic study of organism-environment interactions vs concrete structural-functional unit of nature combining biotic and abiotic components	पारिस्थितिकी / पारिस्थितिक तंत्र
Habitat	Ecological Niche	Physical place or environmental zone where a species resides vs unique functional role, trophic position, and resource utilization pattern (Grinnell)	वास स्थान / पारिस्थितिक निकेत
Population	Ecological Community	Group of interbreeding individuals of the <i>same</i> species in an area vs assemblage of populations of <i>different</i> interacting species in an area	जनसंख्या (समष्टि) / जैविक समुदाय
Gross Primary Productivity (GPP)	Net Primary Productivity (NPP)	Total rate of organic matter synthesized by producers via photosynthesis vs actual biomass available to consumers after plant respiration losses ($NPP = GPP - R$)	सकल प्राथमिक उत्पादकता (GPP) / शुद्ध प्राथमिक उत्पादकता (NPP)
Fundamental Niche	Realized Niche	Full theoretical potential range of environmental conditions and resources a species can use without competition vs actual restricted niche occupied due to biotic competition	मूल (मौलिक) निकेत / वास्तविक निकेत
Primary Succession	Secondary Succession	Ecological succession beginning on barren, lifeless substrate with no pre-existing soil (cooled lava, bare rock) vs succession on disturbed areas where soil already exists (after forest fire, flood)	प्राथमिक अनुक्रमण / द्वितीयक अनुक्रमण
r-Selected Species	K-Selected Species	High reproductive rate, small body, short lifespan, little parental care (insects, rodents, weeds) vs low reproductive rate, large body, long lifespan, high parental care (elephants, whales, humans)	r-रणनीतिकारी / K-रणनीतिकारी प्रजाति
Ecotone	Ecotype	Sharp transition boundary zone between two distinct ecological communities (mangroves, forest-grassland edge) vs genetically distinct geographic variety or sub-population within a species	इकोटोन (संक्रमण क्षेत्र) / इकोटाइप
Sere	Climax Community	Entire chronological transitional sequence of developmental communities during ecological succession vs final, stable, self-perpetuating terminal community in equilibrium with regional climate	क्रमक (Sere) / चरम समुदाय
Bionomics	Ecology	Historical 19th-century synonym of ecology meaning "laws of life" (UPPCS trap: bionomics is NOT management of life) vs modern ecology	बायोनॉमिक्स / पारिस्थितिकी
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2.1 Ecology – Meaning

Level	example
Organism	Camel water conservation in desert
Population	Tiger population in Dudhwa (दुधवा)
Community	Fish, birds, and plants sharing a wetland
Ecosystem	Ganga stretch at Varanasi (वाराणसी) with its biota, water, and nutrient flows
Biome	Tropical deciduous forest of India
Biosphere	Global life-support zone

2.3 Types of Ecosystems

Basis	Categories
Origin	Natural vs Anthropogenic (human-modified)
Medium	Terrestrial (~29% Earth's surface) vs Aquatic (~71%)
Water flow (freshwater)	Lentic (standing) vs Lotic (flowing)
Salinity	Freshwater vs Brackish vs Marine

Pond and lake zonation (lentic)

Zone	Location	Key fact
Littoral	Shallow shore with rooted plants	Warmest, highest light
Limnetic	Open water away from shore	Phytoplankton zone
Profundal	Deep water below light penetration	Low light, decomposers dominate

2.4 Structure, Functions & Components

Four functions of an ecosystem

Function	What it means
Productivity	Rate at which biomass or energy is produced
Decomposition	Breakdown of dead organic matter
Nutrient cycling	Movement of chemical elements through biotic and abiotic pools
Energy flow	One-way transfer of energy through trophic levels

2.8 Ecological Succession

Phases of biotic succession (full sequence)

Phase	What happens
Nudation	Creation of a bare site free of organisms (erosion, fire, glaciation, human clearance)
Migration	Seeds, spores, or other propagules reach the bare or disturbed area
Ecesis	The newcomers successfully establish, germinate, and grow (colonisation)
Reaction	Established species modify the habitat (soil, shade, moisture) for the next community

Phase	What happens
Stabilisation	Change slows and the climax community dominates

2.9 Ecosystem Services

Category	What humans get	Example
Provisioning	Direct material products	Food, water, timber
Regulating	Processes that stabilise the environment	Climate/disease control
Supporting	Services that underpin all other services	Nutrient cycling, soil formation
Cultural	Non-material benefits	Recreation, spiritual value
Preserving (WDR/UPSC five-fold framing)	Long-term diversity maintenance	Maintenance of diversity

2.10 Ecosystem Productivity

Term	Meaning
GPP (Gross Primary Productivity)	Total energy or biomass fixed by producers through photosynthesis
R	Energy lost by producers through respiration
NPP (Net Primary Productivity)	GPP – R = biomass actually available to consumers
NSP	NPP remaining after heterotroph consumption at producer level
Standing crop	Biomass present in an ecosystem at a given instant

2.12 Genetic, Species & Community Ecology

Community level ecology

Interaction	What happens	Example
Competition	Two species vie for the same limited resource	Two grazers on one pasture
Predation	One organism kills and eats another	Tiger hunting deer
Parasitism	One benefits while the host is harmed	Tick on cattle
Mutualism	Both species benefit	Azolla with <i>Anabaena</i> in rice fields
Commensalism	One benefits and the other is unaffected	Cattle egret feeding near grazing cattle

Common Traps — Don't Fall For These

Trap	Correct fact
Odum coined "ecosystem"	Tansley 1935 coined ecosystem
Ecosystem is a closed system	Ecosystem is normally open
Ecesis before Migration	Full: Nudation → Migration → Ecesis → Reaction → Stabilisation
Sere = nutrient cycle	Sere = succession community sequence
Lichens thrive in polluted air	Lichens are sensitive indicators
1% and 10% cannot both be true	Both true in different stages
Energy increases up food chain	Energy decreases (~10% transfer)

Trap	Correct fact
Coral reefs = most stable ecosystem	Coral reefs = high diversity ; marine = high stability
Primary succession has soil already	Primary = no soil ; secondary = soil present
Bionomics = management of life	Bionomics = ecology
Niche coined by Odum	Niche concept linked to Grinnell (1917)
Ecological Transition = Ratzel	John W. Bennett
ESZ under WPA 1972 / total human ban	ESZ under EPA 1986 ; activities regulated, not all banned
Eucalyptus is eco-friendly	Often treated as groundwater hazard
Productivity: oceans first	Decreasing order Mangroves > grasslands > lakes > oceans

Topic 3 — Food Chain & Energy Flow

Current Affairs (this topic)

- Energy-flow and trophic-level traps recycle yearly — 10% rule and pyramid types stay core.
- Bioaccumulation (जैव संचय) / biomagnification (जैव आवर्धन) CA usually rides on pesticide or heavy-metal news; keep the mechanism pair clear.

Confused Pairs

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
Food Chain	Food Web	Single linear sequence of organisms where nutrients and energy transfer through feeding vs complex network of interconnected, interlocking food chains providing alternative feeding pathways	खाद्य श्रृंखला / खाद्य जाल
Grazing Food Chain (GFC)	Detritus Food Chain (DFC)	Begins with living green photosynthetic plants (producers) and drives terrestrial energy vs begins with dead organic matter (detritus) decomposed by fungi and bacteria	चराई खाद्य श्रृंखला / अपरद (अवशेष) खाद्य श्रृंखला
Upright Pyramid of Biomass	Inverted Pyramid of Biomass	Characteristic of terrestrial grasslands and forests where producer biomass far exceeds herbivores vs characteristic of aquatic/pond ecosystems where standing crop of phytoplankton is smaller than zooplankton/fish	सीधा जैवभार पिरामिड (स्थलीय) / उल्टा जैवभार पिरामिड (जलीय/तालाब)
10% Energy Transfer Law	Bioaccumulation	Raymond Lindeman (1942) principle that only ~10% of energy transfers from one trophic level to the next vs progressive buildup of non-biodegradable chemicals within an individual organism over time	10% ऊर्जा नियम (लिंडमैन) / जैव संचय (बायोएक्यूमुलेशन)
Bioaccumulation	Biomagnification	Increase in toxicant concentration within a single organism's tissues through direct absorption from water/food vs increase in toxic chemical concentration (e.g. DDT, mercury) across successive trophic levels of a food chain	जैव संचय (एक जीव में) / जैव आवर्धन (खाद्य श्रृंखला में)
Primary Consumer	Secondary Consumer	Herbivores feeding directly on autotrophic green plants (cow, deer, zooplankton) vs primary carnivores that hunt and feed on herbivores (frog, fox, small fish)	प्राथमिक उपभोक्ता (शाकाहारी) / द्वितीयक उपभोक्ता (मांसाहारी)
Pyramid of Energy	Pyramid of Numbers	Always universally upright in all ecosystems because energy is irretrievably lost as metabolic heat at each level vs can be upright (grassland) or inverted (single large tree supporting thousands of birds/parasites)	ऊर्जा पिरामिड (सदैव सीधा) / संख्या पिरामिड
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3.1 Food Chain

Type	Starts from	UP / textbook example
Grazing	Living producers	Grass → deer → tiger (Dudhwa); Grass → grasshopper → frog → snake
Detritus	Dead organic matter	Leaf litter → earthworm → fungi
Marine	Phytoplankton/diatoms	Diatoms → crustaceans → herrings
Parasitic (special)	Living host tissue	Tree → insects → parasites

3.3 Trophic Levels

Bioaccumulation and Biomagnification

Level	Role	UP / India example
T1	Producer	Terai grass, wheat, Ganga algae
T2	Primary consumer (प्राथमिक उपभोक्ता)	Barasingha, chital, rohu (juvenile)
T3	Secondary consumer (द्वितीयक उपभोक्ता)	Frog, catfish
T4	Tertiary consumer	Python, fishing eagle
T5	Top carnivore	Tiger (Dudhwa), Gangetic dolphin (river apex)

3.4 Energy Flow in Ecosystem

Where energy goes at each level	Key fact
Respiration and heat	Largest share — why pyramid of energy (ऊर्जा पिरामिड) is upright
Movement and metabolism	Not passed to next consumer
Egestion and uneaten parts	Lost from food chain pathway
Passed to next trophic level	~10% (Lindeman 1942)

3.5 Ecological Pyramid (Ecological Pyramids)

Pyramid	Unit	Always upright?	Inverted example
Energy	kcal/m ² /yr	Yes — always	Never inverted
Biomass	g/m ²	Usually	Open ocean (phytoplankton turnover)
Numbers	Individual count	Usually	One tree → many insects/parasites

Common Traps — Don't Fall For These

Trap	Correct fact
Energy flow is cyclic	Unidirectional — nutrients cycle, energy does not
All pyramids always upright	Only energy pyramid is always upright
Grazing chain starts from dead matter	Starts from living green plants
Sun is Trophic Level 1	Sun = energy source ; Producer = T1
10% rule by Darwin	Lindeman 1942
Food web makes energy cyclic	Each pathway is one-way
Biomass pyramid measures energy flow	Biomass = g/m² ; energy = kcal/m²/yr
Open ocean = inverted number pyramid	Open ocean / pond = inverted biomass pyramid
Man is only secondary consumer	Man is primary + secondary consumer
Virus is a decomposer	Decomposers = fungi + bacteria
Food chain within one species	Food chains are across feeding links, not within one species
Highest DDT in grasshopper	Highest in top consumer (e.g. snake)

Topic 4 — Biodiversity

Current Affairs (this topic)

- Track new Ramsar (रामसर) and IUCN (आईयूसीएन) Red List updates against the Must-Score baselines (India Ramsar ~100; UP 12).
- Invasive-species and hotspot stems recycle yearly — keep Western Ghats (पश्चिमी घाट) / Indo-Burma trap pairs stable.

Confused Pairs

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
Genetic Diversity	Species Diversity	Variation of genes and alleles <i>within</i> a single species across its geographic range vs the variety and abundance of <i>different species</i> within a given ecological region	आनुवंशिक विविधता / प्रजाति विविधता
Species Diversity	Ecosystem Diversity	Number of species and their relative abundance (species richness + evenness) vs variety of distinct habitats, biotic communities, and ecological processes in a landscape	प्रजाति विविधता / पारिस्थितिक तंत्र विविधता
Alpha Diversity	Beta Diversity	Species richness measured <i>within</i> a particular local habitat or community vs rate of species turnover / change in species composition <i>between</i> two different habitats across an environmental gradient	अल्फा विविधता (स्थानीय) / बीटा विविधता (आवासीय परिवर्तन)
In-situ Conservation	Ex-situ Conservation	Protecting endangered species within their natural evolutionary habitats (National Parks, Sanctuaries, Biosphere Reserves) vs conserving species outside their natural habitats in artificial environments (Zoological Parks, Botanical Gardens, Gene Banks)	स्व-स्थाने संरक्षण (In-situ) / बाह्य-स्थाने संरक्षण (Ex-situ)
Endangered (EN)	Critically Endangered (CR)	Very high risk of extinction in the wild (e.g. Asiatic Lion, Tiger) vs extremely high imminent risk of extinction in the wild (e.g. Great Indian Bustard, Gharial, Hangul)	लुप्तप्राय (EN) / अति संकटग्रस्त (CR)
Endemic Species	Invasive Alien Species	Native species strictly restricted to a specific unique geographic territory and found nowhere else (e.g. Lion-tailed Macaque in Western Ghats) vs non-native species introduced from outside that outcompete and displace indigenous wildlife (e.g. Lantana camara, Water Hyacinth)	स्थानिक प्रजाति (Endemic) / आक्रामक विदेशी प्रजाति (Invasive)
Biodiversity Hotspot	Hope Spot	Terrestrial regions with exceptionally high endemic species richness under severe threat ($\geq 1,500$ endemic vascular plants and $\geq 70\%$ primary vegetation lost; Norman Myers) vs marine protected areas critical to the health of global oceans (Sylvia Earle / Mission Blue)	जैव विविधता हॉटस्पॉट (स्थलीय) / होप स्पॉट (समुद्री)
People's Biodiversity Register (PBR)	Red Data Book	Mandated under Biological Diversity Act 2002 prepared by local Biodiversity Management Committees (BMCs) documenting traditional ecological knowledge vs official compendium published by IUCN documenting global extinction risks	जन जैव विविधता रजिस्टर (PBR) / रेड डेटा बुक (IUCN)
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4.3 Global & Indian Biodiversity

Region	Megadiverse countries (17)
South America	Brazil, Colombia, Ecuador, Peru, Venezuela
North / Central America	Mexico, United States

Region	Megadiverse countries (17)
Asia	China, India, Malaysia, Indonesia, Philippines
Africa	Democratic Republic of Congo, South Africa, Madagascar
Oceania	Australia, Papua New Guinea

Group	India's rough share of global species
Mammals	~7.6%
Birds	~12.6%
Reptiles	~6.2%
Amphibians	~4.4%
Indian endemism centre	Note
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Western Ghats (पश्चिमी घाट)	Richest among common Indian options
Northeast India	High endemism belt
Himalaya (हिमालय)	Richness from confluence of biogeographic zones
Andaman & Nicobar Islands	Island endemism

4.5 Biodiversity Assessment & Registers

IUCN Red List categories

Category	Meaning
LC	Least Concern
NT	Near Threatened
VU	Vulnerable
EN	Endangered
CR	Critically Endangered
EW	Extinct in the Wild
EX	Extinct

4.8 Biodiversity Hotspots

Hotspot	Indian region	UP link
Himalaya (often called Eastern Himalaya in older stems)	Himalaya + Terai belt; Sikkim is a botanist's paradise here	Dudhwa, Pilibhit Terai
Indo-Burma / Indo-Myanmar	Northeast India + Andaman Islands	—
Western Ghats & Sri Lanka	West coast mountains	—
Sundaland	Nicobar Islands (not Andaman)	Not mainland

4.9 Species Categories

Category	One-line fact	India example
Endemic	Found nowhere else in the world	Asiatic lion (सिंह) (Gir (गीर)), lion-tailed macaque (Western Ghats)
Endangered	IUCN EN — very high extinction risk	Gangetic dolphin, Indian vulture
Threatened	VU + EN + CR together	Broader than endangered alone
Extinct	No surviving individuals (EX) or wild population (EW)	Cheetah in India (1952 wild extinction)
Invasive alien	Non-native + ecological or economic harm	Water hyacinth, lantana, Parthenium
Keystone	Disproportionately large ecosystem impact if removed	Sea otter (kelp forest), tiger (apex regulation)
Flagship	Charismatic species that mobilises public support	Tiger, elephant, one-horned rhino
Umbrella	Protecting its large range saves many other species	Tiger reserve network
Indicator	Reflects environmental quality or change	Lichens (air pollution), frog (wetland health)

Common Traps — Don't Fall For These

Trap	Correct fact
Biodiversity = species count only	Genetic + species + ecosystem
Wilson coined "biodiversity"	Walter G. Rosen (1985) coined; Wilson popularised
Alpha = regional diversity	Alpha = local mean; gamma = regional
Russia is megadiverse	Not in the standard 17 list
India has 6 hotspots	4 — Himalaya, Indo-Burma, WG–Sri Lanka, Sundaland
Sundaland = Andaman + Nicobar	Nicobar under Sundaland; Andaman under Indo-Burma
Eastern Ghats / Vindhyas = hotspot	Not hotspots
Hotspots only in tropics	Temperate hotspots also exist
Threatened = Endangered	Threatened = VU + EN + CR
Botanical garden = in-situ	Ex-situ
Cryo-bank gas = oxygen / CO ₂	Liquid nitrogen
Lichens thrive in pollution	Sensitive indicators — fail in pollution
Ex-situ preferred over in-situ	In-situ preferred
NBA headquarters Delhi	Chennai
BirdLife invented hotspots	Norman Myers ; BirdLife = IBAs
TRAFFIC = UNEP bureau	WWF–IUCN NGO alliance
Coral reefs in Sundarbans	Sundarbans = mangroves ; reefs = GoM, GoK, A&N, Lakshadweep
Vegetarianism threatens biodiversity	Not a threat
NCMB = Mumbai / Bhavnagar	Jamnagar
Silent Valley = Uttarakhand	Kerala ; Valley of Flowers = Chamoli UK
India Ramsar ≈ 26 forever	About 80 ; TN 16 , UP 10 , Punjab 6
Gangetic Plain is a hotspot	Not a global hotspot

Topic 5 — Habitat, Flora & Fauna

Current Affairs (this topic)

- Habitat-loss and corridor news often refresh Terai (तराई) / Himalaya (हिमालय) colour for UP–UK.
- Do not swap endemic vs endangered wording when CA names a new Red List update.

Confused Pairs

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
Habitat	Ecological Niche	Physical geographic environment and specific spatial location where an organism resides vs functional status, behavioral role, and trophic interactions of that species in the community	आवास / पारिस्थितिक निकेत
Flora	Fauna	Total native plant species assemblages inhabiting a particular geographic region or geological era vs total native animal species inhabiting a region or era	वनस्पति (फ्लोरा) / प्राणीजात (फॉना)
Microhabitat	Ecotone	Small, localized physical niche within a broader habitat (e.g., under a rotting log, a tree cavity) vs macro-scale transitional boundary ecosystem between two adjacent distinct biomes (e.g., marsh between dry land and water)	सूक्ष्म आवास (माइक्रोहैबिटेट) / इकोटोन (संक्रमणिका)
Keystone Species	Foundation Species	Species having a disproportionately massive impact on ecosystem structure relative to its biomass (loss triggers ecological collapse; e.g. Fig tree, Sea Otter) vs dominant primary producer that physically creates and structures the habitat (e.g. Corals, Kelp, Mangroves)	कीस्टोन प्रजाति / आधारभूत (फाउंडेशन) प्रजाति
Umbrella Species	Flagship Species	Wide-ranging species whose habitat protection automatically shields hundreds of co-occurring species (e.g. Tiger, Asian Elephant) vs charismatic, culturally iconic species chosen as marketing symbols to rally conservation campaigns (e.g. Giant Panda, Polar Bear)	छत्र (अंब्रेला) प्रजाति / प्रतीक (फ्लैगशिप) प्रजाति
Indicator Species	Invasive Species	Species whose presence, absence, or abundance directly signals environmental stress or health (e.g. Lichens for SO ₂ air pollution, Mayfly larvae for clean water) vs non-native species proliferating aggressively and threatening native ecology	जैव संकेतक प्रजाति / आक्रमक विदेशी प्रजाति
In-situ Conservation	Ex-situ Conservation	On-site conservation in natural ecosystems (Sacred Groves, Wildlife Sanctuaries) vs off-site conservation under artificial human custody (Seed Banks, Tissue Culture Labs)	स्व-स्थाने संरक्षण / बाह्य-स्थाने संरक्षण
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5.1 Habitat

Major habitat types (classification)

Type	Meaning	India example
Terrestrial	Land-based	Terai sal forest, Thar (थार) desert
Aquatic freshwater	Ponds, lakes, rivers	Ganga, Loktak
Marine	Oceans and seas	Arabian Sea coral/ fisheries belt
Wetland	Seasonally or permanently waterlogged	Keoladeo (केवलादेव), UP Ramsar (रामसर) sites
Mangrove / coastal	Tidal saline (लवणीय) coast	Sundarbans (सुंदरबन)
Grassland / savanna (सवाना)	Grass-dominated open tract	Dudhwa grasslands

Type	Meaning	India example
Alpine / montane	High-altitude cold zone	Himalayan snow leopard belt
Desert / arid	Low rainfall extreme temperature	Thar
Arboreal	Tree-dwelling	Lion-tailed macaque
Fossorial	Burrowing underground	Indian pangolin
Cursorial	Open plain runners	Blackbuck
Xericole	Desert-adapted	Camel, desert fox
Amphibious	Both land and water phases	Crocodile, bullfrog

5.2 Habitat Conservation

Key schemes and policies

Scheme / policy	Year	Ministry	Fact
MISHTI	Budget 2023–24	MoEFCC	Mangrove shoreline habitat restoration + community income
NUHHP	2007	MoHUA	Urban housing and sustainable urban habitat
JNNURM	2005	MoHUA	Urban infrastructure renewal
AMRUT	2015	MoHUA	Urban water, sewerage, green spaces
Jal Jeevan Mission	2019	Jal Shakti (शक्ति)	Rural tap water connectivity
CAMPA	2009 / Act 2016	MoEFCC	Compensatory afforestation for diverted forest land
Green India Mission	2014	MoEFCC	Forest cover (वन आवरण) and habitat quality improvement under NAPCC

5.3 Flora and Fauna

UP flora and fauna facts

Category	UP example	Note
State animal	Barasingha (swamp deer)	Terai grassland specialist
State bird	Sarus crane	Wetland and farmland mosaic
Dominant Terai flora	Sal (<i>Shorea robusta</i>)	Dudhwa forest tree
Aquatic fauna	Gangetic dolphin	Ganga–Yamuna stretch
Flagship habitat fauna	Tiger, rhino, elephant	Dudhwa and Pilibhit
Invasive flora	Parthenium, Lantana, water hyacinth	Degrade native habitats

Common Traps — Don't Fall For These

Trap	Correct fact
Habitat = niche	Habitat = place ; niche = role
Mangrove = terrestrial	Mangrove = transitional
MISHTI launched 2019	Budget 2023–24
BSI surveys animals	BSI = plants ; ZSI = animals
Masai in West Africa	Masai = East Africa

Trap	Correct fact
Zoos = in-situ	Zoos = ex-situ
NUHHP before JNNURM	JNNURM 2005 → NUHHP 2007
Mangrove flora in UP Terai	UP has no coastline

Topic 6 — Protected Areas & Conservation

Current Affairs (this topic)

- Track new Ramsar notifications and tiger-reserve counts against the Must-Score totals.
- Keep UP Ramsar **12** and India ~**100** as the teaching baseline unless a stem cites (साइट्स) a later official total.

Confused Pairs

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
National Park (NP)	Wildlife Sanctuary (WLS)	Declared under WPA 1972 Sec 35; highest protection, zero human settlement, grazing banned, boundaries altered only by State Legislature on NBWL advice vs WPA Sec 18; regulated grazing and traditional rights may be permitted by Chief Wildlife Warden	राष्ट्रीय उद्यान (NP) / वन्यजीव अभयारण्य (WLS)
Conservation Reserve	Community Reserve	Established on government-owned lands acting as buffer corridors between PAs under WPA Sec 36A vs established on private or community-owned lands with voluntary community participation under WPA Sec 36C	संरक्षण आरक्षित क्षेत्र / सामुदायिक आरक्षित क्षेत्र
Tiger Reserve	Elephant Reserve	Statutory protected area notified under Wildlife Protection Act 1972 (amended 2006) with legally delineated Core and Buffer zones supervised by NTCA vs administrative/executive designation under Project Elephant without statutory backing under WPA	बाघ अभयारण्य (सांविधिक) / हाथी अभयारण्य (प्रशासनिक)
Biosphere Reserve (MAB)	National Park	Landscape-level conservation framework under UNESCO MAB with 3 distinct concentric zones (Core, Buffer, Transition) integrating human development vs strict statutory preservation excluding human commercial activity	जैवमंडल आरक्षित क्षेत्र / राष्ट्रीय उद्यान
Eco-Sensitive Zone (ESZ)	Protected Area (PA)	Buffer zone up to 10 km notified around National Parks and Sanctuaries under Section 3 of Environment (Protection) Act 1986 to regulate commercial activities vs statutory core area notified under Wildlife Protection Act 1972	पर्यावरण-संवेदनशील क्षेत्र (ESZ) / संरक्षित क्षेत्र (PA)
Ramsar Wetland Site	Natural World Heritage Site	International designation under 1971 Ramsar Convention specifically targeting ecological wise use of waterfowl/wetland habitats vs UNESCO World Heritage designation recognizing universal natural value under 1972 Convention	रामसर आर्द्रभूमि स्थल / प्राकृतिक विश्व धरोहर स्थल
Core Zone (Biosphere)	Transition Zone (Biosphere)	Legally protected untouched sanctum sanctorum where all human activity except non-destructive research is barred vs outermost open area where local sustainable agriculture, settlements, and economic activities are actively fostered	कोर क्षेत्र (पूर्ण सुरक्षित) / संक्रमण क्षेत्र (सहअस्तित्व)
### Protected-area hierarchy — one card			

Category	Legal home	Strictness / use	Classic trap
National Park	WPA Sec 35	Strictest WPA PA	Boundary change needs State Legislature
Wildlife Sanctuary	WPA Sec 18	Limited use with CWLW permission	Not as strict as NP
Tiger Reserve	WPA Sec 38V + NTCA	Core + buffer mandatory	Statutory — unlike Elephant Reserve
Conservation Reserve	WPA Sec 36A	Forest buffer near PA	Not private-land Community Reserve

Category	Legal home	Strictness / use	Classic trap
Community Reserve	WPA Sec 36B	Private/community land + consent	Needs voluntary consent
Biosphere Reserve	UNESCO MAB (executive in India)	Core–buffer–transition	No WPA statutory basis
Elephant Reserve	Project Elephant 1992	Corridor / conflict focus	No WPA statutory basis
Ramsar site	Ramsar Convention 1971	Wise-use wetland	Not a forest or ozone treaty
Natural World Heritage	UNESCO WH Convention	Outstanding universal value	≠ Ramsar ≠ BR WNBR automatically
ESZ	EPA 1986	Buffer restrictions around PA	Not a WPA category

- **UP teaching row:** Dudhwa NP / TR, Pilibhit TR, Ranipur TR, **12 Ramsar**, Terai corridors, **no marine PA**, **no biosphere reserve** inside present UP.
- **Three UNESCO instruments:** World Heritage · Biosphere (MAB / WNBR) · Intangible Cultural Heritage — do not force one site into all three without checking the list.

6.1 Wildlife Sanctuary

Feature	Wildlife Sanctuary	National Park
WPA section	Sec 18–27	Sec 35
Human activity	Allowed with CWLW permission	Largely prohibited
Grazing / forestry	May be permitted	Prohibited
Biomass collection	Limited use may be allowed	Not allowed as in sanctuary
Boundary change	State process	Needs State Legislature resolution
Strictness	Moderate (नरम दल)	Highest among WPA terrestrial PAs

6.3 Biosphere Reserve

#	Biosphere Reserve	State(s)
1	Nilgiri	Tamil Nadu, Karnataka, Kerala
2	Nanda Devi (नंदा देवी)	Uttarakhand
3	Nokrek	Meghalaya
4	Gulf of Mannar	Tamil Nadu
5	Sundarbans	West Bengal
6	Manas	Assam
7	Great Nicobar	Andaman & Nicobar
8	Simlipal	Odisha
9	Dibru-Saikhowa	Assam
10	Dihang-Dibang	Arunachal Pradesh
11	Pachmarhi	Madhya Pradesh
12	Khangchendzonga	Sikkim
13	Agasthyamalai	Kerala, Tamil Nadu
14	Achanakmar-Amarkantak	MP, Chhattisgarh

#	Biosphere Reserve	State(s)
15	Great Rann of Kutch	Gujarat
16	Cold Desert	Himachal (हिमाचल) Pradesh
17	Seshachalam Hills	Andhra Pradesh
18	Panna	Madhya Pradesh

Common Traps — Don't Fall For These

Trap	Correct fact
No human activity in sanctuary	Activity possible with CWLW permission
Nokrek in Manipur	Meghalaya
Sultanpur in UP	Haryana
Rudrasagar in UP	Tripura
ESZ under WPA	EPA 1986
Elephant Reserve under WPA	No — Project Elephant only
Biosphere Reserve under WPA	No — executive/MAB
UP has marine PA	No — landlocked
Grazing allowed in NP	Prohibited in NP

Topic 7 — Wildlife Conservation

Current Affairs (this topic)

- Track Project Tiger / Project Elephant / species project headlines against the statutory vs scheme trap.
- NTCA tiger-reserve count drifts — teach the concept; verify the live total before a count stem.

Confused Pairs

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
Wildlife Conservation	Wildlife Protection	Holistic scientific management and preservation of species, populations, and habitats to prevent extinction vs legal punitive enforcement and regulatory prohibition of hunting/poaching under Wildlife Protection Act 1972	वन्यजीव संरक्षण (वैज्ञानिक प्रबंधन) / वन्यजीव सुरक्षा (कानूनी प्रवर्तन)
Umbrella Species	Flagship Species	Ecologically wide-ranging species whose expansive habitat protection indirectly shelters the broader biological community (e.g. Tiger, Rhino) vs popular charismatic animal used as an emotional figurehead in conservation marketing (e.g. Giant Panda, Snow Leopard)	छत्र प्रजाति / प्रतीक प्रजाति
Keystone Species	Indicator Species	Species whose functional role exerts disproportionate control on community structure (loss causes trophic cascade; e.g. Fig tree, Sea Otter) vs species whose physiological response or presence directly indicates environmental conditions (e.g. Lichen, Frogs)	मुख्य आधार (कीस्टोन) प्रजाति / जैव संकेतक प्रजाति
Project Tiger (1973)	Project Elephant (1992)	Centrally Sponsored Scheme given statutory backing in 2006 under WPA creating the National Tiger Conservation Authority (NTCA) vs Centrally Sponsored Scheme operating through administrative guidelines without statutory authority	प्रोजेक्ट टाइगर (1973, सांविधिक) / प्रोजेक्ट एलिफेंट (1992, गैर-सांविधिक)
Diclofenac	Meloxicam	Non-steroidal anti-inflammatory drug (NSAID) administered to cattle causing lethal visceral gout and acute renal failure in Gyps vultures (banned in 2006) vs safe, vulture-tested veterinary NSAID recommended as its non-toxic replacement	डाइक्लोफेनाक (गिद्ध नाशक) / मेलोक्सिकैम (सुरक्षित विकल्प)
Wildlife Protection Act 1972	Biological Diversity Act 2002	Focuses on species protection, hunting bans, protected area network, and trade schedules (MoEFCC/WCCB) vs focuses on sovereign rights over biological resources, fair & equitable benefit sharing (ABS), and traditional knowledge protection (NBA/SBB/BMC)	वन्यजीव संरक्षण अधिनियम 1972 / जैव विविधता अधिनियम 2002
Central Zoo Authority (CZA)	National Board for Wildlife (NBWL)	Statutory body under WPA Sec 38A overseeing standards, recognition, and animal exchanges in Indian zoos vs apex advisory body chaired by the Prime Minister framing national wildlife policies and approving PA boundary alterations	केंद्रीय चिड़ियाघर प्राधिकरण / राष्ट्रीय वन्यजीव बोर्ड (NBWL)
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7.2 Wildlife Protection

Schedule	Protection	Examples
I	Maximum — hunting banned	Tiger, elephant, rhino, gharial, vulture
II	High	Himalayan black bear, cobra
III–IV	Moderate (नरम दल)/lower fauna (प्राणीजात)	Chital, sambar, hare
V	Vermin — may be hunted	Crow, rat

Schedule	Protection	Examples
VI	Specified plants — licence needed	Pitcher plant and other listed taxa

Common Traps — Don't Fall For These

Trap	Correct fact
WPA enacted in 1973	1972 — Project Tiger = 1973
World Wildlife Day = 22 March	3 March ; Wildlife Week = 2–8 October
Operation Olivia nesting in August	Nov–Feb
Elephant Reserve under WPA	No — executive only; covers wild and captive
All rhinos in Assam	Also WB and Dudhwa UP
Asiatic lion in UP	Gir, Gujarat only
Schedule VI = absolute ban	Cultivation needs a licence
Schedule I tortoise ≠ tiger level	Same maximum protection band
Spider monkey in India	Central/South America only
Kharai cannot be domesticated	Can be domesticated; swims + mangrove graze
AWBI under EPA 1986	Under PCA 1960 (est. 1962)
WWF logo = tiger	Giant Panda
Best gharial site = Pulicat	Chambal
Tiger Summit 2010 = Delhi	Saint Petersburg
Snow leopard in Terai/Gir	High Himalaya only
Crocodile project = gharial only	3 species
Diclofenac ban = human medicine ban	Veterinary use caused crisis

Topic 8 — Forests & Forest Management

Current Affairs (this topic)

- Keep ISFR / FSI function traps stable; new ISFR cycle is the usual forest CA refresh.
- FCA / forest-conservation wording may appear in CA — separate from WPA PA categories.

Confused Pairs

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
Forest Cover	Tree Cover	All land patches with tree canopy density $\geq 10\%$ and area ≥ 1 hectare irrespective of ownership or legal status (FSI remote sensing) vs scattered tree patches, linear trees along canals/roads, and isolated trees < 1 hectare outside recorded forest area	वन आवरण (≥ 1 हे., $\geq 10\%$ छतरी) / वृक्ष आवरण (< 1 हेक्टेयर)
Reserved Forest (RF)	Protected Forest (PF)	Highest degree of protection under Indian Forest Act 1927; all activities (grazing, timber felling) are strictly prohibited unless expressly permitted vs forest where all activities are permitted unless specifically prohibited by notification	आरक्षित वन (RF, पूर्णतः प्रतिबंधित) / संरक्षित वन (PF, सशर्त अनुमत)
Recorded Forest Area (RFA)	Forest Cover	Land legally recorded or notified as forest in government records (RF + PF + Unclassed) regardless of actual tree presence vs physical presence of tree canopy detected by satellite imagery (even on private orchards/tea estates)	अभिलिखित वन क्षेत्र (कानूनी अभिलेख) / वन आवरण (भौतिक वृक्ष आच्छादन)
Forest Conservation Act (FCA 1980)	Forest Rights Act (FRA 2006)	Central regulatory statute preventing de-reservation or diversion of forest land for non-forest purposes without prior central approval vs restorative rights law vesting individual and community forest rights in traditional forest-dwelling tribes and OTFDs	वन संरक्षण अधिनियम 1980 / वन अधिकार अधिनियम 2006
Social Forestry	Agroforestry	Growing trees on non-forest communal lands, roadside strips, and wastelands to meet rural fodder and fuel needs (National Commission on Agriculture 1976) vs deliberate integration of woody perennials (trees/shrubs) with agricultural crops and/or livestock on the same land unit	सामाजिक वानिकी / कृषि वानिकी
Joint Forest Management (JFM)	Community Forest Resource (CFR)	Collaborative institutional partnership between state forest department and village committees to protect and regenerate degraded forests vs statutory community ownership and governance right recognized under Section 3(1)(i) of FRA 2006	संयुक्त वन प्रबंधन (JFM) / सामुदायिक वन संसाधन अधिकार (CFR)
Tropical Wet Evergreen Forest	Tropical Moist Deciduous Forest	Dense multi-layered forest with rainfall > 200 cm, no distinct leafless season (Ebony, Mahogany, Rosewood) vs most widespread forest in India with rainfall 100–200 cm, shedding leaves for 6–8 weeks in dry spring (Teak, Sal, Shisham)	उष्णकटिबंधीय आर्द्र सदाबहार / उष्णकटिबंधीय नम पर्णपाती वन
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8.1 Forests and their Types

IFA class	Protection level
Reserved Forest	Strictest — hunting, grazing, felling banned unless permitted
Protected Forest	Moderate (नरम दल) — state controls but some rights allowed
Unclassed Forest	Not yet formally classified

8.2 Forest Types in India

#	Champion & Seth type	Key region
1	Tropical Wet Evergreen	Western Ghats, NE India
2	Tropical Semi-Evergreen	Transition to evergreen belt
3	Tropical Moist Deciduous	Terai, central India
4	Littoral & Swamp	Sundarbans, coastal Odisha
5	Tropical Dry Deciduous	Largest area (~38%)
6	Tropical Thorn	Rajasthan, Bundelkhand
7	Tropical Dry Evergreen	East coast Karnataka/AP
8	Subtropical Broadleaved Hill	Nilgiri (नीलगिरि), NE hills
9	Subtropical Pine	Chir pine — outer Himalaya
10	Subtropical Dry Evergreen	Siwalik, Jammu
11	Montane Wet Temperate	Nilgiri, eastern Himalaya
12	Himalayan Moist Temperate	Oak, deodar — Uttarakhand, HP
13	Himalayan Dry Temperate	Inner Himalaya conifers
14	Sub-Alpine	Rhododendron, fir — >3000 m
15	Moist Alpine Scrub	High Himalaya
16	Dry Alpine Scrub	Cold desert Himalaya

UP region	Forest type	Key districts
Terai-Duar	Tropical Moist Deciduous	Pilibhit, Bahraich, Shravasti
Vindhyan	Tropical Dry Deciduous	Sonbhadra, Chandauli, Mirzapur (मिर्ज़ापुर)
Bundelkhand	Thorn / Dry Deciduous	Jhansi, Banda, Lalitpur

Common Traps — Don't Fall For These

Trap	Correct fact
Bahraich >20% forest	Chandauli + Shravasti only
Forest cover = RFA only	Cover exists outside notified forest
FCA enacted 1972	1980
JFM = FRA same year	JFM 1990 , FRA 2006
Social forestry = JFM	Social = non-forest land ; JFM = degraded forest
FRA IFR cap 10 ha	4 ha
Agroforestry Policy 2006	2014
ISFR by CPCB	FSI Dehradun
NFP 1952 vs 1988	Current policy = 1988 ; 1952 classes ≠ National Park
CAFRI location	Jhansi , not Banda
MP vs Mizoram	MP = largest area ; Mizoram = highest %
RF vs Protected Forest	RF = strictest IFA class
Red Sanders = rainforest	Dry deciduous South India / Eastern Ghats
Teak = rainforest	Tropical moist deciduous ; MP max teak

Trap	Correct fact
HP = tropical wet evergreen	Pine/temperate belts
Urbanisation = deforestation impact	Urbanisation is a cause
Mangrove safety = canopy shelter only	Extensive roots
Bhitarkanika in West Bengal	Odisha — Brahmani–Baitarni
Gujarat < A&N mangrove	Gujarat 2nd after WB (ISFR 2021 style)
Aluminium green = saves forests	Recyclability/eco-profile; wood-substitute R often false

Topic 9 — Pollution & Waste Management

Current Affairs (this topic)

- NCAP (राष्ट्रीय स्वच्छ वायु कार्यक्रम) / CAQM / solid-waste and e-waste rule updates are the living pollution CA layer.
- Keep primary pollutant vs secondary pollutant and BOD vs COD pairs stable.

Confused Pairs

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
Environmental Pollution	Environmental Degradation	Direct or indirect release of harmful substances, radiation, or thermal energy into air, water, or soil causing toxic injury to living organisms vs broader long-term deterioration of environmental quality, natural capital, and ecosystem productivity	पर्यावरणीय प्रदूषण / पर्यावरणीय क्षरण
Primary Air Pollutant	Secondary Air Pollutant	Emitted directly from identifiable domestic or industrial exhaust sources into the atmosphere (CO, SO ₂ , NO, PM, Lead) vs formed in the atmosphere through chemical or photochemical reactions among primary precursors (Tropospheric Ozone, PAN, Smog, Nitric acid)	प्राथमिक प्रदूषक (सीधे उत्सर्जित) / द्वितीयक प्रदूषक (रासायनिक क्रिया)
Biochemical Oxygen Demand (BOD)	Chemical Oxygen Demand (COD)	Amount of dissolved oxygen needed by aerobic microorganisms to decompose biodegradable organic matter in water at 20°C over 5 days vs total amount of oxygen required to chemically oxidize all organic matter (both biodegradable and non-biodegradable) using a strong chemical oxidant	जैव रासायनिक ऑक्सीजन मांग (BOD) / रासायनिक ऑक्सीजन मांग (COD)
Oligotrophic Lake	Eutrophic Lake	Deep, crystal-clear water body with low nutrient levels, low biological productivity, and high dissolved oxygen vs shallow, murky water body rich in nitrates/phosphates, choked with algal blooms, and experiencing severe hypolimnetic oxygen crashes	अल्पपोषी झील (स्वच्छ/कम पोषक) / सुपोषित झील (अलाल ब्लूम/ऑक्सीजन हास)
Extended Producer Responsibility (EPR)	Polluter Pays Principle	Policy mandate making manufacturers legally and financially responsible for post-consumer collection, recycling, and safe disposal of their products (E-waste, plastic packaging) vs international environmental law doctrine requiring polluters to bear the financial costs of pollution mitigation and cleanup	विस्तारित उत्पादक दायित्व (EPR) / प्रदूषक भुगतान सिद्धांत
Central Pollution Control Board (CPCB)	Central Ground Water Board (CGWB)	Statutory apex pollution regulator established under Water Act 1974 for framing air/water standards and national monitoring vs subordinate scientific office under Ministry of Jal Shakti assessing groundwater resources and aquifer dynamics	केंद्रीय प्रदूषण नियंत्रण बोर्ड (CPCB) / केंद्रीय भूजल बोर्ड (CGWB)
Classical (London) Smog	Photochemical (Los Angeles) Smog	Occurs in cool, humid winter mornings; high concentrations of SO ₂ and coal soot; chemically reducing in nature vs occurs in hot, dry, sunny afternoons; vehicle exhaust NO _x + volatile organic compounds (VOCs); chemically oxidizing in nature with high ozone and PAN	लंदन स्मॉग (सल्फ्यूरस/अपचायक) / प्रकाश रासायनिक स्मॉग (ऑक्सीकारक)
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9.3 Air Pollution

Pollutant	CPCB 24-h standard (µg/m ³)
PM _{2.5}	60
PM ₁₀	100

Pollutant	CPCB 24-h standard ($\mu\text{g}/\text{m}^3$)
SO ₂	80
NO ₂	80

9.4 Water Pollution

Disease	Causative Pollutant / Dust	Industry / Occupational Source	Key feature / Nickname
Silicosis	Free silica (SiO ₂) dust	Stone crushing, quarrying, mining, glass factories	Progressive lung fibrosis; Rajasthan first state to frame Silicosis Policy (2019)
Asbestosis	Asbestos fibers	Roofing sheets, pipe insulation, shipbuilding	Leads to pulmonary fibrosis and Mesothelioma (cancer of pleural lining)
Anthracosis	Coal dust	Coal mining (Jharia, Raniganj, Singrauli)	Black Lung Disease / Coal Workers' Pneumoconiosis
Byssinosis	Cotton fiber dust	Textile mills, ginning factories	Brown Lung Disease ; called " Monday Fever " (symptoms peak after weekend return)
Bagassosis	Sugarcane bagasse dust	Sugar mills, paper & cardboard manufacturing	Fungal spores (<i>Thermoactinomyces</i>) on decomposing crushed sugarcane stalk
Siderosis	Iron / iron oxide dust	Iron foundries, arc welding, iron ore mining	Iron dust accumulation in lung parenchyma
Farmer's Lung	Moldy hay / straw dust	Agriculture, silos, threshing	Hypersensitivity reaction to thermophilic actinomycetes

Parameter	Clean river	Polluted river
BOD (mg/L)	<3	>10
DO (mg/L)	>6	<4
pH	6.5–8.5	<6 or >9

9.12 Biomedical Waste

Colour	Waste type	Treatment
Yellow	Anatomical, soiled, medicines	Incineration/autoclave
Red	Contaminated disposables	Disinfection → recycle
White	Sharps	Autoclave → disposal
Blue	Glass, metal	Disinfection → recycle

Common Traps — Don't Fall For These

Trap	Correct fact
Lichens grow well in polluted air	They die in pollution
CPCB publishes groundwater report	CGWB
O ₃ at ground level is beneficial	Harmful — unlike stratospheric ozone
Yellow BMW bag for sharps	White container
PWM Rules year = SUP ban year	2016 vs 2022

Trap	Correct fact
All marine pollution from ships	~80% from land
BOD measures heavy metals	Organic pollution only
Gomti flows through Kanpur	Lucknow
E-waste Rules still 2016	2022 rules current
Thermal kills fish by heat toxicity alone	Mainly DO reduction
E-waste rank 1st globally	3rd
Complete decay in one half-life	50% decay
Pesticide = biodegradable	Not
AQI includes CO ₂ /CH ₄	Eight pollutants — no
PAN = primary pollutant	Secondary ; eye irritant
Knock-knee = mercury	Fluoride
Eutrophication = air pollution	Water
Cyclone divider = control device	Not
Bhopal = CO/methane	MIC, Dec 1984
Bioremediation cleans all Cd/Pb	Not readily
Green muffler = soil only	Noise tree belts
Most important indoor = outdoor SO ₂	Often radon

Topic 10 — Climate Change

Current Affairs (this topic)

- Keep India NDC / Net Zero 2070 / LiFE as living climate CA; treaty years stay in the ladder below.
- Do not put **Montreal** into a climate-mitigation answer slot when Kyoto (क्योटो) / Paris are present.

Confused Pairs

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
Global Warming	Climate Change	Long-term monotonic increase in Earth's average surface and tropospheric temperature driven by greenhouse gas buildup vs broad spectrum of shifting weather patterns, rising sea levels, altering precipitation, and intensifying extreme climate anomalies	भूमंडलीय तापन (ग्लोबल वार्मिंग) / जलवायु परिवर्तन
Climate Mitigation	Climate Adaptation	Interventions designed to reduce greenhouse gas emissions at the source or enhance carbon sinks (renewable energy, afforestation, carbon capture) vs adjustments in natural or human systems to moderate harm, exploit beneficial opportunities, and cope with actual/expected climate impacts (flood barriers, drought-tolerant seeds)	जलवायु शमन (उत्सर्जन रोकथाम) / जलवायु अनुकूलन (सहनशीलता वृद्धि)
Natural Greenhouse Effect	Enhanced Greenhouse Effect	Essential natural phenomenon keeping Earth's surface habitable at an average ~15°C (without it Earth would freeze at -18°C) vs excessive anthropogenic heat-trapping caused by fossil fuel combustion and industrial emissions driving global warming	प्राकृतिक ग्रीनहाउस प्रभाव / मानवजनित संवर्धित प्रभाव
Global Warming Potential (GWP)	Radiative Forcing	Metric measuring the cumulative heat-trapping ability of 1 kg of a specific GHG over 100 years relative to CO ₂ (=1.0) vs the net change in Earth's radiative energy balance (measured in Watts/m ²) caused by changes in atmospheric gas concentrations	वैश्विक तापन क्षमता (GWP) / विकिरण दबाव
Carbon Sink	Carbon Sequestration	Natural or artificial reservoir that absorbs and stores more carbon than it releases into the atmosphere (oceans, boreal forests, soil) vs the actual physical, biological, or chemical process of capturing and securing carbon dioxide in long-term storage	कार्बन सिंक (अवशोषक भंडार) / कार्बन पृथक्करण (भंडारण प्रक्रिया)
Carbon Credit	Carbon Offset	Formal tradable permit or certificate representing the reduction or removal of 1 metric tonne of CO ₂ equivalent under a compliance cap-and-trade system vs voluntary purchase of emission reduction units to neutralize an individual's or company's carbon footprint	कार्बन क्रेडिट (1 टन CO ₂ e प्रमाण-पत्र) / कार्बन ऑफसेट
Kyoto Protocol (1997)	Paris Agreement (2015)	Top-down legally binding emission reduction targets imposed exclusively on developed nations (Annex-I) with Common But Differentiated Responsibilities vs bottom-up universal framework requiring all countries to submit voluntary Nationally Determined Contributions (NDCs) aiming to cap warming well below 2.0°C/1.5°C	क्योटो प्रोटोकॉल (बाध्यकारी/विकसित देश) / पेरिस समझौता (सार्वभौमिक NDCs)
—			

10.3 Greenhouse Gases (GHGs)

GHG / group	Rough relative share (WMO-style stems)	GWP (100-yr)
CO ₂	~64%	1
CH ₄	~19%	28–36
N ₂ O	~6%	265

GHG / group	Rough relative share (WMO-style stems)	GWP (100-yr)
CFCs + others	~11%	Very high (F-gases)

10.8 Climate Adaptation

#	NAPCC Mission
1	National Solar Mission
2	Enhanced Energy Efficiency
3	Sustainable Habitat (वास स्थान)
4	National Water Mission
5	Sustaining Himalayan Ecosystem
6	Green India
7	Sustainable Agriculture
8	Strategic Knowledge for Climate Change

10.9 Climate Mitigation

Treaty ladder — climate vs ozone (read the stem)

Ladder	Instruments	What it controls	Classic trap
Climate	UNFCCC (1992) (जलवायु परिवर्तन संधि) → Kyoto (क्योटो) (1997) → Paris (2015)	GHG mitigation / NDCs	Montreal is not climate mitigation
Ozone	Vienna (1985) → Montreal (1987) → Kigali (HFCs, 2016)	ODS / HFC phase-down	Kyoto is not ozone

Treaty ladder — climate vs ozone (read the stem)

Treaty	Year	Focus
UNFCCC	1992	Framework
Kyoto Protocol	1997	Binding cuts (developed countries)
Paris Agreement	2015	NDC pledges
Montreal Protocol	1987	Ozone (not climate)

Common Traps — Don't Fall For These

Trap	Correct fact
Montreal = climate treaty	Ozone — Kyoto/Paris = climate
LiFE proposed at COP-25	COP-26 Glasgow
Eliminate greenhouse effect	Natural effect essential — problem is enhanced
Wilson/IPCC invented greenhouse effect	Joseph Fourier (1820s)
CH ₄ highest total contribution	CO₂ (~64% WMO-style / ~76% IPCC forcing)
Water vapour = main Kyoto inventory driver	Most abundant natural GHG / feedback; policy driver = CO₂
Argon / H ₂ / propane / N ₂ are GHGs	Not GHGs
SO ₂ / NO _x warm Earth directly	Not direct GHGs

Trap	Correct fact
India highest per capita emitter	Low per capita , ~3rd total; China #1
Net zero = absolute zero emissions	Emissions balanced by removals
Nuclear power is an NAPCC mission	Not among the eight
GCF founded at Durban	Cancun COP-16 (2010)
Earth Hour = UNEP/UNESCO	WWF
Carbon credit price fixed by UNEP	Market-traded
Adaptation = cut emissions	Mitigation cuts emissions
Kyoto bound India to cuts	Bound Annex I developed countries
Solar flares = primary recent climate cause	Not major (NASA/IPCC framing)
Milankovitch includes solar irradiance	Eccentricity, obliquity, precession only
Blue carbon = petroleum carbon	Ocean/coastal ecosystems

Topic 11 — Ozone Layer

Current Affairs (this topic)

- Montreal / Kigali compliance headlines stay ozone-family CA — do not park them under Paris NDC.
- ODS vs GHG trap is permanent; HFC phase-down is Kigali, not Kyoto (क्योटो).

Confused Pairs

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
Stratospheric Ozone	Tropospheric Ozone	Essential atmospheric shield absorbing harmful solar UV-B radiation (20–30 km altitude) vs toxic secondary pollutant and greenhouse gas causing smog and respiratory ailments	समतापमंडलीय ओजोन (सुरक्षा कवच) / क्षोभमंडलीय ओजोन (हानिकारक प्रदूषक)
Ozone Depletion	Ozone Hole	Widespread gradual thinning of the stratospheric ozone layer globally caused by halogenated gases vs severe, seasonal Antarctic springtime drop where column ozone drops below 220 Dobson Units (DU)	ओजोन क्षरण (वैश्विक स्तर पर) / ओजोन छिद्र (अंटार्कटिक <220 DU)
Vienna Convention (1985)	Montreal Protocol (1987)	Non-binding multilateral framework agreement establishing international cooperation on ozone research and monitoring vs legally binding international treaty mandating precise, phase-out timetables for ozone-depleting substances (ODS)	वियना कन्वेंशन (ढांचागत समझौता) / मॉन्ट्रियल प्रोटोकॉल (बाध्यकारी प्रतिबंध)
Chlorofluorocarbons (CFCs)	Hydrofluorocarbons (HFCs)	Powerful ozone-depleting substances with high chlorine and high GWP (phased out under Montreal) vs zero-ODP fluorinated gases that do not harm ozone but are super greenhouse gases (phased down under 2016 Kigali Amendment)	सीएफसी (ओजोन विनाशक + GHG) / एचएफसी (केवल GHG, किगाली द्वारा नियंत्रित)
Dobson Unit (DU)	Parts Per Million (ppm)	Standard unit measuring the total vertical column abundance of atmospheric ozone (1 DU = 0.01 mm thickness at STP; normal baseline ~300 DU) vs standard volumetric concentration ratio of trace gases in ambient air	डॉबसन यूनिट (स्तंभ ओजोन मोटाई) / पीपीएम (सान्द्रता अनुपात)
Polar Stratospheric Clouds (PSCs)	Cirrus Clouds	High-altitude clouds forming in extreme Antarctic winter cold (-78°C / Type I and II) that catalyze chlorine release from reservoirs vs normal high-altitude wispy ice-crystal weather clouds in the upper troposphere	ध्रुवीय समतापमंडलीय बादल (PSCs) / पक्षाभ मेघ (ट्रोपोस्फेरिक)
Kigali Amendment (2016)	Paris Agreement (2015)	Legally binding amendment to Montreal Protocol committing countries to progressively phase down consumption of hydrofluorocarbons (HFCs) by 80–85% vs UNFCCC climate framework agreement targeting global warming mitigation	किगाली संशोधन (HFC कटौती) / पेरिस समझौता (जलवायु लक्ष्य)
—			

11.1 Ozone Layer

UV type	Wavelength	Effect if unfiltered
UVA	320–400 nm	Ageing, tanning
UVB	280–320 nm	Sunburn, skin cancer, cataracts

UV type	Wavelength	Effect if unfiltered
UVC	100–280 nm	Lethal — normally blocked

11.2 Ozone Depletion

ODS	Main use
CFCs	Refrigerants, aerosols, foam
Halons	Fire extinguishers
Carbon tetrachloride	Solvent
Methyl bromide	Pesticide/fumigant
HCFCs	Transitional refrigerants

11.3 Ozone Hole

Feature	Antarctic	Arctic
Severity	Severe, recurring	Occasional, milder
Season	Sep–Oct (austral spring)	Mar–Apr if event occurs
Polar vortex	Strong, very cold	Weaker, warmer

11.4 Montreal Protocol

Treaty / step	Year	Role
Vienna Convention	1985	Framework
Montreal Protocol	1987	Binding ODS phase-out
London Amendment	1990	Multilateral Fund
Kigali Amendment	2016	HFC phase-down

Common Traps — Don't Fall For These

Trap	Correct fact
Montreal = climate treaty	Ozone / ODS
Ozone layer in troposphere	Stratosphere
Ozone hole = physical hole	Low ozone column < 220 DU
Ozone hole over Arctic always	Antarctic spring
Tropospheric ozone is good	Pollutant
CFCs controlled under Kyoto	Montreal Protocol
Ozone Day = 5 June	16 September
Kigali targets CFCs	HFC phase-down
UV-C reaches surface normally	Blocked by O₃/O₂
Depletion permanent with no recovery	Recovery expected ~2040/2066

Topic 12 — Acid Rain

Current Affairs (this topic)

- Acid-rain CA is rare; keep SO₂ / NO_x chemistry and dry vs wet deposition stable.
- Do not confuse acid rain with ozone hole (ओजोन छिद्र) or greenhouse trapping.

Confused Pairs

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
Normal Rain	Acid Rain	Slightly acidic precipitation with pH ~5.6 caused by dissolved atmospheric carbon dioxide forming weak carbonic acid (H ₂ CO ₃) vs precipitation with pH strictly below 5.6 caused by anthropogenic emissions of sulfur dioxide (SO ₂) and nitrogen oxides (NO _x)	सामान्य वर्षा (pH ~5.6) / अम्ल वर्षा (pH < 5.6)
Sulfuric Acid (H ₂ SO ₄)	Nitric Acid (HNO ₃)	Accounts for ~60–70% of total acid deposition acidity (primarily emitted by coal-fired thermal power plants and metal smelters) vs accounts for ~30–40% of acidity (primarily emitted by automobile exhausts and nitrogenous fertilizer use)	सल्फ्यूरिक अम्ल (60–70% भारांक) / नाइट्रिक अम्ल (30–40% भारांक)
Wet Acid Deposition	Dry Acid Deposition	Direct transfer of acidic pollutants to Earth's surface incorporated into falling rain, fog, sleet, or snow vs settling of acidic gases, sulfate aerosols, and nitrate dust directly onto soil, vegetation, and buildings without moisture	आर्द्र निक्षेपण (वर्षा/कोहरा) / शुष्क निक्षेपण (गैस व धूलिकण)
Marble Cancer	Eutrophication	Chemical weathering and yellowing/spalling of calcium carbonate structures (e.g. Taj Mahal) reacting with sulfuric acid to form soluble gypsum (CaSO ₄) vs excessive nutrient enrichment of water bodies leading to dense algal blooms and dissolved oxygen crashes	संगमरमर कैसर (ताजमहल क्षरण) / सुपोषण (जलीय पोषक संवर्धन)
Gothenburg Protocol (1999)	Montreal Protocol (1987)	UNECE agreement under CLRTAP targeting multi-pollutant abatement of acidification, eutrophication, and ground-level ozone (SO ₂ , NO _x , VOCs, NH ₃) vs UNEP treaty targeting the global phase-out of ozone-depleting substances	गोथेनबर्ग प्रोटोकॉल (अम्लीकरण नियंत्रण) / मॉन्ट्रियल प्रोटोकॉल (ओजोन)
Taj Trapezium Zone (TTZ)	Eco-Sensitive Zone (ESZ)	Specifically demarcated ~10,400 km ² protective geographical zone covering Agra, Firozabad, Mathura, and Bharatpur to safeguard the Taj Mahal from industrial air pollution vs buffer belt up to 10 km declared around National Parks under EPA 1986	ताज ट्रपेज़ियम ज़ोन (10,400 वर्ग किमी) / पारिस्थितिक संवेदनशील क्षेत्र
—			

12.1 Acid Rain

pH	Meaning
< 5.6	Anthropogenic acid rain threshold
~5.6	Normal rain (CO ₂ equilibrium)
7	Neutral distilled water — not natural rain
> 7	Basic/alkaline

12.2 Causes

Pollutant	Main acid	Main sources
SO ₂	H ₂ SO ₄	Coal TPPs, smelters, refineries
NO _x	HNO ₃	Vehicles, high-temp combustion
CO ₂	H ₂ CO ₃ (weak)	Natural baseline only

12.3 Effects

Receptor	Key impact
Lakes/ivers	Fish kills below pH ~5
Soil	Nutrient leaching, Al ³⁺ toxicity
Forests	Conifer needle damage
Marble monuments	Gypsum crust (Taj Mahal)
Lichens	Die in SO ₂ pollution

12.4 Prevention

Measure	Target
FGD scrubbers	SO ₂ from TPP chimneys
Catalytic converters	NO _x from vehicles
BS-VI fuel	SO ₂ + NO _x precursors
CNG/LPG switch	Near-zero SO ₂
Gothenburg Protocol	Transboundary SO ₂ /NO _x
Taj Trapezium	Heritage protection near Agra

Common Traps — Don't Fall For These

Trap	Correct fact
Normal rain pH = 7	~ 5.6 from CO ₂
CO ₂ = main acid rain cause	Weak acid only ; SO ₂ /NO _x = strong acids
Coal acid rain explained by CO ₂ alone	A true (coal → acid rain via SO₂); R true (CO ₂ emitted) but R does not explain A
Acid rain = rain only	Wet + dry deposition
Acidic pH > 7	Acidic < 7
Montreal = acid rain	Gothenburg / CLRTAP
SO ₂ lighter than air	Heavier
Lichens thrive in pollution	Die in SO₂
Tall chimney = prevention	Dispersal only
Taj damage from CO ₂	SO₂ + acid deposition
Acid rain stays in source country	Transboundary

Topic 13 — Desertification & Land Degradation

Current Affairs (this topic)

- UNCCD / LDN and Indian desertification atlas updates are the CA refresh.
- Bundelkhand / Rajasthan land-degradation colour stays high-yield for UP.

Confused Pairs

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
Desertification	Land Degradation	Persistent reduction or loss of biological and economic productivity occurring specifically in Arid, Semi-Arid, and Dry Sub-Humid areas (drylands) vs general decline in the productive capacity and ecosystem health of any land anywhere	मरुस्थलीकरण (शुष्क भूमियों में) / भू-क्षरण (सामान्य भूमि अवनति)
Drought	Desertification	Temporary natural meteorological period of deficient precipitation over months or a few years from which ecosystems can recover vs persistent, semi-permanent ecological degradation and productivity loss driven by unsustainable human use and climate stress	सूखा (अस्थायी मौसमी संकट) / मरुस्थलीकरण (स्थायी भूमि हास)
Sheet Erosion	Gully Erosion	Uniform, imperceptible removal of a thin layer of topsoil over large flat cultivated fields by sheetwash runoff vs cutting of deep, severe channels and ravines by concentrated runoff carving badland topography (e.g. Chambal ravines)	परत (चादर) अपरदन / अवनलिका अपरदन (बीहड़ निर्माण)
Splash Erosion	Rill Erosion	Initial detachment and airborne displacement of soil particles caused directly by the physical impact of falling raindrops vs intermediate erosion stage where tiny, shallow, visible finger-like channels are formed on sloped ground	आघात (स्पलैश) अपरदन / क्षुद्र सरिता (रिल) अपरदन
UNCCD (1994)	UNFCCC (1992)	Sole legally binding international treaty linking environment and development to sustainable land management, desertification, and drought in drylands vs framework convention dedicated to addressing anthropogenic climate change and greenhouse gases	मरुस्थलीकरण रोकथाम संधि (UNCCD) / जलवायु परिवर्तन संधि (UNFCCC)
Land Degradation Neutrality (LDN)	Net Zero Emissions	Target under UN SDG 15.3 ensuring that the quantity and quality of land resources necessary to support ecosystem functions remain stable or increase by 2030 vs balancing greenhouse gas emissions with equivalent removals	भूमि क्षरण तटस्थता (LDN, SDG 15.3) / नेट जीरो उत्सर्जन
—			

13.1 Desertification

Concept	Scope
Desertification	ASDM drylands only
Drought	Rainfall deficit event
Land degradation	All land types, all processes

13.2 Land Degradation

Process	Example region
Salinization	Punjab-Haryana canal belt

Process	Example region
Mining degradation	Sonbhadra, Jharkhand
Waterlogging	Over-irrigated plains
Ravine formation	Chambal basin

13.3 Soil Erosion

Erosion stage	Feature
Sheet	Thin uniform topsoil loss
Rill	Small grooves after rain
Gully	Deep channels, uncrossable by plough
Ravine	Chambal-type badland maze
Agent	Main region
-----	-----
Water	Monsoon slopes, Chambal
Wind	Thar, arid west
Gravity	Himalaya (हिमालय), NE hills

Common Traps — Don't Fall For These

Trap	Correct fact
Desertification = desert expansion only	Any ASDM dryland losing productivity
Land degradation = desertification	Degradation is broader
Afforestation causes degradation	It prevents degradation
Sheet erosion = deep ravines	Sheet is first, mildest stage
Chambal ravines in one state	MP + UP + Rajasthan
UNCCD = climate treaty	UNCCD = desertification; UNFCCC = climate
Normal rain pH 7 stops erosion	Unrelated — erosion is topsoil removal
Wind erosion in Chambal ravines	Water/gully erosion
LDN target 2025	2030 (SDG 15.3)
UNCCD signed at Rio 1992	Adopted 1994, Paris

Topic 14 — Environmental Impact Assessment

Current Affairs (this topic)

- EIA notification amendments and Category A/B traps are the living CA layer.
- Keep screening–scoping–appraisal–monitoring order stable.

Confused Pairs

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
Environmental Impact Assessment (EIA)	Strategic Environmental Assessment (SEA)	Formal analytical process evaluating the potential ecological impacts of a specific <i>project</i> (e.g. thermal plant, highway) before sanction vs higher-level systemic evaluation applied to government <i>policies, plans, and programmes</i>	पर्यावरण प्रभाव आकलन (EIA, परियोजना स्तर) / रणनीतिक पर्यावरण आकलन (SEA, नीति स्तर)
Category A Projects	Category B Projects	Large-scale developmental projects requiring mandatory full-scale EIA appraised centrally at Union level by MoEFCC on recommendations of EAC vs smaller projects appraised at State level by SEIAA on recommendations of SEAC	श्रेणी 'ए' (केंद्रीय मूल्यांकन) / श्रेणी 'ब' (राज्य मूल्यांकन)
Category B1 Projects	Category B2 Projects	State-level projects that undergo complete EIA process and mandatory public hearing due to significant ecological footprint vs smaller Category B projects exempted from EIA study and public consultation, appraised solely on application summary	श्रेणी 'ब1' (EIA अनिवार्य) / श्रेणी 'ब2' (EIA से छूट)
Terms of Reference (ToR)	Environmental Management Plan (EMP)	Comprehensive boundary document outlining all environmental baseline issues and studies that the project proponent must investigate vs actionable mitigation, monitoring, and institutional plan submitted in the final EIA report	संदर्भ की शर्तें (ToR, आरंभिक ढाँचा) / पर्यावरण प्रबंधन योजना (EMP, शमन कार्ययोजना)
Public Hearing (Consultation)	Appraisal	Statutory public stage where local affected communities and citizens voice concerns and objections to the draft EIA vs detailed scientific scrutiny of final EIA and public hearing proceedings by EAC or SEAC	जन सुनवाई (स्थानीय सहभागिता) / वैज्ञानिक मूल्यांकन (EAC/SEAC)
National Green Tribunal (NGT)	Appellate Authority under Air/Water Acts	Specialized judicial tribunal established under NGT Act 2010 hearing original environmental disputes and appeals against environmental clearances (EC) vs administrative appellate authorities designated under Water Act 1974 / Air Act 1981	राष्ट्रीय हरित अधिकरण (न्यायिक अधिकरण) / राज्य अपीलीय प्राधिकरण
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14.1 Environmental Impact Assessment (EIA)

Category	Authority	Screening?	Full EIA?
A	MoEFCC + EAC	No	Usually yes
B1	SEIAA + SEAC	Yes	Yes
B2	SEIAA/DEIAA	Yes	No full report
B + General Condition	MoEFCC (elevated)	—	As Category A
Stage	What happens		
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Screening	B → B1 or B2		

Category	Authority	Screening?	Full EIA?
Scoping	ToR for EIA study		
Public Consultation	Public hearing + comments		
Appraisal	Recommend grant/reject EC		

14.2 Strategic Environmental Assessment (SEA)

Feature	EIA	SEA
Level	Individual project	Policy/plan/programme
Timing	After project concept	Before projects chosen
India law	Mandatory (2006)	No standalone law
Cumulative impacts	Limited	Core focus
Example	EC for one thermal plant	National sector plan before siting
Instrument	Year	Scope
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Rio Principle 17	1992	Project EIA norm
Espoo Convention	1991	Transboundary EIA
SEA Protocol (Kiev)	2003	Plans/programmes
EU SEA Directive	2001/42/EC	EU mandatory SEA
India EIA Notification	2006	Project EC only

Common Traps — Don't Fall For These

Trap	Correct fact
CPCB grants EC	MoEFCC/SEIAA grant EC
Screening for Category A	Screening = Category B only
EIA after construction	EC before construction
EIA = SEA	EIA = project; SEA = plan/policy
India has mandatory SEA law	Project EIA only
EIA for all projects	Listed Schedule projects only
Water Act 1981 base for EIA	EPA 1986
NGT established 1986	NGT Act 2010
EAC grants final EC	EAC recommends; MoEFCC/SEIAA grant
Rio Principle 17 = SEA	Principle 17 = project EIA

Topic 15 — Sustainable Development & Environmental Governance

Current Affairs (this topic)

- SDG progress reports and LiFE / Mission LiFE headlines refresh this chapter's CA.
- Brundtland / Agenda 21 / Rio years stay fixed — do not relocate them into Paris-only answers.

Confused Pairs

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
Brundtland Report (1987)	Limits to Growth (1972)	UN World Commission on Environment & Development (<i>Our Common Future</i>) coining the definition of Sustainable Development vs Club of Rome computer-simulation study modeling impending resource depletion and economic collapse	ब्रंटलैंड रिपोर्ट (1987, सतत विकास) / लिमिट्स टू ग्रोथ (1972)
Millennium Development Goals (MDGs)	Sustainable Development Goals (SDGs)	8 anti-poverty targets for 2000–2015 focused primarily on developing nations funded by rich countries vs 17 universal, integrated economic-social-environmental goals (Agenda 2030) adopted in 2015 applying to all nations equally	सहस्राब्दी विकास लक्ष्य (MDGs) / सतत विकास लक्ष्य (SDGs 2030)
Green Economy	Circular Economy	Macro-economic development paradigm promoting low-carbon growth, resource efficiency, and social equity (UNEP) vs regenerative industrial design philosophy eliminating waste and keeping materials in closed continuous cycles	हरित अर्थव्यवस्था (व्यापक मॉडल) / चक्रीय अर्थव्यवस्था (पुनर्चक्रण लूप)
Linear Economy	Circular Economy	Traditional unsustainable industrial model based on “take, make, use, and dispose” vs restorative circular model based on “reduce, reuse, repair, refurbish, and recycle”	रैखिक अर्थव्यवस्था (कचरा उत्पादक) / चक्रीय अर्थव्यवस्था (शून्य अपशिष्ट)
Mission LiFE (2022)	Agenda 21 (1992)	Mass global citizen movement conceptualized by India to replace mindless consumption with mindful individual sustainable lifestyle actions vs Rio 1992 comprehensive non-binding intergovernmental action plan for sustainable governance	मिशन लाइफ (LiFE, व्यक्तिगत जीवनशैली) / एजेंडा 21 (सरकारी कार्ययोजना)
Natural Capital	Manufactured Capital	World's natural ecological assets including geology, soil, air, water, biodiversity, and ecosystem services vs human-created physical infrastructure, machinery, factories, and financial assets	प्राकृतिक पूंजी (पारिस्थितिक संपदा) / निर्मित पूंजी (मानव-निर्मित संपत्ति)
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15.1 Sustainable Development

Pillar	Focus
Economic	Employment, innovation, infrastructure
Social	Poverty, health, education, equity
Environmental	Pollution, biodiversity, climate
Institutional	Governance, rule of law, participation

15.2 SDGs & SDGs 2030

Goal	Official name
1	No Poverty
2	Zero Hunger
3	Good Health and Well-being
4	Quality Education
5	Gender Equality
6	Clean Water and Sanitation
7	Affordable and Clean Energy
8	Decent Work and Economic Growth
9	Industry, Innovation and Infrastructure
10	Reduced Inequalities
11	Sustainable Cities and Communities
12	Responsible Consumption and Production
13	Climate Action
14	Life Below Water
15	Life on Land
16	Peace, Justice and Strong Institutions
17	Partnerships for the Goals
Feature	MDGs
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Goals	8
Period	2000–2015
Scope	Mainly developing countries

15.3 Environmental Governance, Justice & Ethics

Concept	Fact
Environmental justice	Fair distribution + participation
Anthropocentrism	Human-centred value
Biocentrism	All life has worth
Ecocentrism	Ecosystem (पारिस्थितिकी तंत्र) wholes have worth
Leopold	Land Ethic, 1949
Gandhi	Trusteeship, limit wants

15.4 Green Economy & Circular Economy

Feature	Green economy	Circular economy
Focus	Overall low-carbon inclusive growth	Material/waste loop design
Scope	Broader	Narrower — subset of green
Example	Solar parks, green hydrogen	EPR, plastic recycling

Feature	Green economy	Circular economy
SDG link	7, 8, 13 and more	Mainly SDG 12

15.5 Sustainable Agriculture & NMSA

Idea	Correct fact
Sustainable agriculture	Land quality remains intact
Best strategy set	Mixed cropping + organic + N-fixing + pest-resistant varieties
NMSA	NAPCC mission
Conservation Agriculture	Min tillage + residue cover + rotation

Common Traps — Don't Fall For These

Trap	Correct fact
Agenda 21 = 1995	Rio 1992
Brundtland = Limits to Growth	<i>Our Common Future</i> 1987
SD = environment only	Three pillars balanced
SD = no development	Sustainable, not zero, development
LiFE at COP-25	COP26 2021; launched June 2022
UP SDG report = Finance Dept	Planning Department
Circular = identical to green	Circular $\subset$ Green
MDGs = 17 goals	MDGs = 8; SDGs = 17
SDGs only for developing countries	Universal
Bhopal → Stockholm 1972	Bhopal 1984 → EPA 1986

Topic 16 — Environmental Organizations (India)

Current Affairs (this topic)

- MoEFCC / CPCB / SPCB / NBA HQ and function traps stay stable; scheme rebrands are the CA layer.
- NEERI = Nagpur remains a classic organisation location trap.

Confused Pairs

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
Botanical Survey of India (BSI)	Zoological Survey of India (ZSI)	Apex national taxonomic survey of plant resources (founded 1890, HQ: Kolkata) publishing the Red Data Book of Indian Plants vs apex national survey of faunal resources (founded 1916, HQ: Kolkata) documenting animal biodiversity	भारतीय वनस्पति सर्वेक्षण (BSI, 1890) / भारतीय प्राणी सर्वेक्षण (ZSI, 1916)
Forest Survey of India (FSI)	Forest Research Institute (FRI)	National organization (HQ: Dehradun) responsible for periodic remote-sensing assessment of forest cover and publishing the biennial India State of Forest Report (ISFR) vs premier academic research and deemed university under ICFRE (HQ: Dehradun)	भारतीय वन सर्वेक्षण (FSI, ISFR प्रकाशक) / वन अनुसंधान संस्थान (FRI)
Wildlife Institute of India (WII)	Wildlife Crime Control Bureau (WCCB)	Autonomous research and training institute under MoEFCC (Dehradun) conducting wildlife ecology research and census vs statutory enforcement bureau created under WPA 1972 Sec 38Y (New Delhi) combating organized wildlife poaching and illegal trade	भारतीय वन्यजीव संस्थान (WII, अनुसंधान) / वन्यजीव अपराध नियंत्रण ब्यूरो (WCCB, प्रवर्तन)
Central Pollution Control Board (CPCB)	State Pollution Control Board (SPCB)	Statutory apex body established under Water Act 1974 advising the Union Government, setting national ambient standards, and coordinating inter-state river/air monitoring vs state-level statutory body granting Consent to Establish/Operate to industrial units	केंद्रीय प्रदूषण नियंत्रण बोर्ड (मानक निर्माता) / राज्य प्रदूषण नियंत्रण बोर्ड (अनुमति प्रदाता)
National Green Tribunal (NGT)	Central Empowered Committee (CEC)	Specialized statutory environmental court established under NGT Act 2010 with judicial and expert members deciding civil environmental disputes vs Supreme Court monitoring committee overseeing forest conservation and wildlife sanctuary clearances	राष्ट्रीय हरित अधिकरण (NGT, सांविधिक पीठ) / केंद्रीय अधिकार प्राप्त समिति (CEC)
National Environmental Engineering Research Institute (NEERI)	National Institute of Oceanography (NIO)	CSIR premier lab specializing in water/air treatment, environmental engineering, and monitoring (founded 1958, Nagpur) vs CSIR oceanographic lab researching marine resources and ecology (Dona Paula, Goa)	नीरी (NEERI, नागपुर) / राष्ट्रीय समुद्र विज्ञान संस्थान (NIO, गोवा)
—			

16.1 Ministry of Environment, Forest and Climate Change (MoEFCC)

Function	Body
Category A EC	MoEFCC
Forest clearance	MoEFCC
NAAQS standards	CPCB
Category B EC	SEIAA
EC appeals	NGT

16.2 Central Pollution Control Board (CPCB) (केंद्रीय प्रदूषण नियंत्रण बोर्ड) & State Pollution Control Board (SPCB) (राज्य प्रदूषण नियंत्रण बोर्ड)

SPCB

Term	Body	Purpose
Consent (CTE/CTO)	SPCB	Pollution control for industry operation
Environmental Clearance	MoEFCC/SEIAA	Project impact assessment
Forest Clearance	MoEFCC	Forest land diversion

16.6 Zoological Survey of India (ZSI) (भारतीय प्राणी सर्वेक्षण)

Body	Est.	HQ	Kingdom
BSI	1890	Kolkata	Plants
ZSI	1916	Kolkata	Animals
WII	1982	Dehradun	Wildlife research
FSI	1981	Dehradun	Forest cover

16.7 Forest Survey of India (FSI) (भारतीय वन सर्वेक्षण)

Report	Publisher
ISFR	FSI
Groundwater Quality Report	CGWB
NAAQS bulletin	CPCB
<i>Flora of India</i>	BSI
<i>Fauna of India</i>	ZSI

16.8 National Environmental Engineering Research Institute (NEERI)

Body	HQ	Job
NEERI	Nagpur	Env. engineering research (CSIR)
CPCB	New Delhi	Pollution standards / monitoring
NGT	New Delhi	Environmental adjudication

Common Traps — Don't Fall For These

Trap	Correct fact
CPCB grants EC	MoEFCC/SEIAA
CPCB publishes ISFR	FSI
CPCB prepares groundwater report	CGWB
MoEFCC created 1972	1985
BSI surveys animals	Plants only

Trap	Correct fact
ZSI surveys plants	Animals only
NGT grants EC	Hears appeals
ISFR is annual	Biennial
BSI/ZSI in Dehradun	Kolkata
SPCB grants Category B EC	SEIAA

Topic 17 — Environmental Laws & Policies

Current Affairs (this topic)

- New Acts / Rules / NGT orders are living CA; WPA / EPA / FCA year spine stays fixed.
- Do not confuse NGT with NBA (biodiversity) on acronym stems.

Confused Pairs

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
Environment (Protection) Act (EPA 1986)	Wildlife (Protection) Act (WPA 1972)	Umbrella legislation enacted under Article 253 post-Bhopal disaster enabling the Union to frame rules for EIA, CRZ, ESZ, and hazardous waste vs dedicated legislation protecting wild animal/plant species, regulating hunting, and establishing National Parks/Sanctuaries	पर्यावरण संरक्षण अधिनियम 1986 (छाता कानून) / वन्यजीव संरक्षण अधिनियम 1972
Forest (Conservation) Act (FCA 1980)	Forest Rights Act (FRA 2006)	Central law barring de-reservation or diversion of forest land for non-forest purposes without prior Union MoEFCC approval vs rights-based law administered by Ministry of Tribal Affairs restoring land tenure and customary rights to forest-dwelling STs and OTFDs	वन संरक्षण अधिनियम 1980 (विकास नियंत्रण) / वन अधिकार अधिनियम 2006 (जनजातीय अधिकार)
Water (Prevention and Control of Pollution) Act 1974	Air (Prevention and Control of Pollution) Act 1981	India's first dedicated environmental pollution statute that established CPCB and SPCBs under Article 252 vs legislation enacted under Article 253 to implement 1972 Stockholm decisions that empowered existing Water Boards to regulate air pollution and noise	जल अधिनियम 1974 (प्रदूषण बोर्डों का गठन) / वायु अधिनियम 1981 (स्टॉकहोम अनुपालन)
Biological Diversity Act (BDA 2002)	Protection of Plant Varieties and Farmers' Rights Act (PPVFR 2001)	Implements 1992 CBD via three-tier structure (NBA, SBB, BMC) controlling commercial bio-resource access and benefit sharing (ABS) vs sui generis IPR regime granting legal rights to commercial plant breeders while safeguarding traditional farmers' seed-saving rights	जैव विविधता अधिनियम 2002 / पौधा किस्म व कृषक अधिकार संरक्षण 2001
Genetic Engineering Appraisal Committee (GEAC)	Central Ground Water Authority (CGWA)	Statutory apex regulatory body constituted under Rules 1989 of EPA 1986 approving environmental release of GMOs (e.g. Bt Cotton) vs regulatory authority constituted under Section 3(3) of EPA 1986 regulating commercial groundwater extraction NOCs	जीईएसी (GEAC, जीएमओ नियामक) / केंद्रीय भूजल प्राधिकरण (CGWA)
Coastal Regulation Zone (CRZ)	Eco-Sensitive Zone (ESZ)	Coastal belt up to 500 m from High Tide Line classified into CRZ-I to IV under EPA 1986 to regulate beach and marine development vs buffer belt around protected areas notified under Section 3 of EPA 1986 to curb polluting industrial operations	तटीय विनियमन क्षेत्र (CRZ) / पर्यावरण-संवेदनशील क्षेत्र (ESZ)
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17.1 Environmental Law (Constitutional Framework)

Article	Type	Content
48A	DPSP	State protects environment
51A(g)	Fundamental Duty	Citizen protects environment
21	FR (expanded)	Healthy environment as part of life
48	DPSP	Agriculture — not environment

17.3 Wildlife (Protection) Act, 1972 & Wildlife Protection Laws

Schedule	Content
I	Highest protection — severe penalties
II–IV	Protected species — hunting restricted
V	Vermin — may be hunted
VI	Protected plants

17.4 Water (Prevention and Control of Pollution) Act, 1974 & Air Act, 1981

Air Act 1981

Act	Year	Key output
Water Act	1974	CPCB, SPCB, water consent
Air Act	1981	NAAQS, air pollution control

17.5 Forest (Conservation) Act, 1980 | Forest Rights Act, 2006 | National Forest Policy, 1952

National Forest Policy, 1952

Act	Purpose
FCA 1980	Stop forest diversion without Central approval
FRA 2006	Recognize forest dweller rights
NFP 1952	Early policy — revenue focus

17.6 Biological Diversity Act, 2002 & National Green Tribunal Act, 2010

NGT Act 2010

Act	Year	Key body
BDA	2002	NBA Chennai
NGT Act	2010	NGT Delhi

Common Traps — Don't Fall For These

Trap	Correct fact
Water Act = 1981	Water 1974; Air 1981
EPA = 1972	EPA 1986; WPA 1972
FCA = FRA	1980 diversion vs 2006 rights
Art 48 = environment	Art 48A
Schedule V highest protection	Schedule I
EIA under Water Act	EPA 1986

Trap	Correct fact
NGT grants EC	MoEFCC/SEIAA
NBA HQ Delhi	Chennai
NFP 1988 in syllabus list	Syllabus says 1952 — know both
Consent = EC	SPCB consent ≠ EC

Topic 18 — International Environmental Agreements & Conferences

Current Affairs (this topic)

- COP outcomes and new listings refresh CA; treaty years and Montreal≠Kyoto stay fixed.
- Ramsar / CITES / CMS updates — verify live lists; UP Ramsar baseline remains **12**.

Confused Pairs

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
Stockholm Conference (1972)	Rio Earth Summit (1992)	First UN global conference on the Human Environment that placed environment on the global political agenda and created UNEP vs UN Conference on Environment and Development (UNCED) that produced Rio Declaration, Agenda 21, CBD, UNFCCC, and UNCCD	स्टॉकहोम सम्मेलन 1972 (UNEP उद्भव) / रियो पृथ्वी शिखर सम्मेलन 1992 (रियो संधियां)
Kyoto Protocol (1997)	Paris Agreement (2015)	Annex-I developed nations bound by legal emission targets with punitive compliance mechanisms vs universal bottom-up framework where all 196+ nations self-determine mitigation contributions (NDCs) with five-yearly global stocktakes	क्योटो प्रोटोकॉल (शीर्ष-से-नीचे बाध्यकारी) / पेरिस समझौता (स्वैच्छिक NDCs)
Basel Convention (1989)	Rotterdam Convention (1998)	Restricts and regulates transboundary movements and disposal of hazardous wastes (especially preventing dumping in developing countries) vs establishes legally binding Prior Informed Consent (PIC) procedure for hazardous international trade chemicals and pesticides	बासेल कन्वेंशन (खतरनाक अपशिष्ट आवागमन) / रॉटरडैम कन्वेंशन (पूर्व सूचित सहमति)
Stockholm Convention (2001)	Minamata Convention (2013)	Mandates the global elimination, restriction, and unintentional release reduction of Persistent Organic Pollutants (POPs - "Dirty Dozen") vs global treaty designed to protect human health and the environment from anthropogenic emissions and releases of toxic Mercury (Hg)	स्टॉकहोम संधि (POPs उन्मूलन) / मिनामाता संधि (पारा प्रदूषण नियंत्रण)
Ramsar Convention (1971)	Convention on Migratory Species (CMS / Bonn 1979)	Dedicated specifically to the ecological conservation and sustainable wise use of internationally significant wetlands and waterfowl habitats vs treaty dedicated to the conservation of terrestrial, aquatic, and avian migratory wild animal species across range states	रामसर कन्वेंशन (आर्द्रभूमि संरक्षण) / बॉन कन्वेंशन (प्रवासी प्रजाति संरक्षण)
CITES (1973)	Convention on Biological Diversity (CBD 1992)	International trade treaty strictly regulating commercial import/export of endangered wild flora and fauna through Appendices I, II, III vs comprehensive framework convention covering in-situ/ex-situ conservation, sustainable use, and fair benefit-sharing	साइट्स (CITES, वन्यजीव व्यापार) / जैव विविधता संधि (CBD, समग्र संरक्षण)
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18.1 Conferences — Stockholm, Brundtland, Rio, Agenda 21 & SD Summits

Conference	Year	Key output
Stockholm	1972	UNEP, WED 5 June
Limits to Growth	1972	Club of Rome study
Brundtland	1987	SD definition
Montreal enforcement	Jan 1989	ODS phase-out starts
Rio	1992	UNFCCC, CBD, Agenda 21

Conference	Year	Key output
Earth Summit +5	1997	Rio progress review (NY)
Johannesburg	2002	Rio+10
Rio+20	2012	<i>The Future We Want</i> ; PAGE
SDGs	2015	17 goals, 2030 deadline

18.2 Climate Treaties — UNFCCC, Kyoto Protocol & Paris Agreement

Treaty	Year	Model
UNFCCC	1992	Framework
Kyoto	1997	Binding Annex-I targets
Paris	2015	Universal NDCs

18.4 Biodiversity & Species Treaties — CBD, CITES, CMS & Ramsar

Ramsar

Treaty	Issue	Year
CBD	Biodiversity	1992
CITES	Species trade	1973
CMS	Migratory species	1979
Ramsar	Wetlands	1971

18.5 Chemicals & Waste Treaties — Basel, Rotterdam, Stockholm POPs & Minamata

Treaty	Targets
Basel	Hazardous waste shipment
Rotterdam	Hazardous chemicals trade (PIC)
Stockholm POPs	Persistent organic pollutants
Minamata	Mercury

Mains Answer Framework

UKPCS Prelims 2025

COP session	Location
1. COP 27	A. Belem
2. COP 28	B. Baku
3. COP 29	C. Dubai
4. COP 30	D. Sharm El-Sheikh

Common Traps — Don't Fall For These

Trap	Correct fact
Montreal = climate treaty	Ozone/ODS
Agenda 21 signed 1995	Rio 1992
Brundtland = Limits to Growth	<i>Our Common Future</i> 1987
CITES born at Rio 1992	1973 Washington
Ramsar = forests	Wetlands
CMS = CITES	Migration vs trade
Vienna = binding ODS phase-out	Framework; Montreal = binding
Kyoto = Paris model	Annex binding vs universal NDCs
Stockholm POPs = Stockholm 1972	2001 POPs treaty — different
UNEP HQ Geneva	Nairobi

Topic 19 — Climate & Environmental Institutions

Current Affairs (this topic)

- IPCC AR cycle and UNEP / WMO headlines are institutional CA.
- Keep UNFCCC (यूएनएफसीसीसी) Secretariat vs IPCC role split clear.

Confused Pairs

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
Intergovernmental Panel on Climate Change (IPCC)	UN Framework Convention on Climate Change (UNFCCC)	Scientific assessment panel established in 1988 by WMO and UNEP that reviews peer-reviewed literature (does not conduct research or negotiate policy) vs international legal treaty and negotiating secretariat that coordinates annual Conference of Parties (COPs)	आईपीसीसी (वैज्ञानिक समीक्षा निकाय) / यूएनएफसीसीसी (वार्ता व संधि सचिवालय)
United Nations Environment Programme (UNEP)	United Nations Development Programme (UNDP)	Apex global environmental authority coordinating UN environmental activities, hosting MEA secretariats (HQ: Nairobi, Kenya) vs UN global development network focusing on poverty reduction, sustainable growth, and publishing the Human Development Report (HQ: New York)	यूएनईपी (नैरोबी, पर्यावरण एजेंसी) / यूएनडीपी (न्यूयॉर्क, विकास एजेंसी)
International Union for Conservation of Nature (IUCN)	World Wide Fund for Nature (WWF)	World's oldest and largest global conservation union composed of governments and civil society (1948, Gland) publishing the IUCN Red List vs independent international private conservation NGO (1961, Gland) famous for the Giant Panda symbol and Earth Hour	आईयूसीएन (रेड लिस्ट प्रकाशक) / डब्ल्यूडब्ल्यूएफ (पांडा प्रतीक एनजीओ)
Global Environment Facility (GEF)	Green Climate Fund (GCF)	Multilateral financial partnership established in 1991 serving as financial mechanism for 5 MEAs (CBD, UNCCD, Stockholm, Minamata, UNFCCC) vs dedicated operating financial mechanism of the UNFCCC established at COP16 Cancun focusing specifically on climate projects in developing nations	जीईएफ (GEF, बहु-संधि कोष) / जीसीएफ (GCF, हरित जलवायु कोष)
Food and Agriculture Organization (FAO)	World Meteorological Organization (WMO)	Specialized UN agency leading international efforts to defeat hunger, maintain agricultural biodiversity, and soil health (HQ: Rome) vs specialized UN agency dedicated to atmospheric science, climatology, and operational hydrology (HQ: Geneva)	एफएओ (रोम, खाद्य व कृषि) / डब्ल्यूएमओ (जिनेवा, मौसम विज्ञान)
International Solar Alliance (ISA)	International Renewable Energy Agency (IRENA)	Intergovernmental treaty-based organization launched jointly by India and France at COP21 Paris to promote solar energy deployment (HQ: Gurugram, India) vs intergovernmental organization supporting countries in their transition to all renewable energy types (HQ: Abu Dhabi, UAE)	अंतरराष्ट्रीय सौर गठबंधन (ISA, गुरुग्राम) / इरेना (IRENA, अबू धाबी)
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19.1 IPCC

Body	Role
IPCC	Science assessment
UNFCCC	Treaty negotiations

19.3 UNDP

Report	Publisher
Human Development Report	UNDP
Emissions Gap Report	UNEP

19.5 WWF

Org	Output
IUCN	Red List
WWF	Living Planet Report

Common Traps — Don't Fall For These

Trap	Correct fact
IPCC sets NDCs	UNFCCC/COP
IPCC = UNFCCC	Science vs treaty
UNEP HQ Geneva	Nairobi
Emissions Gap = IPCC/UNDP	UNEP
HDR = UNEP	UNDP
HDI team included Priesner	Haq + Sen
IUCN regulates trade	CITES
WWF = IUCN	Same city, different roles
FAO HQ Nairobi	Rome
GEF donors only developed	Stmt 2 false in Q145

Topic 20 — Biodiversity Conservation Methods

Current Affairs (this topic)

- 30×30 / Kunming–Montreal CA sits here with in-situ vs ex-situ method traps.
- Sacred grove and seed-bank examples stay teaching baselines.

Confused Pairs

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
In-situ Conservation	Ex-situ Conservation	On-site conservation of wild populations within their natural ecosystems where evolutionary processes continue uninterrupted (National Parks, Wildlife Sanctuaries, Sacred Groves) vs off-site conservation of threatened components outside natural habitats in human custody (Botanical Gardens, Zoological Parks, Cryobanks)	स्व-स्थाने संरक्षण (प्राकृतिक आवास) / बाह्य-स्थाने संरक्षण (मानव-निर्मित व्यवस्था)
Botanical Garden	Herbarium	Ex-situ facility maintaining curated collections of <i>living</i> plant species for scientific research, propagation, and conservation display vs reference repository of <i>dried, pressed, and mounted</i> plant specimens preserved on archival sheets for taxonomic identification	वनस्पति उद्यान (जीवित पादप संग्रह) / पादप संग्रहालय (हर्बेरियम, शुष्क नमूने)
Zoological Park (Zoo)	Safari / Wildlife Sanctuary	Ex-situ facility where wild animals are maintained in artificial enclosures under human management for captive breeding and public education vs in-situ natural protected habitat where free-ranging wild animals live in untamed wild conditions	चिड़ियाघर (कृत्रिम बाड़ा) / वन्यजीव अभयारण्य / सफारी
Orthodox Seeds	Recalcitrant Seeds	Desiccation-tolerant seeds that can survive dehydration to low moisture content ($\leq 5\%$) and sub-zero freezing, ideal for conventional long-term seed banking (wheat, rice, legumes) vs desiccation-sensitive seeds that lose viability rapidly if dried or frozen, requiring cryopreservation (mango, rubber, jackfruit, tea)	सामान्य (रूढ़िवादी) बीज / कठिन (असहनीय) बीज
Cryopreservation	Seed Bank	Ultra-low temperature preservation of living cells, gametes, embryos, and tissues suspended in liquid nitrogen at -196°C stopping all metabolic decay vs storage of dried orthodox seeds in vault freezers at standard temperatures of -18°C to -20°C	क्रायो-संरक्षण (-196°C द्रवित नाइट्रोजन) / बीज बैंक (-18°C शुष्क भंडारण)
Sacred Groves	Biosphere Reserves	Community-protected virgin forest patches preserved through traditional spiritual beliefs and religious taboos (e.g. Deorais, Orans, Kavu, Law Kyntang) vs modern scientific statutory conservation landscape demarcated by government under UNESCO MAB	पवित्र उपवन (पारंपरिक संरक्षण) / जैवमंडल आरक्षित क्षेत्र (वैज्ञानिक प्रारूप)
National Bureau of Plant Genetic Resources (NBPGR)	National Bureau of Animal Genetic Resources (NBAGR)	Premier ICAR institute managing the National Gene Bank for agricultural crop and wild relative germplasm (HQ: New Delhi) vs premier ICAR institute maintaining registrations and genetic databases of indigenous livestock breeds (HQ: Karnal, Haryana)	एनबीपीजीआर (नई दिल्ली, पादप जीन बैंक) / एनबीएजीआर (करनाल, पशु आनुवंशिकी)
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20.1 In-situ Conservation

Method	Type	India example
National Park	In-situ	Dudhwa (UP)
Wildlife Sanctuary	In-situ	Katerniaghat (UP)
Biosphere Reserve	In-situ	Similipal
Sacred Grove	In-situ	Tribal (आदिवासी) UP belts
Seed Bank	Ex-situ	NBPGR Delhi

20.2 Ex-situ Conservation

Feature	In-situ	Ex-situ
Location	Natural habitat	Captive/stored
Priority	Primary	Supplementary
Examples	Dudhwa NP, sacred grove	Delhi Zoo, NBPGR
trap	NP, sanctuary, BR	Zoo, seed bank, botanic garden

20.3 Gene Bank

Institution	Location	Conserves
ICAR-NBPGR	New Delhi	Plant seeds, pollen, tissue, DNA
National Gene Bank	New Delhi (NBPGR)	Long-term base collection
NBAGR	Karnal, Haryana	Livestock semen, embryos, DNA
Svalbard Global Seed Vault	Norway	Duplicate global seed backup

20.4 Seed Bank

Type	Storage	Examples
Orthodox	Conventional seed bank (-18°C)	Wheat, rice, chickpea
Recalcitrant	Living collection or cryo	Mango, coconut, jackfruit
Feature	Seed Bank	Gene Bank
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Material	Seeds primarily	Seeds + DNA + tissue + sperm + embryos
Scope	Narrower	Broader
Both are	Ex-situ	Ex-situ

20.5 Botanical Garden

Garden	Location	Notable
Indian Botanic Garden	Howrah	1787; Great Banyan; largest in India
Lalbagh	Bengaluru	1760; heritage trees
NBRI Garden	Lucknow	Medicinal plant research

20.6 Zoological Park

Facility	Location	Type
National Zoological Park	New Delhi	Zoo (ex-situ)
CZA	New Delhi	Regulator (MoEFCC)
Kukrail Crocodile Centre	Lucknow, UP	Captive breeding (ex-situ)
Katerniaghat WLS	UP	Sanctuary (in-situ contrast)

Common Traps

Trap	Correct
Zoo = in-situ	Ex-situ — captive, off-site
Seed bank = in-situ	Ex-situ — cold storage off-site
Botanical garden = in-situ	Ex-situ — managed living collection
NBPGR = Kolkata	New Delhi — Kolkata = BSI/ZSI
NBAGR = Delhi or NBPGR	Karnal — animals only
NBPGR stores animal semen	NBAGR Karnal
BSI = botanical garden	BSI surveys flora; garden is a facility
Zoos are primary conservation	In-situ PAs preferred
Mango in standard seed bank	Recalcitrant — living collection/cryo
Svalbard vault in India	Norway — duplicate backup only
Sanctuary = zoo	Sanctuary = wild in-situ ; zoo = captive ex-situ
Gene bank = in-situ farmer field	Formal gene bank = ex-situ storage

Topic 21 — Species & Ecology

Current Affairs (this topic)

- New IUCN category moves for flagship species (प्रतीक (फ्लैगशिप) प्रजाति) are the usual CA refresh.
- Keystone / umbrella / flagship / indicator definitions stay concept-fixed.

Confused Pairs

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
IUCN Red List	Red Data Book of India	Global extinction risk assessments maintained by IUCN (HQ: Gland, Switzerland) categorizing global species from LC to EX vs national taxonomic inventories published by BSI (plants) and ZSI (animals) cataloguing Indian threatened flora/fauna	आईयूसीएन रेड लिस्ट (वैश्विक) / भारतीय रेड डेटा बुक (राष्ट्रीय - BSI/ZSI)
Threatened Species	Endangered Species (EN)	Broader collective umbrella category encompassing three distinct risk levels: Vulnerable (VU) + Endangered (EN) + Critically Endangered (CR) vs a specific single IUCN conservation status indicating very high risk of extinction	संकटग्रस्त प्रजातियां (VU + EN + CR समग्र) / लुप्तप्राय प्रजाति (EN विशिष्ट वर्ग)
Mangrove Ecosystem	Coral Reef Ecosystem	Salt-tolerant halophytic coastal wetland forests adapted to intertidal muddy sediments featuring pneumatophores (Rhizophora, Avicennia) vs marine calcium carbonate structures formed by symbiotic coral polyps and zooxanthellae in clear, shallow, warm waters	मैंग्रोव पारितंत्र (श्वसन मूल वन) / प्रवाल भित्ति पारितंत्र (उष्ण जलीय प्रवाल)
Pneumatophores	Stilt Roots	Specialized vertical negative geotropic aerial roots that emerge above mud to facilitate oxygen gas exchange in waterlogged swamp soils (Avicennia) vs adventitious support roots emerging from lower trunk/branches providing mechanical anchor against coastal tidal surges (Rhizophora)	वात-मूल (श्वसन मूल, न्यूमेटोफोर) / अवस्तंभ मूल (सहारा जड़ें)
Coral Bleaching	Ocean Acidification	Stress response where coral polyps expel their symbiotic photosynthetic zooxanthellae due to elevated sea surface temperatures turning white vs chemical lowering of ocean pH caused by oceanic absorption of excess atmospheric CO ₂ dissolving calcium carbonate aragonite structures	प्रवाल विरंजन (ऊष्मा तनाव से शैवाल निष्कासन) / महासागरीय अम्लीकरण (pH ह्रास)
Neritic Marine Zone	Oceanic (Pelagic) Zone	Shallow coastal marine zone extending from low-tide mark to the edge of the continental shelf (depth up to 200 m) with abundant light and high marine productivity vs deep open marine waters beyond the continental shelf edge where nutrients and light drop precipitously	नेरिटिक मंडल (महाद्वीपीय मग्नतट) / खुला महासागरीय मंडल (पेलैजिक)
Zooxanthellae	Coral Polyp	Photosynthetic single-celled dinoflagellate microalgae living symbiotically within coral tissues providing up to 90% of their organic nutrients vs tiny, soft-bodied marine invertebrate animal belonging to Phylum Cnidaria that secretes a hard calcium carbonate skeleton	ज़ूज़ैथेली (सहजीवी प्रकाश-संश्लेषी शैवाल) / प्रवाल पॉलीप (कैल्शियम सावी जंतु)
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21.1 Red Data Book

Feature	Red Data Book (India)	IUCN Red List
Scope	National	Global
Publisher	BSI (plants), ZSI (animals)	IUCN (Gland)

Feature	Red Data Book (India)	IUCN Red List
Function	National threat inventory	Standardised risk categories

21.2 IUCN Red List Categories

Code	Category
EX	Extinct
EW	Extinct in Wild
CR	Critically Endangered
EN	Endangered
VU	Vulnerable
NT	Near Threatened
LC	Least Concern
DD	Data Deficient
NE	Not Evaluated

21.3 Wetlands

Type	India example
Freshwater marsh	Keoladeo (केवलादेव) Ghana (घन)
Riverine wetland	Upper Ganga (UP)
High-altitude wetland	Chandra Taal, Tso Moriri
Coastal/mangrove wetland	Sundarbans

21.4 Mangroves

Species	Feature
Rhizophora	Stilt/prop roots
Avicennia	Salt-secreting leaves
Sonneratia	Pneumatophores

21.6 Grasslands

Type	Location	Key species
Terai grassland	UP foothills	Barasingha, tiger — Dudhwa
Floodplain grassland	Assam	Rhino — Kaziranga
Semi-arid savanna	Deccan	Blackbuck
Alpine meadow	High Himalaya	Medicinal herbs

21.7 Himalayan Ecosystem

Zone	Altitude (approx.)	Vegetation
Tropical foothills	<1,000 m	Sal, riverine forest
Temperate	2,000–3,000 m	Deodar, fir
Alpine meadow	3,600–4,500 m	Herbs, dwarf shrubs
Nival	>4,500 m	Lichens, permanent snow

21.8 Marine Ecosystem

Zone	Character	Productivity
Intertidal	Tide-exposed shore	Moderate (नरम दल)
Neritic	Continental shelf	High
Pelagic	Open ocean	Low–moderate
Benthic	Seafloor	Variable

Common Traps

Trap	Correct
Threatened = Endangered only	VU + EN + CR
Red Data Book = IUCN Red List	National vs global
BSI = animal Red Data Book	BSI = plants; ZSI = animals
IUCN = CITES	Assessment vs trade
Sultanpur Ramsar in UP	Haryana (Gurugram)
Rudrasagar Ramsar in UP	Tripura
MISHTI launched 2019	2023–24 Budget
Coral bleaching from cooling	Heat (+2°C) stress
Mangroves = freshwater wetland only	Coastal saline
Gangetic dolphin = marine	Freshwater river
Zoo/sanctuary same as wetland	Different ecosystems
EW = EX	EW = captive survival remains

Topic 22 — Renewable Energy

Current Affairs (this topic)

- India's renewable capacity and green-hydrogen headlines refresh energy CA.
- Keep renewable vs clean vs conventional classification traps stable.

Confused Pairs

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
Solar Photovoltaic (PV)	Concentrated Solar Power (CSP)	Semiconductor solar cells that convert incident sunlight photon energy directly into direct-current electricity via the photoelectric effect vs optical mirrors/lenses that concentrate solar thermal energy to heat a fluid and drive a steam turbine	सौर फोटोवोल्टिक (सीधा विद्युत उत्पादन) / सेंकेंद्रित सौर ताप (CSP, टरबाइन चालित)
Green Hydrogen	Grey Hydrogen / Blue Hydrogen	Produced by zero-carbon electrolysis of water powered entirely by renewable electricity (solar/wind) vs produced via steam methane reforming of natural gas releasing unabated CO ₂ (Grey) or with carbon capture and storage (Blue)	हरित हाइड्रोजन (शून्य कार्बन जल अपघटन) / धूसर व नीला हाइड्रोजन (जीवाश्म स्रोत)
Biomass Energy	Biogas	Direct thermal or electrical energy produced by combusting solid organic matter (bagasse, rice husk, wood pellets) vs gaseous fuel mixture consisting primarily of methane (50–70%) and CO ₂ generated through anaerobic microbial digestion of organic waste	बायोमास ऊर्जा (ठोस जैविक ईंधन) / बायोगैस (अवायवीय पाचन, मीथेन प्रधान)
Geothermal Energy	Ocean Thermal Energy (OTEC)	Harnesses natural high-temperature hydrothermal heat and steam from Earth's deep crustal magma reservoirs (e.g. Puga Valley, Ladakh; Manikaran, HP) vs exploits the temperature differential between warm tropical surface ocean water and freezing deep seawater ($\geq 20^{\circ}\text{C}$ gradient)	भू-तापीय ऊर्जा (पृथ्वी का आंतरिक ताप) / महासागरीय तापीय ऊर्जा (OTEC, सतही-गहन ताप भेद)
Small Hydro Power (SHP)	Large Hydro Power	Hydropower installations with individual installed capacity up to and including 25 MW administered under the Ministry of New and Renewable Energy (MNRE) vs large dam-based hydropower plants >25 MW administered under the Ministry of Power	लघु जल विद्युत (≤ 25 MW, MNRE) / बड़ी जल विद्युत परियोजनाएं (>25 MW)
International Solar Alliance (ISA)	International Renewable Energy Agency (IRENA)	Treaty-based intergovernmental solar coalition initiated by India and France at COP21 Paris with global HQ in Gurugram, India vs global renewable energy intergovernmental agency covering all renewables founded in 2009 with HQ in Abu Dhabi, UAE	अंतरराष्ट्रीय सौर गठबंधन (ISA, गुरुग्राम) / इरेना (IRENA, अबू धाबी)
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22.1 Solar Energy

Scheme	Year	Fact
National Solar Mission	2010	NAPCC pillar / JNNSM
ISA	2015	HQ Gurugram; Summit 2018 Delhi
PM-KUSUM	2019	Solar pumps
PLI Solar PV	2021	Domestic manufacturing

22.2 Wind Energy

Feature	Onshore	Offshore
Cost	Lower	Higher
Wind quality	Moderate (नरम दल)	Stronger, steadier
India status	Mature	Early policy stage

22.6 Small Hydro Projects

Category	Capacity
Small hydro (SHP)	≤25 MW
Large hydro	>25 MW
Ramganga	~198 MW (large)

22.7 Green Hydrogen

Type	Source	CO ₂
Green	Renewable electrolysis	Very low
Grey	Natural gas	High
Blue	Gas + CCS	Moderate

Common Traps

Trap	Correct
Biogas = butane + CO ₂	Methane + CO₂
ISA HQ Paris/New Delhi	Gurugram
ISA exactly 38 members / all UN	Outdated / false
Rajasthan leads wind	Tamil Nadu wind; Rajasthan solar
Geothermal from Sun	Earth interior heat
Geothermal in Ganga Delta	Not a listed province
Tidal = wave energy	Moon gravity vs wind waves
Tidal potential in Thar	Khambhat/Kutch coast
SHP = 50 MW	≤25 MW
Nepal border = zero Indian hydro	False — Q31
Grey hydrogen = green	Grey = fossil gas
Largest solar park = always Gujarat	Bhadla, Rajasthan
Fuel cell = AC	DC; heat + water
DCR = TV licences	Domestic solar PV content
India #3 silicon wafers	Not a wafer leader
Green H ₂ Mission 2019	January 2023
Nuclear = renewable in MCQs	Usually non-renewable category
Singrauli = hydro	Thermal power hub

Topic 23 — Disaster & Environment

Current Affairs (this topic)

- Flood / landslide / heat-wave seasons refresh disaster–environment CA for UP–Himalaya (हिमालय).
- Sendai Framework year (**2015**) stays fixed — link hazards to land-use and climate stress.

Confused Pairs

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
Meteorological Drought	Agricultural Drought	Situation where actual seasonal rainfall over an area is significantly less than the long-term climatological mean (>25% deficient = moderate; >50% deficient = severe) vs condition where soil moisture deficiency severely impedes crop growth, triggering wilting and agrarian loss	मौसमी (मौसम विज्ञान) सूखा / कृषि सूखा (मृदा नमी संकट)
Hydrological Drought	Socio-Economic Drought	Prolonged meteorological drought leading to depletion of surface water reservoirs, dried-up streams, and falling groundwater aquifer tables vs severe water scarcity disrupting economic goods, public water supply, and livelihoods	जल-वैज्ञानिक सूखा (जलाशय हास) / सामाजिक-आर्थिक सूखा (आजीविका संकट)
Cloudburst	Urban Flash Flood	Extreme, highly localized torrential rainfall event exceeding 100 mm per hour over a small geographic area (~20–30 km ²) causing catastrophic flash torrents vs rapid inundation of built-up city areas caused by impermeable concrete surfaces and choked storm drains	बादल फटना (≥100 मिमी/घंटा अतिवृष्टि) / नगरीय आकस्मिक बाढ़
Storm Surge	Tsunami	Abnormal coastal sea-level rise and massive sea surge pushed onshore by the fierce winds and extreme low atmospheric pressure of a tropical cyclone vs series of immense sea waves generated by underwater seaquake, submarine volcanic eruption, or seafloor tectonic dislocation	तूफानी महोर्मि (चक्रवात जनित उभार) / सुनामी (अंतःसमुद्री भूकंप जनित तरंग)
National Disaster Management Authority (NDMA)	National Institute of Disaster Management (NIDM)	Apex statutory policy-making and executive body for disaster management established under DM Act 2005, chaired ex-officio by the Prime Minister vs premier national statutory institute responsible for capacity building, research, training, and documentation in disaster management	राष्ट्रीय आपदा प्रबंधन प्राधिकरण (NDMA, नीति पीठ) / राष्ट्रीय आपदा प्रबंधन संस्थान (NIDM, प्रशिक्षण)
Hazard	Vulnerability	Dangerous physical event, phenomenon, or human activity that has the potential to cause loss of life, injury, or property damage (e.g. earthquake, cyclone) vs conditions determined by physical, social, economic, and environmental factors increasing susceptibility to hazards	आपदा जोखिम/खतरा (Hazard) / सुभेद्यता (Vulnerability)
—			

23.1 Floods

Type	Speed	UP example
Riverine	Days–weeks	Ganga at Prayagraj (प्रयागराज)/Varanasi (वाराणसी)
Flash	Hours	Terai foothills
Urban	Hours	Lucknow (लखनऊ), Kanpur (कानपुर) waterlogging

23.4 Cyclones

Feature	Eye	Eyewall
Winds	Calm	Strongest
Pressure	Lowest	Rising outward

23.5 Heat Waves

Feature	Heat wave	Urban heat island
Scale	Regional weather event	Local city microclimate
Duration	Days–weeks	Persistent year-round bias
Cause	Atmospheric pattern	Built surfaces, lost greenery

Common Traps

Trap	Correct
DPAP = 1952	1973 (CDP = 1952)
Strongest winds in cyclone eye	Eyewall
Landslides purely natural	Deforestation/road cutting trigger
Heat wave = urban heat island	Different scale/cause
All drought = only rainfall deficit	Also agricultural/hydrological
Piral Lao = UP scheme	Uttarakhand
India disaster-free	False
Arabian Sea > Bay for cyclones	Bay of Bengal more
Floods only coastal	UP riverine/Terai major
Mangroves worsen cyclones	Reduce surge/wind damage
NDMA Chair = Home Minister	Prime Minister
Sendai = ozone treaty	Disaster risk reduction 2015

Topic 24 — Current Environmental Issues

Current Affairs (this topic)

- This chapter is the CA layer — keep NbS, blue economy, and plastics stems current.
- UP is landlocked: do not invent statewide blue-economy projects.

Confused Pairs

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
Primary Microplastics	Secondary Microplastics	Microscopic plastic particles (<5 mm) intentionally manufactured for commercial applications (microbeads in facial scrubs, toothpaste, virgin industrial resin pellets) vs microplastics formed by physical weathering, photo-degradation, and fragmentation of macro-plastics in the environment	प्राथमिक माइक्रोप्लास्टिक (निर्मित मनके) / द्वितीयक माइक्रोप्लास्टिक (अपघटित कण)
Carbon Credit	Green Credit (2023)	Tradable compliance unit representing 1 metric tonne of CO ₂ equivalent emission reduction or removal registered under carbon markets vs incentives under India’s Green Credit Programme (2023) rewarding diverse eco-actions (afforestation, water harvesting, mangrove conservation) beyond carbon	कार्बन क्रेडिट (1 टन CO ₂ e मानक) / ग्रीन क्रेडिट (पर्यावरण कार्य प्रोत्साहन)
Net Zero Emissions	Gross Zero Emissions	Balancing residual greenhouse gas emissions across all sectors with equivalent anthropogenic carbon removals and sequestration by 2050/2070 vs absolute elimination of all greenhouse gas emissions to absolute zero without relying on offsets or sinks	नेट जीरो (उत्सर्जन-अवशोषण संतुलन) / सकल शून्य (पूर्ण उत्सर्जन शून्यता)
Blue Economy	Blue Carbon	Sustainable utilization of ocean, coastal, and marine resources for economic growth, improved livelihoods, and jobs while preserving marine ecosystem health vs organic carbon captured, sequestered, and stored over centuries by coastal marine ecosystems (mangroves, seagrasses, salt marshes)	नीली अर्थव्यवस्था (महासागरीय संधारणीय विकास) / नीला कार्बन (तटीय मैंग्रोव संचित कार्बन)
Green Finance	Greenwashing	Structured financial investments, bonds, and banking loans directed specifically towards environmentally sustainable development and clean energy projects vs deceptive marketing, publicity, or PR spin falsely exaggerating a company’s environmental credentials and sustainability	हरित वित्त (टिकाऊ निवेश) / ग्रीनवॉशिंग (झूठा पर्यावरण प्रचार)
Mission LiFE (2022)	Lifestyle for Environment	Global citizen movement launched in October 2022 at Kevadia by PM Modi and UN Secretary-General mobilizing individuals to become ‘Pro-Planet People’ (P3) through 75 actionable behavioral changes vs abstract corporate environmental pledges	मिशन लाइफ (LiFE, P3 नागरिक आंदोलन) / जीवनशैली पर्यावरण
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24.1 Microplastics

Type	Origin	Example
Primary	Manufactured small	Microbeads, nurdles
Secondary	Large plastic breaks down	Bag fragments, tyre particles

24.3 Carbon Trading

Concept	What it is
Carbon credit	Unit = 1 tCO ₂ e
Carbon trading	Market exchanging credits/allowances
Carbon offset (कार्बन ऑफ़सेट)	Using credits to compensate own emissions

24.4 Net Zero Emissions

Country	Net zero year
EU / UK	2050
China	2060
India	2070

24.5 Green Finance

Instrument	India milestone
Sovereign Green Bonds	FY 2022–23
SEBI Green Bond Framework	2023
Loss and Damage Fund	COP28 2023

24.6 Blue Economy

Sector	Scheme
Mangrove restoration	MISHTI 2023
Ports/shipping	Sagarmala
Deep-sea exploration	Deep Ocean Mission

24.7 LiFE Mission

Field	Detail
Idea	COP26 Glasgow 2021
Launch	June 2022
Nodal	MoEFCC
Actions	75 pro-planet behaviours

Common Traps

Trap	Correct
LiFE idea at COP-25 Madrid	COP26 Glasgow 2021
LiFE launched 2019	June 2022
MISHTI launched 2019	2023–24 Budget
India net zero 2050	2070

Trap	Correct
Microplastics <5 cm	<5 mm
Microplastics biodegrade quickly	Persist decades
Carbon credit = carbon trading	Unit vs market
Green Credit = carbon credit	Different schemes
Green finance = Green Credit	Capital vs incentive certificate
Montreal = climate treaty	Ozone — Kyoto/Paris = climate
PAT trades carbon credits directly	ESCerts — energy efficiency
Net zero = gross zero tomorrow	Balanced with removals
MISHTI for Ganga/UP	Coastal mangroves

Topic 25 — Global Environmental Geography

Current Affairs (this topic)

- Biome (जीवोम) and realm stems stay geographic; CA rarely moves the Köppen (कोपेन) or biome map.
- Bridge biome teaching with Ecology (पारिस्थितिकी) & Ecosystem (पारिस्थितिकी तंत्र) without duplicating full cards.

Confused Pairs

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
Biome	Ecosystem	Continental-scale geographical region characterized by distinct climate regimes, soil types, and climax vegetation communities (e.g. Tundra, Taiga, Savanna) vs localized structural and functional ecological unit where living organisms interact with abiotic components (e.g. a pond, forest patch)	जीवोम (बायोम, महाद्वीपीय स्तर) / पारिस्थितिकी तंत्र (स्थानीय कार्यात्मक इकाई)
Taiga (Boreal Forest)	Tundra	World's largest terrestrial biome dominated by evergreen needleleaf coniferous trees (Spruce, Fir, Pine, Larch) in subarctic latitudes south of the tree-line vs cold, treeless arctic and alpine biome dominated by mosses, lichens, dwarf shrubs, underlain by permafrost	टैगा (बोरियल शंकुधारी वन) / टुंड्रा (वृक्षविहीन शीत मरुस्थल)
Savanna	Steppe / Prairie	Tropical grassland biome characterized by continuous tall grass cover interspersed with scattered, drought-deciduous drought-resistant trees (e.g. Acacia, Baobab) vs temperate mid-latitude semi-arid grassland biome with treeless expanses of short to tall grasses	सवाना (उष्णकटिबंधीय सवृक्ष घास मैदान) / स्टेपी/प्रेयरी (शीतोष्ण घास मैदान)
Selva	Puszta	Dense equatorial tropical rainforest biome of the Amazon River basin characterized by evergreen multi-tiered canopy and extreme biodiversity vs temperate flat grassland plain of the Pannonian Basin in Hungary characterized by continental climate and fertile chernozem soils	सेल्वा (अमेज़न वर्षावन) / पुस्ज़टा (हंगरी शीतोष्ण घास मैदान)
Permafrost	Mollisols (Chernozem)	Subsurface ground (soil, sediment, or rock) that remains continuously frozen below 0°C for at least two consecutive years, trapping massive ancient methane and organic carbon deposits vs deep, dark, humus-rich organic soils characteristic of temperate grasslands with highest agricultural fertility	स्थायी हिम मृदा (पर्माफ्रॉस्ट) / चेर्नोज़म (काली उपजाऊ घास मृदा)
Biodiversity Hotspot	Ecological Island	Large biogeographic region meeting Norman Myers' criteria ($\geq 1,500$ endemic plant species and $\geq 70\%$ primary habitat lost) vs isolated terrestrial habitat surrounded by completely different ecosystems or human urbanization functioning like an oceanic island	जैव विविधता हॉटस्पॉट (हासग्रस्त समृद्ध क्षेत्र) / पारिस्थितिक द्वीप (पृथक वास)
### Biome ↔ ecosystem bridge (read with Ecology chapter)			

- A **biome** is the large climate-defined vegetation–fauna belt (Amazon Selva, taiga, savanna).
- An **ecosystem** (पारिस्थितिकी तंत्र) is the local functional unit of energy flow and nutrient cycling inside that belt.
- Global textbook biome counts (often ~14) are **not** India's **ten** biogeographic zones (Rodgers and Panwar).
- UP plains sit mainly in the **Gangetic Plain** biogeographic colour with Terai (तराई)–deciduous transitions — not a tropical rainforest biome answer.

25.2 Major Global Biomes

Biome	Climate	Global example
Tropical rainforest	Hot wet year-round	Amazon Selva
Desert	Arid <25 cm	Sahara, Gobi, Thar
Savanna	Wet-dry grassland + trees	East Africa
Temperate grassland	Moderate (नरम दल) rain	Pampas, Puszta
Taiga	Cold coniferous	Siberia, Canada
Tundra	Very cold, permafrost	Arctic

25.3 Climate Zones (Köppen)

Group	Name	India / example
A	Tropical	India Am
B	Arid	Thar BWh, Gobi BWk
C	Temperate	Mediterranean climates
D	Cold	Siberia taiga
E	Polar	Antarctica

25.4 World Deserts & Critical Ecosystems

Desert	Region	Fact
Sahara	N Africa	Largest hot desert
Atacama	Chile	Driest
Gobi	Mongolia + China	Cold desert
Kalahari	S Africa	Bushman/San
Thar	India-Pakistan	Hot desert

25.5 Floral Regions & Vegetation Matching

Pair	Fact
Planting Hearth	C.O. Sauer
World Floral Regions	Adolf Engler
Cryophyte Steppe	Arctic Tundra
Puszta	Hungary
Pampas	Argentina

25.6 Biodiversity Hotspots & Degradation Belts

India hotspot	Region
Himalaya	Mountain endemism
Western Ghats	Peninsular rainforest

India hotspot	Region
Indo-Burma	NE India
Sundaland	Nicobar

Common Traps

Trap	Correct
Northern Australia temperate	Tropical
Gobi in Russia/Kazakhstan as main	Mongolia + China
Masai = West Africa	East Africa
Selva false / Terra Roxa false	Both TRUE
Engler = Planting Hearth	Sauer = Hearth; Engler = Floral
Puszta = Argentina	Hungary (Pampas = Argentina)
Permafrost in savanna	Taiga + Tundra
Mangrove purely terrestrial	Transitional coastal
India has 14 biogeographic zones	10 zones
Hotspot = richness alone	Endemism + habitat loss

Topic 26 — Water Resources & Water Conservation

Current Affairs (this topic)

- Jal Jeevan / Namami Gange (नमामि गंगे) / wetland CA refresh water chapters.
- Keep watershed vs command-area and BOD water-quality traps stable.

Confused Pairs

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
Integrated Water Resources Management (IWRM)	Watershed Management	Coordinated macro-framework promoting the balanced development of water, land, and related resources across an entire river basin to maximize economic and social welfare equitably vs micro-level land and water conservation technique focusing on a ridge-to-valley catchment draining to a single outlet	एकीकृत जल संसाधन प्रबंधन (IWRM) / वाटरशेड (जलग्रहण) प्रबंधन
Unconfined Aquifer	Confined (Artesian) Aquifer	Permeable water-bearing geological formation whose upper boundary is the free atmospheric water table directly recharged by infiltrating rainfall vs deep aquifer bounded above and below by impermeable aquitards where groundwater is stored under hydrostatic artesian pressure	अप्रतिबंधित जलभृत (मुक्त भूजल स्तर) / प्रतिबंधित (आर्टेशियन) जलभृत
Sewage Treatment Plant (STP)	Common Effluent Treatment Plant (CETP)	Treatment facility designed specifically to clean domestic municipal sewage and blackwater from residential settlements vs collective facility built to treat hazardous industrial toxic effluents generated by a cluster of small- and medium-scale industrial factories	मलजल उपचार संयंत्र (STP, घरेलू सीवेज) / साझा उत्प्लावक उपचार संयंत्र (CETP, औद्योगिक अपशिष्ट)
Rainwater Harvesting (RWH)	Artificial Groundwater Recharge	Collection, conveyance, and storage of runoff rainwater from rooftops or ground surfaces into surface storage tanks for immediate usage vs purposeful diversion and injection of harvested surface runoff deep into depleted subterranean aquifers through recharge shafts/borewells	वर्षा जल संचयन (सतही उपयोग) / कृत्रिम भूजल पुनर्भरण (भूजल रीचार्ज)
Central Ground Water Board (CGWB)	Central Ground Water Authority (CGWA)	Apex national scientific multidisciplinary organization under Ministry of Jal Shakti assessing groundwater resources, aquifer mapping, and monitoring vs statutory regulatory authority constituted under Section 3(3) of EPA 1986 issuing mandatory NOCs and regulating industrial groundwater extraction	केंद्रीय भूजल बोर्ड (CGWB, वैज्ञानिक आकलन) / केंद्रीय भूजल प्राधिकरण (CGWA, कानूनी नियमन)
Jal Jeevan Mission (JRM / Har Ghar Jal)	Atal Bhujal Yojana (ABHY)	Centrally sponsored flagship mission launched in August 2019 ensuring 55 lpcd functional household tap connections (FHTC) to all rural households by 2024 vs World Bank assisted community-led groundwater management scheme focused on water-stressed districts across 7 states	जल जीवन मिशन (हर घर जल, नल कनेक्शन) / अटल भूजल योजना (सामुदायिक भूजल प्रबंधन)
—			

26.2 Integrated Water Resources Management (IWRM) (एकीकृत जल संसाधन प्रबंधन)

Traditional	IWRM
Departmental silos	River basin planning
Supply focus	Supply + demand + quality
Environment (पर्यावरण) ignored	Ecological flows included

26.3 Rainwater Harvesting

Method	Function
Rooftop tank	Direct use
Recharge pit/well	Aquifer recharge
Check dam (ढ़ाँढ़ -)	Slow runoff + percolation
Johad/taanka	Traditional arid storage

26.5 River Conservation

Programme	Year	Fact
Central Ganga Authority / GAP	1985	First Ganga STPs
NRCA rename	1995	Broader river mandate
YAP launch	1993	Yamuna bilateral Japan–India track
NGRBA	2009	PM-chaired basin authority
Namami Gange	2014	NMCG nodal
YAP-III	2016	Under Namami Gange
NRCP	1995	Other rivers

26.7 Groundwater Conservation

Body	Role
CGWB	Groundwater quantity/quality assessment
CGWA	Groundwater regulation (EPA 1986)
CPCB	Surface water & air pollution
CWC	Surface water/irrigation data

26.8 Drinking Water Quality

Parameter	Limit (IS 10500)
pH	6.5–8.5
Fluoride	≤1.5 mg/L
Arsenic	≤0.01 mg/L

Common Traps

Trap	Correct
Rivers second after oceans	Ice then GW then rivers
Groundwater report = CPCB	CGWB
CGWA = CGWB	EPA regulator ≠ assessment board
JJM launched 2014	2019
Namami Gange = 1985	2014 (GAP = 1985)

Trap	Correct
Freshwater = biotic resource	Not — abiotic
NWM = river linking + free borewells	Recycling / coastal tech / +20% efficiency yes; linking/reimbursement no
UV removes odour + turbidity	Only microbial kill among classic claims
Chilika under NLCP	Not
NGRBA chair = rotating CM	Prime Minister
Dolphin decline = more crocodiles	Dams, nets, agrochemicals
Lisbon = freshwater glaciers	Oceans
Hariyali = Jal Kranti	Watershed programme
Namami Gange = 1985	2014 (GAP = 1985)
Gomti = Kanpur	Lucknow
Sultanpur in UP	Haryana
Watershed increases erosion	Reduces erosion
IWRM = dams only	Basin + quality + ecology
CWC = groundwater quality	Surface water data
Drinking water unlimited fluoride	≤1.5 mg/L

Topic 27 — Renewable & Non-Renewable Energy

Current Affairs (this topic)

- Capacity and fuel-mix CA overlaps the Renewable Energy chapter — keep classification traps consistent.
- Nuclear is low-carbon but conventionally non-renewable in many Prelims frames.

Confused Pairs

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
Renewable Energy	Non-Renewable Energy	Energy derived from natural regenerative sources that replenish naturally on a human timescale without depleting stocks (solar, wind, small hydro, geothermal, biomass) vs finite geological energy reserves that take millions of years to form and cannot be replenished once consumed (coal, petroleum, natural gas, uranium)	नवीकरणीय ऊर्जा (अक्षय स्रोत) / अनवीकरणीय ऊर्जा (परिमित जीवाश्म भंडार)
Conventional Energy	Non-Conventional Energy	Traditional energy sources that have been historically established and extensively utilized for commercial power generation for centuries (coal, oil, gas, large hydro, commercial nuclear) vs alternative, modern clean renewable sources developed to replace fossil fuels (solar, wind, tidal, ocean thermal, fuel cells)	पारंपरिक ऊर्जा स्रोत / गैर-पारंपरिक (वैकल्पिक) ऊर्जा स्रोत
Bioethanol	Biodiesel	Pure ethanol fuel produced through the biochemical fermentation of sugary and starchy feedstocks (sugarcane juice, molasses, damaged grains, corn) blended with petrol (E20 mandate) vs fatty acid methyl esters (FAME) produced via the transesterification of non-edible plant vegetable oils (Jatropha curcas, Pongamia, animal fats) blended with diesel	बायो-इथेनॉल (किण्वन, पेट्रोल संवर्धन) / बायो-डीजल (ट्रांसएस्टरीकरण, डीजल संवर्धन)
Perform, Achieve and Trade (PAT)	Renewable Purchase Obligation (RPO)	Market-based regulatory mechanism under the National Mission for Enhanced Energy Efficiency (BEE / EC Act 2001) issuing tradable Energy Saving Certificates (ESCerts) to designated energy-intensive industries vs statutory mandate under Electricity Act 2003 requiring power discoms to purchase a minimum percentage of electricity from renewable sources	निष्पादन, उपलब्धि और व्यापार (PAT, ऊर्जा बचत प्रमाण पत्र) / नवीकरणीय खरीद दायित्व (RPO)
Bureau of Energy Efficiency (BEE)	Petroleum Conservation Research Association (PCRA)	Statutory body established under the Energy Conservation Act 2001 (Ministry of Power) promoting star rating standards for electric appliances and energy conservation codes vs registered society under the Ministry of Petroleum and Natural Gas promoting conservation and fuel efficiency in the petroleum sector	ऊर्जा दक्षता ब्यूरो (BEE, स्टार लेबलिंग) / पेट्रोलियम संरक्षण अनुसंधान संघ (PCRA)
Biogas	Compressed Biogas (CBG)	Unrefined raw gaseous fuel mixture generated from anaerobic digestion of cattle dung and agro-waste containing ~55–60% methane and ~40% CO ₂ vs purified, upgraded methane gas (≥90% CH ₄) scrubbed of CO ₂ and H ₂ S and compressed to ~250 bar for use as clean green automotive transport fuel (SATAT initiative)	बायोगैस (कच्ची गोबर गैस) / संपीड़ित बायोगैस (CBG, परिष्कृत 90%+ मीथेन)
—			

27.3 Conventional Sources of Energy

Conventional	Non-conventional
Coal, oil, gas	Solar, wind, biomass

Conventional	Non-conventional
Large hydro	Small hydro ≤ 25 MW
Nuclear	Tidal, geothermal, ocean

27.8 Biofuels

Fuel	Feedstock	Blend use
Bioethanol	Molasses, maize, 2G waste	Petrol E10/E20
Biodiesel	Jatropha, Pongamia, UCO	Diesel B5/B10
Biogas/CBG	Dung, organic waste	Cooking / transport

27.14 Ocean Energy

Type	Driver
Tidal	Moon gravity
Wave	Wind on sea surface
OTEC	Warm–cold water ΔT

27.16 Renewable Energy Scenario in India

Source	Key fact
Solar	Fastest capacity growth
Wind	TN/Gujarat strongholds
SHP	Himalayan streams ≤ 25 MW
Biomass	Bagasse, biogas, waste-to-energy

27.19 Energy Conservation

Item	Year/body
Energy Conservation Act	2001
BEE	Under Power Ministry
Star labeling	Appliance efficiency

27.20 Energy Efficiency

Scheme	What it trades/does
PAT	ESCerts (energy efficiency)
CCTS	Carbon credits (CO ₂ e)
UJALA	LED bulbs at scale

Common Traps

Trap	Correct
Nuclear = renewable	Usually non-renewable
Non-conventional = non-renewable	= renewable
Coal = biomass	Fossil, not biomass
Uranium = fossil fuel	Nuclear fuel , not fossil
Biogas = butane	CH₄ + CO₂
Tidal = OTEC/wave	Different drivers
PAT = carbon credits	ESCCerts
BEE = MNRE	Power vs Renewables
Large hydro = non-conventional	Usually conventional
E20 already forever since 2010	Target ~2025–26
SPR only in Delhi	Vizag, Mangalore, Padur
Energy crisis = only hydro shortage	Fossil fuel exhaustion risk
Biofuels always cheap	Often not cost-effective
Maize = biodiesel crop	Ethanol ; Jatropha = biodiesel
Fuel cell = AC output	DC ; by-products heat + water
Gasification gas = CO ₂ only	Producer gas CO + H ₂ (+ CH ₄)

Topic 28 — Environmental Research & Institutions

Current Affairs (this topic)

- FRI / FSI / WII / IIRS function traps are evergreen Dehradun (देहरादून) CA colour.
- Do not swap ISFR (FSI) with FRI research mandate.

Confused Pairs

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
Botanical Survey of India (BSI)	Zoological Survey of India (ZSI)	Premier taxonomic institution (HQ: Kolkata, established 1890) undertaking floristic surveys, documenting plant species, and managing the Central National Herbarium vs premier institution (HQ: Kolkata, established 1916) surveying faunal diversity, animal taxonomy, and systematic collections	भारतीय वनस्पति सर्वेक्षण (BSI, 1890) / भारतीय प्राणी सर्वेक्षण (ZSI, 1916)
Forest Survey of India (FSI)	Forest Research Institute (FRI)	Subordinate organization under MoEFCC (HQ: Dehradun, established 1981) conducting national forest resource inventory via satellite imagery and publishing biennial ISFR vs deemed university and premier research institution under ICFRE (HQ: Dehradun, established 1906 as Imperial Forest Research Institute)	भारतीय वन सर्वेक्षण (FSI, द्विवार्षिक वन रिपोर्ट) / वन अनुसंधान संस्थान (FRI, वानिकी अनुसंधान)
National Biodiversity Authority (NBA)	State Biodiversity Board (SBB)	Autonomous statutory regulatory authority established in 2003 under Biological Diversity Act 2002 (HQ: Chennai) regulating foreign nationals' access to Indian bioresources and advising Union vs state-level statutory bodies regulating commercial utilization of bioresources by Indian citizens	राष्ट्रीय जैव विविधता प्राधिकरण (NBA, चेन्नई) / राज्य जैव विविधता बोर्ड (SBB)
National Bureau of Plant Genetic Resources (NBPGR)	National Bureau of Agricultural Insect Resources (NBAIR)	Nodal ICAR organization (HQ: New Delhi) managing the National Gene Bank preserving orthodox seeds and germplasm of food and agricultural crops vs premier ICAR research bureau (HQ: Bengaluru) dedicated to insect taxonomy, molecular characterization, and biocontrol insect resources	राष्ट्रीय पादप आनुवंशिक संसाधन ब्यूरो (NBPGR, दिल्ली) / कृषि कीट संसाधन ब्यूरो (NBAIR, बेंगलुरु)
Wildlife Institute of India (WII)	Indira Gandhi National Forest Academy (IGNFA)	Autonomous scientific research and training institute under MoEFCC (HQ: Dehradun) conducting wildlife ecological research and Tiger Census vs premier apex staff college (HQ: Dehradun) providing two-year professional induction training to directly recruited Indian Forest Service (IFS) officers	भारतीय वन्यजीव संस्थान (WII, अनुसंधान) / इंदिरा गांधी राष्ट्रीय वन अकादमी (IGNFA, IFS प्रशिक्षण)
India State of Forest Report (ISFR)	Basic Animal Husbandry Statistics	Biennial statutory publication by Forest Survey of India (FSI, Dehradun) reporting state-wise forest cover, tree cover, mangrove cover, and forest carbon stocks vs annual publication by Ministry of Fisheries, Animal Husbandry & Dairying reporting livestock census, milk, egg, and meat production	भारत वन स्थिति रिपोर्ट (ISFR, द्विवार्षिक) / बुनियादी पशुपालन सांख्यिकी (वार्षिक)
—			

28.4 Forest Research Institute (FRI) (वन अनुसंधान संस्थान)

Feature	FRI	FSI
Year	1906	1981
Role	Research	Cover survey

Feature	FRI	FSI
Output	Silviculture protocols	ISFR

28.6 National Biodiversity Authority (NBA) (राष्ट्रीय जैव विविधता प्राधिकरण)

Level	Body	Seat	Role
National	NBA	Chennai	Foreign access, ABS
State	SBB	State capital	Domestic commercial access
Local	BMC	Panchayat/ward	PBR

Common Traps

Trap	Correct
BSI = animals	Plants
ZSI = plants	Animals
FRI publishes ISFR	FSI publishes ISFR
NBA HQ Delhi	Chennai
NBA prepares PBR	BMC prepares PBR
SBB = SPCB	Biodiversity ≠ pollution
NBPGR = Kolkata	New Delhi
NBPGR = animal bank	NBAGR Karnal
Bahraich >20% forest	False
ISFR annual	Biennial (2 years)

Topic 29 — Indian Environmental Movements

Current Affairs (this topic)

- Movement years and states are fixed history (इतिहास); CA rarely renames Chipko / Appiko / NBA.
- Keep NBA (Andolan) ≠ National Biodiversity Authority acronym trap ready.

Confused Pairs

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
Bishnoi Movement (1730)	Chipko Movement (1973)	Amrita Devi / Khejri tree sacrifice in Khejarli, Rajasthan vs Gaura Devi / Sunderlal Bahuguna anti-commercial logging in Chamoli (Uttarakhand)	बिश्नोई आंदोलन (1730) / चिपको आंदोलन (1973)
Chipko Movement (1973)	Appiko Movement (1983)	Himalayan forest conservation in Chamoli (Bahuguna/Chandi Prasad Bhatt) vs Western Ghats forest hugging in Uttara Kannada, Karnataka (Pandurang Hegde)	चिपको (उत्तराखंड) / अप्पिको (कर्नाटक)
Silent Valley Movement (1970s)	Narmada Bachao Andolan (1985+)	Kunthipuzha river anti-hydro dam protecting lion-tailed macaque in Kerala vs anti-large dam displacement on Narmada (Medha Patkar, Baba Amte)	साइलेंट वैली (केरल) / नर्मदा बचाओ आंदोलन
Jungle Bachao Andolan (1982)	Chipko Movement (1973)	Singhbhum (Jharkhand) tribals opposing replacement of natural sal with commercial teak ("Greed Game Political Populism") vs tree-hugging in UP hills	जंगल बचाओ (सिंहभूम) / चिपको आंदोलन
Appiko Movement (1983)	Silent Valley (1970s)	Forest conservation by tree embrace in Karnataka vs grassroots campaign stopping hydro-electric project in Kerala rainforest	अप्पिको (कर्नाटक) / शांत घाटी (केरल)
Tehri Dam Protest	Narmada Bachao Andolan	High-seismic zone anti-dam fast by Sunderlal Bahuguna on Bhagirathi vs multi-state anti-submergence movement across MP/Gujarat/Maharashtra	टिहरी बांध विरोध / नर्मदा बचाओ
Gadgil Commission (2011)	Kasturirangan Committee (2013)	WGEEP designated entire Western Ghats as ecologically sensitive (ESZ 1, 2, 3) vs HLWG recommending only 37% area as Eco-Sensitive Area (ESA)	गाडगिल समिति / कस्तूरिंगन समिति
—			

29.7 Save Western Ghats Movement

Cause, Course, Result

Report	Year	Tone
Gadgil WGEEP	2011	Stronger ESZ / conservation
Kasturirangan HLWG	2013	Narrower ESA / moderated

Common Traps

Trap	Correct
Appiko = Uttarakhand/UP	Karnataka
Chipko = Karnataka	Uttarakhand 1973
Silent Valley = Tamil Nadu	Kerala

Trap	Correct
Bishnoi = 1973 Chipko	1730 Rajasthan
Medha Patkar = Chipko	Narmada Bachao
Appiko = Save W Ghats	1983 vs 2011+
NBA = Biodiversity Authority	Context: movement = Narmada
Jungle Bachao = Garhwal	Singhbhum Jharkhand
Gaura Devi = Appiko	Chipko Reni
Hegde = Chipko	Appiko

Topic 30 — Environmental Literature & Awareness

Current Affairs (this topic)

- Book–author pairs (Carson, Fukuoka, etc.) stay fixed; new award themes are light CA.
- Silent Spring ≠ Silent Valley (साइलेंट वैली) (movement).

Confused Pairs

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
Environmental Education	Environmental Awareness	Structured pedagogical curriculum in schools/colleges (Eco-Clubs, NGC) vs general public sensitization & behavioral change (Mission LiFE)	पर्यावरण शिक्षा / पर्यावरण जागरूकता
Silent Spring (1962)	Tragedy of the Commons (1968)	Rachel Carson's book exposing chemical pesticide (DDT) bioaccumulation vs Garrett Hardin's economic essay on shared resource degradation	साइलेंट स्प्रिंग / ट्रैजेडी ऑफ़ द कॉमन्स
Limits to Growth (1972)	Brundtland Report (1987)	Club of Rome computer model projecting economic/ecological collapse vs UN WCED report (<i>Our Common Future</i>) coining Sustainable Development	लिमिट्स टू ग्रोथ (1972) / ब्रंटलैंड रिपोर्ट (1987)
Small Is Beautiful (1973)	Silent Spring (1962)	E.F. Schumacher's critique of mass production ("economics as if people mattered") vs Carson's chemical toxicity expose	स्मॉल इज ब्यूटीफुल / साइलेंट स्प्रिंग
Eco-mark (1991)	Greenwashing	Official BIS earthen pot label for certified environment-friendly consumer goods vs deceptive corporate marketing claiming false eco-credentials	इको-मार्क (मिट्टी का घड़ा) / ग्रीनवॉशिंग
Greendex	Ecological Footprint	National Geographic consumer sustainability index vs Global Footprint Network measure of nature's demand vs biocapacity	ग्रीनडेक्स / पारिस्थितिक पदचिह्न
National Green Corps (NGC)	Mission LiFE (2022)	MoEFCC Eco-club programme in school networks vs international citizen-movement promoting lifestyle for environment	नेशनल ग्रीन कॉर्प्स / मिशन लाइफ (LiFE)
—			

30.1 Environmental Literature

Work	Author/Body	Year
Silent Spring	Rachel Carson	1962
Tragedy of the Commons	Garrett Hardin	1968
Limits to Growth	Club of Rome	1972
Our Common Future	Brundtland / WCED	1987
Green Development	W.M. Adams	1990

30.2 Environmental Education

Year	Milestone
1972	Stockholm — EE on agenda
1977	Tbilisi Declaration

Year	Milestone
1992	Agenda 21 Ch. 36
~2001–02	NGC / ECO Club national push
2017	GSDP
2020	NEP 2020 ESD integration

30.4 Environmental Communication

Channel	Example
Magazine	Down To Earth
Children's	Gobar Times
Scheme IEC	Namami Gange / Swachh Bharat
People's science	KSSP

30.5 Environment Awards (high-yield)

Identity: Indira Gandhi Paryavaran | Tyler | Global 500 → Champions of the Earth | Rajiv Gandhi | Borlaug

- **Indira Gandhi Paryavaran Puraskar (1987, MoEF)** honours significant measurable contribution to environment protection and improvement — not only wildlife research or wasteland alone.
- **Tyler Prize for Environmental Achievement** (University of Southern California) is often called the “Nobel of Environment” — environmental science, health, and energy. Notable awardees include Madhav Gadgil (2015 pairing), Partha Dasgupta (2016), Pavan Sukhdev & Gretchen Daily (2020).
- **Rajiv Gandhi Environment Award** recognises clean/new technology and industrial pollution-reduction practice (MoEFCC track).
- **Global 500 Award** (UNEP, from **1987**) honoured environmental achievement; later succeeded in spirit by **Champions of the Earth** (from **2005**). Narendra Modi received Champions of the Earth **2018**.
- **Borlaug Award** is for outstanding agricultural research/contribution. Norman Borlaug's Nobel was the **Peace Prize** (Green Revolution (हरित क्रांति)), not a medicine Nobel.

Logic: Tyler ≠ drama/child-rights prize. Global 500 = environment, not anti-terror. AGMARK ≠ Eco-mark; AGMARK ≠ FAO mark.

Common Traps

Trap	Correct
Silent Spring = Hardin	Carson 1962
Brundtland = Limits to Growth	Our Common Future 1987
Limits to Growth = Brundtland	Club of Rome 1972
LiFE idea at COP-25	COP26 2021
Agenda 21 = only EE	Primarily Sustainable Development
ECO Club = CPCB	MoEFCC NGC
Awareness = only school EE	Also campaigns + Art 51A(g)
Bharat Nirman = NEP scheme	Infrastructure, not NEP EE

Trap	Correct
IPCC = MoEFCC awareness campaign	Science assessment body
Van Mahotsav = February	July

Topic 31 — Environmental Days

Current Affairs (this topic)

- Annual themes for WED / Earth Day / IDF refresh; calendar dates stay fixed (WED = **5 June**).
- Do not invent UP-only international day dates.

Confused Pairs

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
World Wetlands Day	World Wildlife Day	2 February (signing of Ramsar Convention 1971) vs 3 March (adoption of CITES 1973)	विश्व आर्द्रभूमि दिवस (2 फरवरी) / विश्व वन्यजीव दिवस (3 मार्च)
World Forestry Day	World Water Day	21 March (UN International Day of Forests) vs 22 March (UN World Water Day)	विश्व वानिकी दिवस (21 मार्च) / विश्व जल दिवस (22 मार्च)
World Earth Day	World Environment Day	22 April (Gaylord Nelson, 1970) vs 5 June (Stockholm Conference opening, 1972)	पृथ्वी दिवस (22 अप्रैल) / विश्व पर्यावरण दिवस (5 जून)
International Biodiversity Day	World Earth Day	22 May (Nairobi agreed text of CBD 1992) vs 22 April (Earth Day)	जैव विविधता दिवस (22 मई) / पृथ्वी दिवस (22 अप्रैल)
World Ozone Day	World Environment Day	16 September (signing of Montreal Protocol 1987) vs 5 June (Stockholm UNEP foundation)	विश्व ओजोन दिवस (16 सितंबर) / पर्यावरण दिवस (5 जून)
Earth Day	Earth Hour	Fixed annual civic observance on 22 April vs WWF voluntary last Saturday of March 1-hour lights-off event	पृथ्वी दिवस (22 अप्रैल) / अर्थ आवर (मार्च अंतिम शनिवार)
World Soil Day	World Day to Combat Desertification	5 December (FAO celebration of King Bhumibol's birthday) vs 17 June (UNCCD adoption anniversary)	विश्व मृदा दिवस (5 दिसंबर) / मरुस्थलीकरण रोकथाम दिवस (17 जून)
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31.10 Other High-Yield Days

Identity: Toilet | No Tobacco | Habitat | Cities | Population (जनसंख्या) | Health | Overshoot

- **World Toilet Day = 19 November** (UNGA 2013; sanitation for all / end open defecation messaging).
- **World No Tobacco Day = 31 May** (WHO) — not 5 May. Themes rotate (e.g. tobacco–environment / grow food not tobacco).
- **World Habitat Day = first Monday of October** (UN; not a fixed calendar date like 5 October).
- **World Cities Day = 31 October. World Population Day = 11 July. World Health Day = 7 April.**
- **Earth Overshoot Day** marks when humanity’s annual demand exceeds Earth’s regenerative capacity — an environmental-degradation awareness marker (Global Footprint Network partnership framing).
- **Swachh Bharat Mission** launched **2 October 2014**. Sierra Leone’s national cleanliness day is the last Saturday of each month.

Logic: Anti-tobacco ≠ 5 May. Habitat Day ≠ fixed 5 Oct.

Common Traps

Trap	Correct
Wildlife Day = 22 March	3 March (Water is 22 March)
Wetlands Day = 2 / 3 March	2 February
Earth Day = 5 June	22 April ; WED = 5 June
Earth Day = 22 March	22 April ; Water = 22 March
Biodiversity Day = 22 March / April	22 May
Forests Day = 22 March	21 March
Ozone Day = 5 June	16 September
WED = Earth Day	Different dates and origins
Earth Hour = Earth Day	Hour ≠ Day
Population Day = 5 June	11 July ; 5 June = WED

Topic 32 — National Parks & Protected Areas (Advanced)

Current Affairs (this topic)

- New NP / TR notifications refresh advanced PA CA — cross-check with the Protected Areas hierarchy.
- Location-match traps stay high-yield; UP has no marine PA.

Confused Pairs

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
National Park (NP)	Wildlife Sanctuary (WLS)	WPA Sec 35; highest protection, no human settlement/grazing, boundaries altered only by State Legislature on NBWL advice vs WPA Sec 18; regulated grazing/traditional rights allowed	राष्ट्रीय उद्यान (NP) / वन्यजीव अभयारण्य (WLS)
Community Reserve	Conservation Reserve	Established on private or community-owned lands under WPA Sec 36C vs established on government-owned contiguous lands under WPA Sec 36A	सामुदायिक रिज़र्व / संरक्षण रिज़र्व
First Marine NP	First National Park in India	Gulf of Kutch Marine NP (1982, Gujarat) vs Hailey / Jim Corbett NP (1936, Uttarakhand)	प्रथम समुद्री उद्यान (कच्छ की खाड़ी) / प्रथम राष्ट्रीय उद्यान (कॉर्बेट 1936)
Dudhwa National Park	Valmiki National Park	Only National Park in Uttar Pradesh (Lakhimpur Kheri) vs only National Park in Bihar (West Champaran)	दुधवा राष्ट्रीय उद्यान (यूपी) / वाल्मीकि राष्ट्रीय उद्यान (बिहार)
Keibul Lamjao NP	Khangchendzonga NP	World's only floating park on Loktak Lake, Manipur (Sangai deer) vs India's only Mixed World Heritage Site in Sikkim	केइबुल लामजाओ (मणिपुर) / कंचनजंगा (सिक्किम)
Hemis National Park	South Button Island NP	India's largest National Park (Ladakh, snow leopard) vs India's smallest National Park (Andaman & Nicobar, 0.03 km²)	हेमिस (लद्दाख, सबसे बड़ा) / साउथ बटन (अंडमान, सबसे छोटा)
Biosphere Reserve Core	National Park	Untouched strictly protected sanctuary within UNESCO MAB zoning vs statutory strictly protected area under Wildlife Protection Act, 1972	बायोस्फीयर कोर क्षेत्र / राष्ट्रीय उद्यान
—			

32.2 Wildlife Sanctuaries of Uttar Pradesh

All 25 Wildlife Sanctuaries of UP

#	Sanctuary	District(s)	Key fact
1	Bakhira	Sant Kabir (कबीर) Nagar	Ramsar waterfowl
2	Chandraprabha	Chandauli	Early WLS (1957)
3	Hastinapur	Meerut–Muzaffarnagar–Ghaziabad–Bijnor–Amroha	Largest
4	Kaimur	Mirzapur (मिर्ज़ापुर), Sonbhadra	Vindhyan forest
5	Katarniaghat	Bahraich	1975 ; Terai Arc
6	Kishanpur	Lakhimpur Kheri	Dudhwa corridor
7	Lakh Bahosi	Kannauj (कन्नौज)	Lake birds
8	Mahavir Swami (स्वामी)	Lalitpur	Bundelkhand
9	National Chambal	Agra, Etawah (+ MP, RJ)	Gharial/dolphin
10	Nawabganj	Unnao	Ramsar birds

#	Sanctuary	District(s)	Key fact
11	Okhla	GB Nagar, Ghaziabad	NCR wetland
12	Parvati Arga	Gonda	Ramsar
13	Patna (पटना) Bird	Etah	Waterfowl
14	Ranipur	Chitrakoot (चित्रकूट), Banda	Ranipur TR 2023
15	Saman	Mainpuri	Ramsar
16	Samaspur	Rae Bareli	IN UP (Q58)
17	Sandi	Hardoi	Ramsar
18	Shekha Jheel	Aligarh (अलीगढ़)	Ramsar
19	Sohagibarwa	Maharajganj	Indo-Nepal
20	Sohelwa	Shravasti, Balrampur, Gonda	Bhabar (भाबर)–Terai
21	Sur Sarovar (Keetham)	Agra	Ramsar
22	Surha Tal	Ballia	Eastern wetland
23	Turtle	Varanasi (वाराणसी)	Gangetic turtles
24	Vijai Sagar	Mahoba	Bundelkhand lake
25	Sarsai Nawar	Etawah	IN UP (Q58)

32.3 National Parks of India

High-yield NP ↔ Leading states

National Park	State/UT	Extra fact
Jim Corbett	Uttarakhand	India first (1936); rivers Ramganga + Kosi
Dudhwa	Uttar Pradesh	Only UP NP; rhino rehab 1984+
Valmiki	Bihar	NOT UP
Sultanpur / Kalesar	Haryana	Sultanpur trap
Dachigam / Salim Ali	J&K	Hangul; Salim Ali = ex–City Forest
Keoladeo Ghana	Rajasthan	Birds + WH — not lion
Kanha / Bandhavgarh / Panna / Van Vihar	Madhya Pradesh	Kanha hard-ground barasingha / Bhoorsingh ; Bandhavgarh white tiger; Van Vihar = Bhopal
Gir	Gujarat	Asiatic lion
Kaziranga / Manas	Assam	Rhino / WH; Kaziranga rivers Brahmaputra–Diphlu
Periyar / Silent Valley / Eravikulam	Kerala	Elephant–tiger; Kunthipuzha ; Nilgiri Tahr (“cloud goat”)
Bandipur / Nagarhole (Rajiv Gandhi)	Karnataka	Western Ghats
Hemis	Ladakh	Largest NP
Keibul Lamjao	Manipur	Loktak floating / Sangai
Mollem	Goa	—
Indravati	Chhattisgarh	Bastar; NP 1981 / TR 1983
Betla (Palamu)	Jharkhand	Project Tiger (प्रोजेक्ट टाइगर)
Simlipal / Bhitarkanika	Odisha	Simlipal ≠ MP; saltwater crocodile
Sundarbans / Buxa	West Bengal	Mangrove + WH; Buxa Jalpaiguri

National Park	State/UT	Extra fact
Desert / Ranthambore (रणथंभौर) / Sariska	Rajasthan	GIB; both tiger reserves; Sariska Alwar
Great Himalayan / Pin Valley	Himachal Pradesh	GHNP = Kullu WH 2014; Pin = Lahaul–Spiti (not J&K)
Rajaji / Govind / Valley of Flowers	Uttarakhand	Rajaji TR 2015 ; Govind = Tons source; VoF alpine
Khangchendzonga	Sikkim	Mixed WH 2016
Namdapha	Arunachal Pradesh	Eastern Himalaya (हिमालय) hotspot; tropical→arctic climate belts
Gulf of Mannar	Tamil Nadu	Marine NP 1980
Intanki / Ntangki	Nagaland	—
Guindy	Tamil Nadu	—
Papikonda	Andhra Pradesh	—
Phawngpui	Mizoram	Blue Mountain

River / species / wrong-pair keys

Stem	Correct lock
Corbett river	Ramganga + Kosi — not Ganga (गंगा)
Kaziranga river	Brahmaputra / Diphlu — not Manas
Silent Valley river	Kunthipuzha — not Kaveri (कावेरी)
Ghana / Bharatpur	Birds — not lion
Eravikulam	Nilgiri Tahr — not Hoolock
Hoolock Gibbon	NE India / Assam
Periyar	Elephant / tiger — not Hangul
Desert NP	GIB; two districts ; habitation exists
Nokrek / Loktak / Namdapha	Garo (गारो) Hills / Manipur lake / Arunachal — only Nokrek–Garo matched among classic three-pair stems

32.4 Marine National Parks

#	Marine NP	State/UT	Fact
1	Gulf of Kutch	Gujarat	First
2	Gulf of Mannar	Tamil Nadu	Dugong / GoM BR
3	Mahatma Gandhi Marine	Andaman	Wandoor
4	Rani Jhansi Marine	Andaman	Ritchie's Archipelago

32.5 Important National Parks of the World

Park	Country	Fact
Yellowstone	USA	World first 1872
Banff	Canada	1885
Serengeti	Tanzania	Savanna (सवाना) mega-fauna
Kruger	South Africa	Classic African NP

Park	Country	Fact
Kakadu	Australia	Wetland/rock art landscape
Galápagos	Ecuador	Evolution / WH

32.6 National Parks as Natural Heritage

Site	State	WH note
Kaziranga / Manas	Assam	1985 batch
Keoladeo	Rajasthan	Also Ramsar
Sundarbans	West Bengal	Mangrove WH
Nanda Devi + VoF	Uttarakhand	Alpine
Western Ghats	Multi-state	Serial 2012
Great Himalayan	Himachal	2014
Khangchendzonga	Sikkim	Mixed 2016

32.7 Coastal National Parks

Coastal / marine park	Coast	Fact
Sundarbans NP	East	Mangrove + WH
Bhitarkanika NP	East	Mangrove / croc
Gulf of Mannar Marine NP	East	Coral islands
Gulf of Kutch Marine NP	West	First Marine NP
MG / Rani Jhansi Marine	Andaman	Island marine

Common Traps

Trap	Correct
Valmiki / Sultanpur in UP	Bihar / Haryana
Dudhwa = Pilibhit district	Lakhimpur Kheri
First Marine NP = Sundarbans	Gulf of Kutch
World first NP = Corbett	Yellowstone 1872
Simlipal = MP	Odisha
Keoladeo = UP	Rajasthan
Samaspur / Sarsai Nawar not in UP	Both are in UP
Gahirmatha = Marine NP	Marine sanctuary
UP has UNESCO natural WH	None
Hangul = Gir	Dachigam
NP boundary needs Parliament	State Legislature
Corbett river = Ganga	Ramganga + Kosi
Silent Valley river = Kaveri	Kunthipuzha
Ghana = lion	Birds (Keoladeo)

Trap	Correct
Pin Valley = J&K	Himachal (Lahaul–Spiti)
Eravikulam = Hoolock	Nilgiri Tahr
Periyar = Hangul	Elephant / tiger
Desert NP = no people	Habitation exists ; GIB yes
Nagarhole = Andhra	Karnataka (Rajiv Gandhi NP)
Bandipur = Tamil Nadu	Karnataka

Topic 33 — Biosphere Reserves

Current Affairs (this topic)

- WNBR new listings drift the UNESCO count; India national BR list stays the teaching spine.
- Biosphere ≠ WPA statutory category.

Confused Pairs

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
Biosphere Reserve (BR)	National Park (NP)	Landscape-level conservation integrating human livelihoods (MAB 3-tier zoning) vs species/habitat preservation strictly excluding human activity	जैवमंडल आरक्षित क्षेत्र / राष्ट्रीय उद्यान
National BRs in India	WNBR (UNESCO) Listed	Total 18 Biosphere Reserves designated by MoEFCC vs 12 inducted into UNESCO's World Network of Biosphere Reserves	18 राष्ट्रीय बायोस्फीयर / 12 यूनेस्को WNBR सूची
First BR in India	Latest BR added to WNBR	Nilgiri (1986, TN/KER/KAR) vs Panna (2020, Madhya Pradesh)	प्रथम बायोस्फीयर (नीलगिरि 1986) / नवीनतम यूनेस्को (पन्ना 2020)
Core Zone	Buffer Zone	Legally protected untouched sanctum sanctorum vs zone surrounding core where research, education, and limited tourism are permitted	कोर क्षेत्र (पूर्णतः सुरक्षित) / बफर क्षेत्र (सीमित गतिविधि)
Buffer Zone	Transition Zone	Managed area buffering the core vs outermost area where local communities, agriculture, and settlements sustainably coexist	बफर जोन / संक्रमण क्षेत्र (बस्तियां/खेती)
Nokrek Biosphere Reserve	Loktak Lake	Located in Garo Hills, Meghalaya (Citrus indica) vs freshwater wetland in Manipur	नोकरेक (मेघालय) / लोकटक (मणिपुर)
Uttar Pradesh BR count	Madhya Pradesh BR count	Zero Biosphere Reserves in UP vs 3 Biosphere Reserves in MP (Pachmarhi, Achanakmar-Amarkantak, Panna)	यूपी (शून्य बायोस्फीयर) / म.प्र. (3 बायोस्फीयर)
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33.2 Major Biosphere Reserves of India

All 18 Biosphere Reserves of India

#	Biosphere Reserve	State(s)	Notified	WNBR note
1	Nilgiri	TN, KA, KL	1986	First; WNBR 2000
2	Nanda Devi	Uttarakhand	1988	WNBR 2004
3	Nokrek	Meghalaya	1988	Smallest; WNBR 2009
4	Gulf of Mannar	Tamil Nadu	1989	Marine; WNBR 2001
5	Sundarbans	West Bengal	1989	WNBR 2001
6	Manas	Assam	1989	Removed UNESCO 2011
7	Great Nicobar	A&N	1989	WNBR 2013
8	Simlipal	Odisha (ओडिशा)	1994	WNBR 2009
9	Dibru-Saikhowa	Assam	1997	National only
10	Dihang-Dibang	Arunachal Pradesh	1998	National only

#	Biosphere Reserve	State(s)	Notified	WNBR note
11	Pachmarhi	Madhya Pradesh	1999	WNBR 2009
12	Khangchendzonga	Sikkim	2000	WNBR 2018
13	Agasthyamalai	Kerala, Tamil Nadu	2001	WNBR 2016
14	Achanakmar–Amarkantak	MP, Chhattisgarh	2005	WNBR 2012
15	Great Rann of Kutch	Gujarat	2008	Largest
16	Cold Desert	Himachal Pradesh	2009	Recent WNBR addition
17	Seshachalam Hills	Andhra Pradesh	2010	National only
18	Panna	Madhya Pradesh	2011	Newest; WNBR 2020

33.3 Biosphere Reserves of Uttar Pradesh

State	BR count	Fact
Uttar Pradesh (उत्तर प्रदेश)	0	Any “UP BR” option is false
Uttarakhand	1	Nanda Devi
Madhya Pradesh	3	Pachmarhi, Achanakmar–Amarkantak, Panna
Bihar	0	Valmiki = NP, not BR

33.4 Biosphere Reserves of the World

World BR / landscape	Country	Fact
Yellowstone	USA	Iconic (साकार) MAB / NP landscape
Serengeti–Ngorongoro	Tanzania	Savanna (सवाना) mega-fauna
Galápagos	Ecuador	Evolution / WH overlap
Lake Baikal	Russia	Freshwater biodiversity
Doñana	Spain	Wetland BR classic
Mount Kenya	Kenya	Afro-alpine BR

Common Traps

Trap	Correct
Nokrek = Manipur	Meghalaya
Simlipal = Madhya Pradesh	Odisha
UP has a Biosphere Reserve	Zero
Nanda Devi = UP	Uttarakhand
Manas no longer a BR	Still national BR; UNESCO removed 2011
BR = WPA category	Executive / MAB , not WPA Sec 35
Core allows farming	No exploitative use in core
Dudhwa / Valmiki = UP BR	NP / Bihar NP — not UP BR
Agasthyamalai = Assam	Kerala (+ TN)
First BR = Nanda Devi	Nilgiri 1986

Trap	Correct
BR only for a few species	Conserves all forms of life + sustainable use
Largest BR = Nokrek	Great Rann of Kutch (national); Gulf of Mannar often largest UNESCO-certified
Khangchendzonga = HP	Sikkim ; Cold Desert = HP
Bundala = India	Sri Lanka
Nallamalai = Indian BR	Not a notified BR
Transition = no people	Transition = settlements / livelihoods

Topic 34 — Climate Change (Advanced)

Current Affairs (this topic)

- Advanced climate CA mirrors Climate Change — NDC / net zero / extreme-event science updates.
- Keep adaptation vs mitigation and Montreal≠Paris traps identical to the core climate chapter.

Confused Pairs

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
Global Warming	Climate Change	Long-term rise in Earth's average surface temperature vs broader spectrum of changes including shifts in precipitation, weather extremes, and sea levels	भूमंडलीय तापन / जलवायु परिवर्तन
Climate Mitigation	Climate Adaptation	Actions to reduce or prevent GHG emissions (renewable energy, carbon taxes) vs actions to adjust and build resilience to actual or expected climate impacts (seawalls, drought crops)	शमन (उत्सर्जन कटौती) / अनुकूलन (अनुकूलनशीलता)
Carbon Sink	Carbon Sequestration	Natural or artificial reservoir absorbing more carbon than it releases (oceans, forests) vs the process/technique of capturing and storing carbon dioxide	कार्बन सिंक (अवशोषक) / कार्बन पृथक्करण (भंडारण प्रक्रिया)
Carbon Credit	Carbon Tax	Tradable permit allowing emission of 1 tonne of CO ₂ equivalent under cap-and-trade vs direct price levied by government per tonne of GHG emitted	कार्बन क्रेडिट (1 टन CO ₂ e) / कार्बन कर
IPCC	UNFCCC	Scientific assessment body established in 1988 by WMO/UNEP (doesn't conduct research) vs multilateral legal treaty framework adopted at Rio 1992	आईपीसीसी (वैज्ञानिक निकाय) / यूएनएफसीसीसी (कानूनी संधि)
Emissions Gap Report	Adaptation Gap Report	Annual UNEP report comparing actual NDC pledges vs Paris 1.5°C targets vs annual UNEP assessment of adaptation finance and planning shortfalls	उत्सर्जन अंतराल रिपोर्ट / अनुकूलन अंतराल रिपोर्ट
Kyoto Protocol (1997)	Paris Agreement (2015)	Top-down binding targets on developed countries (Annex-I) vs bottom-up voluntary Nationally Determined Contributions (NDCs) applicable to all	क्योटो प्रोटोकॉल (1997) / पेरिस समझौता (2015)
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34.15 Climate Change Policy of India

Year	Fact
2008	NAPCC
2015	Paris + first NDC + ISA
2021	Net zero 2070 (COP-26)
2022	Updated NDC; LiFE launch
2023	CCTS / Green Credit

34.16 IPCC Assessment Reports

Report	Year	Fact
AR1	1990	Helped lead to UNFCCC

Report	Year	Fact
AR4	2007	Nobel year
AR5	2014	Informed Paris
AR6 SYR	2023	Latest synthesis

Common Traps

Trap	Correct
Emissions Gap = IPCC	UNEP
Montreal = climate mitigation	Ozone ; Kyoto = climate
Carbon credit = Montreal	Kyoto
Green Credit = carbon credit	Not identical
IPCC sets emission targets	Science only
Absolute emitter rank = 10th	~ 3rd absolute; 2023 Q109 keyed 10th
Argon is a GHG	Not
Fourier missing / modern invent	Joseph Fourier (1820s)
GCF = Durban	Cancun 2010
Earth Hour = UNEP	WWF
Net zero = absolute zero	Balance with removals
Nuclear = NAPCC mission	Not listed
Milankovitch includes solar irradiance	No — eccentricity/obliquity/precession
CDM projects only in Annex-I	Projects in developing countries
Argon is a GHG	Not
India net zero 2050	2070
LiFE idea at COP-25	COP-26
PAT = carbon credit trading	ESCert s (efficiency)

Topic 35 — Atmosphere

Current Affairs (this topic)

- Atmosphere layer and composition facts are stable; WMO GHG / ozone bulletins are light CA.
- Troposphere weather vs stratosphere ozone remains the classic layer trap.

Confused Pairs

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
Troposphere	Stratosphere	Lowest weather layer, temperature decreases with altitude (normal lapse rate 6.5°C/km) vs calm dry layer where temperature increases with height due to UV absorption by ozone	क्षोभमंडल (मौसम परत) / समतापमंडल (ओजोन परत)
Stratosphere	Mesosphere	Contains protective ozone layer; stable air suited for jet aircraft vs coldest atmospheric layer (-90°C) where incoming meteors burn upon entry	समतापमंडल / मध्यमंडल (उल्कापिंड दहन)
Mesosphere	Thermosphere (Ionosphere)	Temperature plummets to lowest levels vs temperature surges past 1500°C; contains ionized gas reflecting terrestrial radio waves and hosting auroras	मध्यमंडल / तापमंडल (आयनमंडल)
Homosphere	Heterosphere	Lower ~85 km where atmospheric gases remain uniformly mixed in proportion vs upper layer where gases stratify by molecular weight (N ₂ , O, He, H)	सममंडल (समान मिश्रण) / विषममंडल (गुरुत्वाकर्षण परते)
Insolation	Terrestrial Radiation	Shortwave electromagnetic solar radiation received by Earth vs outgoing longwave infrared radiation re-radiated by Earth's heated surface	सूर्यताप (लघुतरंग) / भौमिक विकिरण (दीर्घतरंग)
Good Ozone	Bad Ozone	Naturally occurring stratospheric ozone shielding Earth from solar UV-B vs ground-level tropospheric ozone acting as secondary pollutant and respiratory irritant	अच्छा ओजोन (समतापमंडल) / बुरा ओजोन (क्षोभमंडल)
—			

35.2 Atmospheric Composition

Gas	Approx. %	Key fact
N ₂	78%	Not GHG
O ₂	21%	Respiration
Ar	0.93%	Inert; not GHG
CO ₂	0.04%	GHG
H ₂ O	0–4%	Variable

35.3 Atmospheric Layers

Layer	Approx. height	Temp trend	Signature
Troposphere	0–8/18 km	↓ upward	Weather
Stratosphere	~12–50 km	↑ upward	Ozone
Mesosphere	~50–85 km	↓ upward	Meteors; coldest

Layer	Approx. height	Temp trend	Signature
Thermosphere	~85–600 km	↑ upward	Aurora / ISS
Exosphere	~600 km+	Transition	Space merge

Common Traps

Trap	Correct
Ozone layer in troposphere	Stratosphere
Atmosphere heated mainly by shortwave	Longwave from surface
N ₂ is a GHG	Not ; N ₂ O is
Temp falls in every layer	Tropo↓ Strato↑ Meso↓ Thermo↑
Weather in stratosphere	Troposphere only
Meteors burn in troposphere	Mesosphere
Max ozone in mesosphere/exosphere	Stratosphere
Argon is a GHG	Not
Thermosphere “feels” very hot	High T, low density
Composition % = climate importance	Trace CO ₂ still critical

Topic 36 — Ozone Layer

Current Affairs (this topic)

- Twin of Ozone Layer core — keep Montreal / Kigali as ozone-family CA only.
- Prefer one consistent ODS story across both ozone chapters.

Confused Pairs

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
Stratospheric Ozone	Tropospheric Ozone	Beneficial atmospheric filter blocking lethal solar UV radiation vs harmful photochemical smog component damaging human lungs and crop yields	समतापमंडलीय ओजोन (सुरक्षा कवच) / क्षोभमंडलीय ओजोन (प्रदूषक)
Ozone Depletion	Ozone Hole	Global thinning of the stratospheric ozone layer by ODS vs severe seasonal Antarctic drop in total column ozone below 220 Dobson Units (DU)	ओजोन क्षरण / ओजोन छिद्र (<220 DU)
Vienna Convention (1985)	Montreal Protocol (1987)	Non-binding framework agreement establishing ozone research/cooperation vs binding international protocol mandating specific phase-out schedules for ODS	वियना कन्वेंशन (1985) / मॉन्ट्रियल प्रोटोकॉल (1987)
Chlorofluorocarbons (CFCs)	Hydrofluorocarbons (HFCs)	Chlorine-bearing potent ozone depleters + GHGs (phased out under Montreal) vs zero ODP non-ozone-depleting refrigerants that are super greenhouse gases (phased down under Kigali)	सीएफसी (ओजोन क्षयकारी) / एचएफसी (ग्रीनहाउस गैस)
Ozone Depletion Potential (ODP)	Global Warming Potential (GWP)	Measure of substance's ability to destroy O ₃ relative to CFC-11 (=1.0) vs measure of heat-trapping efficiency relative to CO ₂ (=1.0) over 100 years	ओजोन रिक्तीकरण क्षमता (ODP) / वैश्विक तापन क्षमता (GWP)
Dobson Unit (DU)	Parts Per Million (ppm)	Standard measure of atmospheric column ozone thickness (1 DU = 0.01 mm at STP; normal = ~300 DU) vs concentration ratio of gases in volume	डॉबसन यूनिट (DU) / पीपीएम (ppm)
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36.4 Ozone Depleting Substances (ODS)

ODS family	Typical use	Fact
CFCs	Cooling, aerosols, foam, electronics cleaners	Main CI source
Halons (1211, 1301)	Fire fighting	Br; high ODP
CCl ₄ / methyl chloroform / CH ₃ Br	Solvent / fumigant	Controlled ODS
HCFCs	Transitional coolants	Lower ODP; being phased out
HFCs	Coolants	Not ODS; Kigali / GWP

36.6 Vienna Convention

Treaty	Year	Role
Vienna Convention	1985	Framework / cooperation
Montreal Protocol	1987	Binding ODS controls

Treaty	Year	Role
Kigali Amendment	2016 (force 1 Jan 2019)	HFC phase-down under Montreal

Common Traps

Trap	Correct
Montreal = climate mitigation	Ozone / ODS ; Kyoto = climate
Ozone layer in troposphere	Stratosphere (~90% of O ₃)
Ozone hole over India like Antarctica	Antarctic spring feature
Vienna = detailed CFC ban alone	Framework ; Montreal = phase-out
HFC = classic ODS	High GWP ; Kigali track
Ozone Day = 5 June	16 September
Hole = physical hole	Column O ₃ < 220 DU
Good ozone at ground level	Ground O ₃ = bad pollutant
CFC not a greenhouse gas	CFCs are GHGs and ODS
CFC main source = automobiles	Refrigerants / foam / aerosols / electronics cleaners
CFC use = tubeless tyres	Not a classic CFC use
Fridge gas = methane	Mafron /halocarbon; large plants may use ammonia
Ozone absorbs IR as UV-shield role	Protective role = UV
UV harm key = liver/brain cancer	Classic key = skin cancer
Carbon trading = Montreal	Kyoto
Agenda 21 = ozone treaty	Sustainable Development
Kyoto = save water / ozone	GHG / climate
Ionosphere in homosphere	Homosphere ≈ tropo + strato + meso; ionosphere not included
Methyl chloroform not an ODS	It is Montreal-controlled ODS

Topic 37 — Greenhouse Gases

Current Affairs (this topic)

- WMO GHG Bulletin and methane CA refresh this twin of Climate Change (जलवायु परिवर्तन).
- GWP order traps (CO_2 / CH_4 / N_2O / fluorinated) stay core.

Confused Pairs

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
Global Warming Potential (GWP)	Radiative Forcing	Heat absorption efficiency of 1 kg of a gas over 100 years relative to CO_2 (=1) vs net change in energy balance of Earth (W/m^2) caused by that gas	वैश्विक तापन क्षमता (GWP) / विकिरण दबाव
Carbon Dioxide (CO_2)	Water Vapour (H_2O)	Primary anthropogenic driver of long-term climate change (atmospheric residence 100+ yrs) vs dominant natural greenhouse gas acting as a fast temperature feedback	कार्बन डाइऑक्साइड (मानवजनित) / जलवाष्प (प्राकृतिक फीडबैक)
Methane (CH_4)	Nitrous Oxide (N_2O)	GWP ~28–30 (residence ~12 yrs; rice paddies, enteric livestock, wetlands) vs GWP ~265–273 (residence ~114 yrs; chemical fertilizers, adipic acid)	मीथेन (धान/पशुपालन) / नाइट्रस ऑक्साइड (उर्वरक)
Sulfur Hexafluoride (SF_6)	Carbon Dioxide (CO_2)	Most potent GHG known with GWP >23,500 (used in electrical switchgear) vs benchmark baseline GHG with GWP = 1	सल्फर हेक्साफ्लोराइड (GWP 23,500) / CO_2 (मानक 1)
Kyoto Six GHGs	Montreal Phase-Out Gases	CO_2 , CH_4 , N_2O , HFCs, PFCs, SF_6 (later NF_3 added) targeted for climate reduction vs CFCs, Halons, CCl_4 , HCFCs targeted for stratospheric ozone protection	क्योटो 6 ग्रीनहाउस गैसों / मॉन्ट्रियल ओजोन गैसों
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37.7 Hydrofluorocarbons (HFCs)

Gas	ODS?	Climate role	Treaty home
CFC	Yes	High GWP GHG	Montreal
HFC	No	High GWP GHG	Kyoto + Kigali
HCFC	Yes (lower)	Transitional	Montreal phase-out

Common Traps

Trap	Correct
Argon / N_2 / O_2 are GHGs	Not
Hydrogen / propane are GHGs	Not
Paddy gas = CO_2 or CO	Methane
Laughing gas = ammonia / CFC	N_2O
Biogas = butane / CO	CH_4 + CO_2
Highest GWP = most total warming	CO_2 dominates by volume
Water vapour = primary human driver	Feedback / most abundant natural GHG
CFC is not a GHG	It is (and ODS)

Trap	Correct
SO ₂ warms Earth directly	Indirect / acid-rain role
Fourier missing	Joseph Fourier (1820s)
HFC = ODS	GHG only ; Kigali
Stratospheric O ₃ = smog pollutant	UV shield ; bad O ₃ is tropospheric
CNG = butane	Methane

Topic 38 — Pollution (Advanced)

Current Affairs (this topic)

- Advanced pollution CA mirrors Pollution & Waste — AQI / CAQM / waste-rule updates.
- Keep criteria-pollutant chemistry pairs consistent with the core pollution chapter.

Confused Pairs

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
Primary Air Pollutant	Secondary Air Pollutant	Emitted directly from identifiable sources (CO, SO ₂ , NO, PM) vs formed in atmosphere through photochemical reactions (Ozone, PAN, Smog, Acid Rain)	प्राथमिक प्रदूषक (सीधे उत्सर्जित) / द्वितीयक प्रदूषक (रासायनिक क्रिया)
Biological Oxygen Demand (BOD)	Chemical Oxygen Demand (COD)	Amount of dissolved O ₂ required by aerobic bacteria to decompose organic matter vs amount of oxygen required to chemically oxidize all biodegradable and non-biodegradable matter	जैव रासायनिक ऑक्सीजन मांग (BOD) / रासायनिक ऑक्सीजन मांग (COD)
Classical (London) Smog	Photochemical (Los Angeles) Smog	Occurs in cool, humid winter mornings; sulfurous, reducing, coal-smoke based vs occurs in dry, sunny afternoons; oxidizing, auto-exhaust NO _x + VOCs generating ground ozone and PAN	लंदन स्मॉग (सल्फ्यूरस, अपचायक) / प्रकाश रासायनिक स्मॉग (ऑक्सीकारक)
Central Pollution Control Board (CPCB)	Central Ground Water Board (CGWB)	Statutory body under Water Act 1974 for overall air/water quality monitoring vs apex subordinate office under Jal Shakti Ministry for groundwater management	सीपीसीबी (प्रदूषण नियंत्रण) / सीजीडब्ल्यूबी (भूजल बोर्ड)
SAFAR	NCAP	Ministry of Earth Sciences real-time metropolitan air quality forecasting system vs MoEFCC national targeted programme to cut PM _{2.5} /PM ₁₀ by 20–30% across 131 non-attainment cities	सफर (सटीक पूर्वानुमान प्रणाली) / राष्ट्रीय स्वच्छ वायु कार्यक्रम
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38.3 Air Quality Index (AQI)

AQI	Category
0–50	Good
51–100	Satisfactory
101–200	Moderate
201–300	Poor
301–400	Very Poor
401–500	Severe

38.16 Heavy Metal Pollution

Metal / pollutant	Classic disease / effect
Cr	Kanpur tanneries
Hg	Minamata / neurological
Pb	Neurotoxin; IQ / CNS

Metal / pollutant	Classic disease / effect
Cd	Itai-itai / bone–kidney
As	Black Foot / cancers
F	Knock-knee / fluorosis
Asbestos	Lung scarring / cancer

Common Traps

Trap	Correct
Lichens thrive in polluted air	Sensitive; die back
Fly ash = secondary pollutant	Primary ; smog is secondary
Fly ash = SiO ₂ /CaO only, no toxics	Also Pb/As/Cu -type metals
BOD = blood oxygen test	Aquatic pollution assay
Groundwater report = CPCB	CGWB
Pesticide = biodegradable	Not
Photochemical smog without O ₃	Always contains ozone
PAN = primary stack gas	Secondary ; eye irritant
AQI includes CO ₂ /CH ₄	Eight pollutants — not those GHGs
Most important indoor = SO ₂	Classic answer often radon
Knock-knee = mercury	Fluoride ; Minamata = Hg
Eutrophication = air pollution	Water nutrient overload
Cyclone divider = control device	Not a standard name
CCAC = unique G20 initiative	Voluntary partnership; focus CH₄ / BC / HFCs
Bioremediation cleans all Cd/Pb	Not readily/completely
Noisiest UP city distractors	Moradabad (Frontier 2022)
Arsenic only = India exclusive	Classic Bangladesh fact
NCAP = SAFAR	Programme vs forecast system
Industrial noise day = 55 dB	75 dB (CPCB day)
Green muffler = soil fix	Noise (tree belts)
Bhopal gas = CO/SO ₂	MIC , Dec 1984

Topic 39 — Acid Rain

Current Affairs (this topic)

- Twin of Acid Rain core — chemistry fixed; rare CA only.
- Acid rain ≠ photochemical smog ≠ ozone depletion (ओजोन क्षरण).

Confused Pairs

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
Normal Rain	Acid Rain	Slightly acidic with pH ~5.6 due to dissolved atmospheric CO ₂ forming weak carbonic acid (H ₂ CO ₃) vs precipitation with pH < 5.6 caused by anthropogenic SO ₂ and NO _x	सामान्य वर्षा (pH 5.6) / अम्ल वर्षा (pH < 5.6)
Sulfuric Acid (H ₂ SO ₄)	Nitric Acid (HNO ₃)	Contributes ~60–70% of total acid precipitation (emitted mainly by coal thermal power plants) vs contributes ~30–40% (emitted by automobile engines and fertilizers)	सल्फ्यूरिक अम्ल (60-70%) / नाइट्रिक अम्ल (30-40%)
Wet Acid Deposition	Dry Acid Deposition	Acidic compounds deposited on Earth via rain, fog, sleet, or snow vs acidic gases, sulfate, and nitrate particles settling directly on soil and buildings without precipitation	आर्द्र निक्षेपण (वर्षा/कोहरा) / शुष्क निक्षेपण (धूल/गैस कण)
Marble Cancer	Eutrophication	Yellowing and pitting of calcium carbonate monuments (Taj Mahal) reacting with sulfuric acid to form soluble gypsum vs excess nutrient loading in water bodies causing algal blooms	संगमरमर कैंसर (ताजमहल क्षरण) / सुपोषण (शैवाल प्रस्फुटन)
Göteborg Protocol (1989)	Montreal Protocol (1987)	UNECE protocol to abate acidification, eutrophication, and ground-level ozone (SO ₂ , NO _x , VOCs, NH ₃) vs UNEP protocol phasing out ozone-depleting chlorofluorocarbons	गोथेनबर्ग प्रोटोकॉल (अम्लीकरण) / मॉन्ट्रियल प्रोटोकॉल (ओजोन)
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39.1 Acid Rain

pH	Meaning
< 5.6	Acid rain threshold
~5.6	Normal rain (CO ₂)
7	Neutral (not natural rain)
> 7	Basic

39.3 Effects

Receptor	Effect
Lakes/rivers	Fish kills below ~pH 5
Soil	Nutrient leach + Al ³⁺
Forests	Conifer damage
Marble	Gypsum crust (Taj)
Lichens	Die in high SO ₂

39.4 Prevention

Tool	Cuts
FGD	SO ₂
Catalytic converter / BS-VI	NO _x + fuel S
Fuel switch	Both
Taj Trapezium	Local heritage protection
Gothenburg/CLRTAP	Transboundary acidification

Common Traps

Trap	Correct
Normal rain pH = 7	~5.6
Acidic pH > 7	Acidic < 7
CO ₂ = main severe acid rain gas	Weak only ; SO ₂ /NO _x drive strong acids
Montreal = acid rain	Ozone ; Gothenburg/CLRTAP for acidifying
Acid rain = only liquid rain	Includes dry deposition
SO ₂ lighter than air	Heavier
Taj damage from CO ₂ alone	SO₂ + acid → gypsum
Lichens love polluted SO ₂ air	Die / indicate pollution
FGD removes only NO _x	Mainly SO₂
Neutral rain = natural rain	Natural rain already slightly acidic

Topic 40 — Desertification

Current Affairs (this topic)

- Twin of Desertification & Land Degradation — UNCCD / LDN CA refresh.
- Keep desertification ≠ drought-only wording clear.

Confused Pairs

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
Desertification	Land Degradation	Land degradation specifically occurring in arid, semi-arid, and dry sub-humid areas (drylands) from climatic variations and human activities vs any reduction in productive potential of land everywhere	मरुस्थलीकरण (शुष्क क्षेत्र) / भूमि क्षरण (सामान्य भूमि)
Drought	Desertification	Temporary natural period of deficient precipitation over months/years vs persistent, long-term degradation and loss of biological/economic productivity in dryland soils	सूखा (अस्थायी जल संकट) / मरुस्थलीकरण (स्थायी भूमि ह्रास)
Sheet Erosion	Gully Erosion	Uniform, imperceptible removal of thin topsoil layer by surface runoff on gentle slopes vs formation of deep pronounced ravines and gorges rendering land unfit for cultivation (Chambal Badlands)	परत अपरदन (अदृश्य मिट्टी ह्रास) / अवनलिका अपरदन (बीहड़ निर्माण)
UNCCD (1994)	UNFCCC (1992)	Only legally binding international agreement linking environment and development to sustainable land management in drylands vs framework treaty for combating greenhouse gases	मरुस्थलीकरण संधि (UNCCD) / जलवायु संधि (UNFCCC)
Land Degradation Neutrality (LDN)	Net Zero Emissions	Target under SDG 15.3 ensuring no net loss of healthy land resources by 2030 vs balancing anthropogenic GHG emissions with equivalent carbon removals	भूमि क्षरण तटस्थता (LDN) / नेट जीरो उत्सर्जन
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40.2 Land Degradation

Process	Meaning
Water/wind erosion	Topsoil loss
Salinization	Excess irrigation salts
Waterlogging	Poor drainage
Mining / pollution	Direct land damage

40.4 Soil Erosion

Type	Feature
Sheet	Uniform thin topsoil loss
Rill	Small channels
Gully	Deeper channels
Ravine	Deep badlands (Chambal)
Wind	Deflation + abrasion

Common Traps

Trap	Correct
Desertification = only desert expansion	ASDM productivity loss
UNCCD = climate treaty	Desertification ; UNFCCC = climate
UNCCD signed at Rio 1992	Adopted 1994
Afforestation causes degradation	Prevents it
Gully = sheet	Different stages
Drought = desertification	Related, not identical
LDN year 2025/2040	2030
Chambal only in one state	MP–UP–Rajasthan
DPAP = 1994	1973–74
Normal erosion = focus	Accelerated erosion is

Topic 41 — Environmental Monitoring

Current Affairs (this topic)

- CPCB (केंद्रीय प्रदूषण नियंत्रण बोर्ड) / SPCB monitoring network and satellite products are living CA.
- Keep ambient vs emission monitoring distinction clear.

Confused Pairs

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
Environmental Indicator	Environmental Monitoring	Specific quantitative metric reflecting state of an ecosystem (e.g., Lichens for SO ₂ , E. coli for fecal water contamination) vs continuous systematic collection and analysis of environmental data over time	पर्यावरणीय संकेतक / पर्यावरणीय निगरानी
Environmental Impact Assessment (EIA)	Environmental Audit	Prospective preventive study predicting ecological impacts of a project <i>before</i> environmental clearance vs retrospective verification of legal compliance and emissions <i>after</i> commissioning	पर्यावरण प्रभाव आकलन (EIA) / पर्यावरण लेखापरीक्षा
National Air Monitoring Programme (NAMP)	Continuous Ambient Air Quality (CAAQMS)	Nationwide manual sampling network monitoring SO ₂ , NO ₂ , PM10, PM2.5 twice weekly vs automated sensors measuring 12 criteria pollutants with real-time online data	राष्ट्रीय वायु निगरानी (NAMP) / सतत परिवेशी वायु निगरानी (CAAQMS)
Bioindicator	Biomarker	Organism whose presence/absence indicates environmental quality (Lichens for air, Diatoms/Mayfly for fresh water) vs cellular/biochemical alteration measured in an organism due to chemical exposure	जैव संकेतक (लाइकेन/ मेफ्लाई) / जैव मार्कर
Ringelmann Scale	Decibel (dB)	Visual chart scale (0 to 5) used to measure the apparent density/opacity of industrial smoke emissions vs logarithmic acoustic unit measuring environmental sound levels	रिंगेलमैन पैमाना (धुआं घनत्व) / डेसिबल (ध्वनि तीव्रता)
—			

41.1 Environmental Indicators

Indicator	Medium	High value means
PM2.5 / AQI	Air	Worse air
BOD	Water	More organic pollution
DO	Water	Low = stressed aquatic life
dB	Noise	Louder / possible breach
Ringelmann	Smoke	Denser plume

41.2 Environmental Monitoring

System	Nodal	Medium
NAMP / CAAQMS	CPCB–SPCB	Air
NWMP	CPCB	Surface water
WQMN / GW Quality Report	CGWB	Groundwater

System	Nodal	Medium
SAFAR	IITM	Air + forecast
Noise monitoring	CPCB–SPCB	Sound

41.3 Environmental Audit

Tool	Role
Form V statement	Annual statutory disclosure to SPCB
ISO 14001	Voluntary EMS certification
EC compliance report	Project clearance conditions
SPCB inspection	Enforcement audit on ground

Common Traps

Trap	Correct
BOD = blood/altitude oxygen	Aquatic pollution assay
Ringelmann = fog/noise/water	Smoke
GW Quality Report = CPCB	CGWB
Monitoring = audit	Measure ≠ verify compliance
Lichens thrive in dirty air	Sensitive indicators
SAFAR = IMD alone / CPCB alone	IITM (+ MoEFCC support)
Form V = ISO certificate	Statutory statement to SPCB
COD < BOD always	COD ≥ BOD
NAMP = real-time only	Mostly manual ; CAAQMS = continuous
High BOD = clean water	Dirty organics

Topic 42 — International Environmental Organizations

Current Affairs (this topic)

- UNEP / IUCN / IPCC / GEF role headlines refresh institutional CA.
- HQ and mandate traps stay fixed teaching.

Confused Pairs

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
UNEP	UNDP	UN environmental coordinator established post-1972 Stockholm Conference (HQ: Nairobi, Kenya) vs UN development network producing the Human Development Index (HQ: New York)	यूएनईपी (नैरोबी, पर्यावरण) / यूएनडीपी (न्यूयॉर्क, विकास)
IPCC	UNFCCC	Scientific review panel established in 1988 by WMO/UNEP to assess peer-reviewed climate data (HQ: Geneva) vs legally binding treaty secretariat coordinating annual COPs (HQ: Bonn, Germany)	आईपीसीसी (वैज्ञानिक समीक्षा) / यूएनएफसीसीसी (वार्ता सचिवालय)
IUCN	WWF	Oldest global conservation union (1948, Gland) publishing the Red List of Threatened Species vs private international conservation NGO (1961, Gland) famous for Giant Panda logo and Earth Hour	आईयूसीएन (रेड लिस्ट) / डब्ल्यूडब्ल्यूएफ (पांडा प्रतीक)
CITES	CMS (Bonn Convention)	Regulates commercial cross-border trade in endangered species of wild flora and fauna (1973) vs promotes conservation and habitat protection for migratory wild animal species (1979)	साइट्स (वन्यजीव व्यापार) / बॉन कन्वेंशन (प्रवासी प्रजातियां)
Global Environment Facility (GEF)	Green Climate Fund (GCF)	Multilateral financial mechanism (World Bank 1991) serving CBD, UNCCD, Stockholm, Minamata, and UNFCCC vs dedicated operating financial entity of UNFCCC created at COP16 Cancun	जीईएफ (वैश्विक पर्यावरण कोष) / जीसीएफ (हरित जलवायु कोष)
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42.8 WMO, Greenpeace & US EPA

Body	HQ / year	Job
WMO	Geneva	Weather/climate cooperation; IPCC parent
Greenpeace	Amsterdam	Environmental NGO
US EPA	Est. 1970	US federal environment agency
UNEP	Nairobi	UN environment programme

Org ↔ Report ↔ HQ Card

Org	HQ	Flagship fact
UNEP	Nairobi	Emissions Gap / GEO
UNDP	New York	HDR / HDI
FAO	Rome	SOFO; Rotterdam (+UNEP)
UNESCO	Paris	World Heritage; MAB
IUCN	Gland	Red List

Org	HQ	Flagship fact
WWF	Gland	Living Planet
IPCC	Geneva	Assessment Reports
WMO	Geneva	Meteorology; IPCC co-parent
Greenpeace	Amsterdam	Environmental NGO

Common Traps

Trap	Correct
Emissions Gap = IPCC/UNDP	UNEP
HDR = UNEP	UNDP
UNEP HQ Geneva/NY	Nairobi
IPCC sets NDCs	UNFCCC/COP
IUCN regulates trade	CITES
WWF = IUCN	Same city, different roles
FAO HQ Nairobi	Rome
GEF donors = developed only	False
HDI team = Priesner	Haq-Sen lineage
Limits to Growth = UNESCO	Club of Rome

Topic 43 — International Environmental Agreements

Current Affairs (this topic)

- Twin of Agreements & Conferences — COP / listing CA; treaty ladder identical.
- Ramsar India ~**100** / UP **12** baseline unless a stem cites a later official total.

Confused Pairs

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
Kyoto Protocol (1997)	Montreal Protocol (1987)	Legally binding treaty targeting reduction of 6 greenhouse gases causing climate change vs universally ratified protocol phasing out ozone-depleting chlorofluorocarbons	क्योटो प्रोटोकॉल (जलवायु परिवर्तन) / मॉन्ट्रियल (ओजोन क्षरण)
Kyoto Protocol (1997)	Paris Agreement (2015)	Top-down legally binding reduction targets solely for developed Annex-I countries vs bottom-up voluntary Nationally Determined Contributions (NDCs) applicable to all countries	क्योटो (शीर्ष से नीचे बाध्यकारी) / पेरिस (स्वैच्छिक NDCs)
Basel Convention (1989)	Rotterdam Convention (1998)	Regulates transboundary movements and disposal of hazardous wastes vs enforces Prior Informed Consent (PIC) procedure for hazardous international trade chemicals	बासेल (खतरनाक अपशिष्ट आवागमन) / रॉटरडैम (पूर्व सूचित सहमति)
Stockholm Convention (2001)	Minamata Convention (2013)	Mandates global elimination and phase-out of Persistent Organic Pollutants (POPs - "Dirty Dozen") vs protects human health and environment from anthropogenic emissions of toxic mercury (Hg)	स्टॉकहोम (पीओपी कीटनाशक) / मिनामाता (पारा नियंत्रण)
Ramsar Convention (1971)	CITES (1973)	Intergovernmental treaty dedicated specifically to the conservation and wise use of wetlands of international importance vs treaty controlling illegal commercial wildlife trade	रामसर कन्वेंशन (आर्द्रभूमि) / साइट्स (वन्यजीव व्यापार)
Stockholm Conference (1972)	Rio Earth Summit (1992)	UN Conference on the Human Environment that led to creation of UNEP vs UN Conference on Environment & Development giving birth to Agenda 21, CBD, UNFCCC, and UNCCD	स्टॉकहोम सम्मेलन 1972 / रियो पृथ्वी शिखर सम्मेलन 1992
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Treaty ↔ Issue Matrix

Treaty	Year	Issue
Ramsar	1971	Wetlands
CITES	1973	Wildlife trade
CMS (Bonn)	1979	Migratory species
Basel	1989	Hazardous waste
UNFCCC	1992	Climate framework
CBD	1992	Biodiversity
Kyoto	1997	Climate (Annex-I)
Rotterdam	1998	PIC chemicals
Stockholm	2001	POPs
Minamata	2013	Mercury
Paris	2015	Climate (NDCs)

Common Traps

Trap	Correct
Montreal = climate mitigation	Ozone ; Kyoto/Paris = climate
Kyoto = save water	Climate / GHG
Stockholm Convention = 1972 conference	POPs 2001 vs UNEP conference 1972
CITES = IUCN Red List	Trade rules vs risk assessment
CMS = CITES	Migration vs trade
Ramsar = forests/ozone	Wetlands
CITES born at Rio 1992	1973 Washington
Paris = Kyoto Annex-only model	Universal NDCs
Basel = wildlife trade	Hazardous waste
Minamata = all hazardous waste	Mercury

Topic 44 — Current Environmental Issues

Current Affairs (this topic)

- Primary living CA chapter — NbS, plastics, heat, Himalaya (हिमालय) cryosphere, LiFE.
- Cross-link facts to Climate / Biodiversity / Pollution without relocating treaty years.

Confused Pairs

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
Carbon Footprint	Ecological Footprint	Total amount of greenhouse gases (in tonnes of CO ₂ e) emitted directly/indirectly by an entity vs total biocapacity area (in global hectares, gha) required to sustain a lifestyle	कार्बन पदचिह्न (CO ₂ e उत्सर्जन) / पारिस्थितिक पदचिह्न (भूमि/संसाधन मांग)
Net Zero Emissions	Gross Zero Emissions	Balancing residual carbon emissions with equivalent carbon removals via sinks and DACCS vs completely eliminating all greenhouse gas emissions across all sectors without offsets	नेट जीरो (उत्सर्जन-अवशोषण संतुलन) / सकल शून्य (पूर्ण उत्सर्जन शून्यता)
Green Economy	Circular Economy	Broad macro-framework promoting low-carbon, resource-efficient, and socially inclusive growth vs industrial economic model eliminating waste through closed-loop recycling and repair	हरित अर्थव्यवस्था / चक्रीय अर्थव्यवस्था
Green Hydrogen	Grey Hydrogen / Blue Hydrogen	Produced by electrolysis of water powered entirely by renewable electricity vs produced from natural gas without carbon capture (Grey) or with carbon capture & storage (Blue)	हरित हाइड्रोजन (नवीकरणीय) / धूसर व नीला हाइड्रोजन
Primary Microplastics	Secondary Microplastics	Tiny plastic beads (<5mm) intentionally manufactured for cosmetics and industrial pellets vs microscopic plastic fragments resulting from physical weathering and breakdown of larger plastic debris	प्राथमिक माइक्रोप्लास्टिक (निर्मित मनके) / द्वितीयक माइक्रोप्लास्टिक (अपघटित टुकड़े)
Nature-based Solutions (NbS)	Geoengineering	Ecosystem protection, restoration, and sustainable management to address societal challenges (e.g. mangrove restoration) vs large-scale technological manipulation of Earth's systems (e.g. solar radiation management)	प्रकृति-आधारित समाधान (NbS) / भू-इंजीनियरिंग
Mission LiFE Genesis	Mission LiFE Official Launch	Conceptualized and introduced by PM Modi at COP26 Glasgow in November 2021 vs officially launched jointly with UN Secretary-General António Guterres at Ekta Nagar (Kevadia), Gujarat in October 2022	मिशन लाइफ विचार (ग्लासगो 2021) / औपचारिक शुभारंभ (केवडिया 2022)
—			

44.1 Carbon Footprint

Scope	Type	Example
Scope 1	Direct	Factory (फैक्टरी) boiler
Scope 2	Purchased energy	Grid electricity
Scope 3	Other indirect	Supplier emissions, flights

44.2 Ecological Footprint

Metric	Unit	Rough India
Ecological footprint	gha	~1.0–1.2 / person
Carbon footprint	tCO ₂ e	~2 / person

44.3 Net Zero Emissions

Country/bloc	Net zero year
EU / UK	2050
China	2060
India	2070

44.5 Blue Economy

Sector	Fact
Mangroves	MISHTI 2023
Ports	Sagarmala
Deep sea	Deep Ocean Mission

44.6 Circular Economy

Linear	Circular
Virgin in → dump out	Recycled/regenerative loops
Waste = disposal problem	Waste = design flaw

44.7 Green Hydrogen

Colour	Method
Green	Electrolysis + renewables
Grey	Natural gas reforming
Blue	Grey + CCS
Black/brown	Coal gasification

44.8 Microplastics

Type	Origin
Primary	Made small
Secondary	Fragmented from large items

44.9 Climate Refugees

Type	Example
Sudden-onset	Cyclone / flash flood displacement
Slow-onset	Drought / sea-level rise
Legal refugee (1951)	Persecution — not climate alone

44.11 LiFE Mission

Field	Detail
Idea	COP26 2021
Launch	June 2022
Nodal	MoEFCC
Actions	~75

Common Traps

Trap	Correct
LiFE idea at COP-25 Madrid	COP26 Glasgow 2021 ; launch June 2022
MISHTI initiated 2019	Budget 2023–24
India net zero 2050	2070
Net zero = emissions to absolute zero	Balance with removals
Emissions Gap = IPCC	UNEP
Ecological footprint in tCO ₂ e	Unit = gha
Microplastics <5 cm	<5 mm
Green H ₂ = grey H ₂	Grey = natural gas reforming
Green economy = circular identically	Circular = material loops ; green = broader
Climate refugee = 1951 legal status	Not a formal Convention category
NbS = geoengineering	Ecosystem protect/manage/restore
MISHTI for Ganga basin inland	Coastal mangroves
Highest India per capita carbon footprint	High total , low per capita

Uttarakhand — 01 — Natural Resources & Climate Contribution

Current Affairs (this topic)

- Keep FSI / FRI / WII function traps and Chipko **1973 Reni** stable.
- Himalayan Mission and spring / landslide seasons refresh climate–resource CA.

Confused Pairs

Term	One-line difference
FSI	Forest cover / ISFR — Dehradun
FRI	Forestry research — Dehradun
WII	Wildlife research + training — Dehradun
Chipko	1973 Reni — forest protection movement
Weathering	In-situ rock breakdown
Erosion	Transport of weathered material
NEERI	Nagpur 1958 — CSIR env. engineering
Natural capital	Forests, water, minerals — not roads

1.4 Dehradun Institution Triangle

Institute	Job
FSI	Forest cover / ISFR
FRI	Forestry research
WII	Wildlife science + training

Uttarakhand — 02 — Biodiversity & Protected Areas

Current Affairs (this topic)

- Keep NP notification years and sanctuary–district pairs stable.
- Nanda Devi / Valley of Flowers (फूलों की घाटी) WHS + BR fauna (प्राणीजात) set recycles every year.

Quick Revision — Spine Only

Head	Key
Biogeographic zone	Himalaya (हिमालय) — high endemism; altitudinal belts
Forest richness A/R	A true; R (topography + climate) explains A
Nanda Devi BR	UNESCO WHS (यूनेस्को) (with Valley of Flowers); snow leopard, musk deer, blue sheep
Askot	Pithoragarh (पिथौरागढ़)
Binsar	Almora (अल्मोड़ा)
Govind	Uttarkashi (उत्तरकाशी)
Nandhaur	Nainital (नैनीताल)
Corbett NP	1936
Rajaji (राजाजी) NP	1983
Gangotri (गंगोत्री) NP	1989
Govind NP	1990 (not 1989)

Confused Pairs

Pair	Key
Askot vs Binsar	Pithoragarh vs Almora
Govind 1989 vs 1990	1990 keyed; Gangotri keeps 1989
Nanda Devi fauna set	All three named species in
A true / R not explain	UKPCS keys R explains A
NP vs WLS vs BR	Strict NP / regulated WLS / BR zonation — different legal and management ideas

2.3 Sanctuary–district ratta

Sanctuary	District
Askot	Pithoragarh
Binsar	Almora
Govind	Uttarkashi
Nandhaur	Nainital

Common Traps

1. **Govind NP = 1989 → 1990.**
2. **Askot = Almora** → Pithoragarh.
3. **R does not explain forest richness** → keyed that it does.
4. **BR = only another NP name** → BR uses core–buffer–transition zonation.

Uttarakhand — 03 — Climate Vulnerability & Governance

Current Affairs (this topic)

- Keep SCCC **2011** and UAPCC department trap stable.
- Cloudburst / GLOF seasons refresh vulnerability without changing the institutional keys.

Quick Revision — Spine Only

Head	Key
National climate role	Carbon sink (कार्बन सिंक) · water tower · Himalayan Mission (NAPCC)
Vulnerability	Highly vulnerable mountain state
Documented effects	Water stress · biodiversity / habitat (वास स्थान) pressure
SCCC	State Council for Climate Change (जलवायु परिवर्तन) — 2011
UAPCC prep	Not State Revenue Department — forest/environment (पर्यावरण) set-up
Hazards link	Landslides, cloudbursts, floods, glacial risk
National Panchamrit	COP26 Glasgow climate package — national climate chapters

Confused Pairs

Pair	Key
UAPCC by Revenue Dept	False — forest / environment set-up
SCCC 2011 vs 2009/2013	2011 keyed
Only vulnerability true	2025 keys both vulnerability and water-stress/biodiversity statements
UK separate UNFCCC track	No — contributes via India's NDCs / ecosystem services

Common Traps

1. **UAPCC = Revenue Department** → false.
2. **Rejecting biodiversity impact of climate change** → keyed true with vulnerability.
3. **SCCC from statehood 2000** → **2011**.